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(19) **United States**(12) **Patent Application Publication** (10) **Pub. No.: US 2025/0275542 A1**
Lundkvist et al. (43) **Pub. Date: Sep. 4, 2025**(54) **BAKED AND PAR-BAKED PRODUCTS WITH
THERMOSTABLE AMG VARIANTS FROM
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Ozcomlekci**, Istanbul (TR)(73) Assignee: **NOVOZYMES A/S**, Bagsvaerd (DK)(21) Appl. No.: **18/250,397**(22) PCT Filed: **Nov. 2, 2021**(86) PCT No.: **PCT/EP2021/080317**

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(57)

ABSTRACT

The invention relates to methods of producing a baked or par-baked product, said method comprising a first step of providing a dough comprising a mature thermostable variant of a parent glucoamylase at least 70% identical to SEQ ID NO:1, SEQ ID NO:6, SEQ ID NO:7 or SEQ ID NO:8; and a second step of baking or par-baking the dough to produce a baked or par-baked product, as well as baking compositions comprising said variant and uses of said variant.

Specification includes a Sequence Listing.

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P_oxalicum      RPDPKGGNLTFFIHKEGERSLQGILDNLGGRGKKTPGTAAGLFIASPNTENPNYYYYTWTR
AMG_NL          RPDPKGGNLTFFIHKEGERSLQGILDNLGGRGKKTPGTAAGLFIASPNTENPNYYYYTWTR
AMG_anPAV498    RNDSKGGNLTFFIHKEGERSLQGILDNLGGRGKKTPGTAAGLFIASPNTENPNYYYYTWTR
JPO001          ANDSKGGNLTFFIHKEGERSLQGILDNLGGRGKKTPGTAAGLFIASPNTENPNYYYYTWTR
JPO124          ANDSKSTNLTTFFIHKEGERSLQGILDNLGGFGKYTPGTAAGLFIASPNTENPNYYYYTWTR
JPO172          ANDSKSTNLTTFFIHKEGERSLQGILDNLGGFGKYTPGTAAGLFIASPNTRNPNNYYYTWTR
P_miczynskii   NPTSKDAKMASYISKEGQRSIVGITENLGKGSKTPTGAAGLFIASPNMANPNYYYYTWTR
P_russellii     ---SKDGNLASYSIAKEGQRSIVGITENLGKGSKTPTGAAGLFIASPNMANPNYYYYTWTR
P_glabrum       ---NPFNSLDSEFILKEGARSYQGIIDNLGNKVAPGTAAGLFVASPNTANPDYFYTWTR
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P_oxalicum	DSALTAKCLIDLFDERSRAKFPIDRKYLETGIRDYVSSQAILQSVSNPSTGLKDGSGLGEP
AMG_NL	DSALTAKCLIDLFDERSRAVFPIDRKYLETGIRDYVSSQAILQSVSNPSTGLKDGSGLGEP
AMG_anPAV498	DSALAAKCLIDLFDERSRAVFPIDRKYLETGIRDYVSSQAILQSVSNPSTGLKDGSGLGEP
JPO001	DSALAAKCLIDLFDERSRAVFPIDRKYLETGIRDYVSSQAILQSVSNPSTGLKDGSGLGEP
JPO124	DSALAAKCLIDLFDERSRAVFPIDRKYLETGIRDYVSSQAILQNVSNPSTGLKDGSGLGEP
JPO172	DSALAAKCLIDLFDERSRAVFPIDRKYLETGIRDYVSSQAILQNVSNPSTGLKDGSGLGEP
P_miczynskii	DSALTFKCLIDLFDSDQDY-ISRKELETDIRNYVSSQAVLQNVSNPSTGLKDGSGLGEP
P_russellii	DSALTFKCLIDLFDSDQDY-ISRKELETDIRNYVSSQAVLQNVSNPSTGLKDGSGLGEP
P_glabrum	DSALTFKCLIDLFDGGSTEFGLNSELETDIRNYVSSQAVLQNVSNPSTGLEDTGLGEP
	*****. *****. . **** *.******.* *****.....*****.

P_oxalicum	KFEIDLNPFSGAWGRPQRDGPALRATAMITYANYLISHGQKSDVSQVMWPIIANDLAYVG
AMG_NL	KFEIDLNPFSGAWGRPQRDGPALRATAMITYANYLISHGQKSDVSQVMWPIIANDLAYVG
AMG_anPAV498	KFEIDLNPFSGAWGRPQRDGPALRATAMITYANYLISHGQKSDVSQVMWPIIANDLAYVG
JPO001	KFEIDLNPFSGAWGRPQRDGPALRATAMITYANYLISHGQKSDVSQVMWPIIANDLAYVG
JPO124	KFEIDLNPFSGPWGRPQRDGPALRATAMITYANYLISHGQKSDVSQVMWPIIANDLAYVG
JPO172	KFEIDLNPFSGPWGRPQRDGPALRATAMITYANYLISHGQKSDVSQVMWPIIANDLAYVG
P_miczynskii	KFEIDLNPFSGSWGRPQRDGPALRATAMITYADWLISHGSKSEANIMWPIIANDLAYVG
P_russellii	KFEIDLNPFSGSWGRPQRDGPALRATAMITYADWLIVSHGQKSEATNIMWPIIANDLAYVG
P_glabrum	KFEVDLNPFTGSWGRPQRDGPALRATALITYTNYLLSQGKSEAVNIMWPIISNDLAYVG
	*****.* *****.******.*.....*..* * ..**

P_oxalicum	QYWNNTGFDLWEEVDGSSFFTTIAVQHRALVEGSQLAKKLGKSCDACDSQPPQILCFLQSF
AMG_NL	QYWNNTGFDLWEEVDGSSFFTTIAVQHRALVEGSQLAKKLGKSCDACDSQPPQILCFLQSF
AMG_anPAV498	QYWNNTGFDLWEEVDGSSFFTTIAVQHRALVEGSQLAKKLGKSCDACDSQPPQILCFLQSF
JPO001	QYWNNTGFDLWEEVDGSSFFTTIAVQHRALVEGSQLAKKLGKSCDACDSQPPQILCFLQSF
JPO124	QYWNNTGFDLWEEVDGSSFFTTIAVQHRALVEGSQLAKKLGKSCDACDSQPPQILCFLQSF
JPO172	QYWNNTGFDLWEEVDGSSFFTTIAVQHRALVEGSQLAKKLGKSCDACDSQPPQILCFLQSF
P_miczynskii	QYWNNTGFDLWEEVDGSSFFTLAVQHRSLVQGASLAKKLGKSCPACVSQAPQILCFLQSF
P_russellii	QYWNKTGFDLWEEVDGSSFYTLAVQHRALVQGASLAKKLGKSCCTACVSSQAPQILCFLQSF
P_glabrum	QYWNDTGFDLWEETDGSSFFTLAVQHRALVQGATLAKKLGKSCAACSSQAPQILCFLQSF

P_oxalicum WNGKYITSNINTQASRSGIDLDSVLGSIHTFDPEAACDDATFQPFC SARALANHKVYVDSF
AMG_NL WNGKYITSNINTQASRSGIDLDSVLGSIHTFDPEAACDDATFQPFC SARALANHKVYVDSF
AMG_anPAV498 WNGKYITSNINTQASRSGIDLDSVLGSIHTFDPEAACDDATFQPFC SARALANHKVYVDSF
JPO001 WNGKYITSNINTQASRSGIDLDSVLGSIHTFDPEAACDDATFQPFC SARALANHKVYVDSF
JPO124 WNGKYITSNINTQASRSGIDLDSVLGSIHTFDPEAACDDATFQPFC SARALANHKVYVDSF
JPO172 WNGKYITSNINTQASRSGIDLDSVLGSIHTFDPEAACDDATFQPFC SARALANHKVYVDSF
P_miczynskii WNGNYITANINLDTSRSGIDLNSILGSIHTFDPEASCDDSTFQPFC SARALANHKVYVD AF
P_russellii WNGNYITANINLDTSRSGIDLNSILGSIHTFDPEASCDDSTFQPFC SARALANHKVYVD AF
P_glabrum WNGKYITANINLDTSRTGIDANTLLGSIHTFDPEAACDDSTFQPFC SARALANHKVYVD AF

P_oxalicum	LWFAFVEFIPAGTTPFEYKYYKVEPNGDI-TWEKGNRVFVAPTGCPVQPHSNDVWQF
AMG_NL	LWFAFVEFIPAGTTPFEYKYYKVEPNGDI-TWEKGNRVFVAPTGCPVQPHSNDVWQF
AMG_anPAV498	LWFAFVEFIPAGTTPFEYKYYKVEPNGDI-TWEKGNRVFVAPTGCPVQPHSNDVWQF
JPO001	LWFAFVEFIPAGTTPFEYKYYKVEPNGDI-TWEKGNRVFVAPTGCPVQPHSNDVWQF
JPO124	LWFAFVEFIPAGTTPFEYKYYKVEPNGTI-VWEKGNRVFVAPTGCPVQPHSNDVWRS
JPO172	LWFAFVEFIPAGTTPFEYKYYKVEPNGTI-VWEKGNRVFVAPTGCPVQPHSNDVWRS
P_miczynskii	LWFAFVELVPAGTTPFEYKYYKVEPNGTV-IWENGENDRVYVAPTGCPIQPSQTDVWR-
P_russellii	LWFAFVELVPAGTTPFEYKYYKVEPNGTV-IWENGENDRVYVAPTGCPIQPSQTDIWR-
P_glabrum	LWFGTLELVTAGTTPFEYKYYKIEPNGTVTIWESGDNRVSVVPTGCPIQPSLHDVWRS
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BAKED AND PAR-BAKED PRODUCTS WITH THERMOSTABLE AMG VARIANTS FROM PENICILLIUM

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a 35 U.S.C. 371 national application of international application no. PCT/EP2021/080317 filed Nov. 2, 2021, which claims priority or the benefit under 35 U.S.C. 119 of Danish application nos. PA 2020 01238 and PA 2021 00367 filed Nov. 2, 2020 and Apr. 12, 2021, the contents of which are fully incorporated herein by reference.

REFERENCE TO SEQUENCE LISTING

[0002] This application contains a Sequence Listing in computer readable form, which is incorporated herein by reference.

FIELD OF THE INVENTION

[0003] The invention relates to methods of producing a baked or par-baked product, said method comprising a first step of providing a dough comprising a mature thermostable variant of a parent glucoamylase at least 70% identical to SEQ ID NO:1, SEQ ID NO:6, SEQ ID NO:7 or SEQ ID NO:8; and a second step of baking or par-baking the dough to produce a baked or par-baked product, as well as baking compositions comprising said variant and uses of said variant.

BACKGROUND OF THE INVENTION

[0004] World-wide, baked products (breads, biscuits, etc.) containing sugar is one of the most popular product segments. The recipe amount of sugar will typically be 1-25% of total flour weight.

[0005] However, due to increased market price for sugar, shortage in sugar availability in some parts of the world as well as health-concerns, there is a need for methods of producing baked products that contain a reduced amount of added sugar without sacrificing the quality of the baked product and perhaps even improving it.

[0006] WO 2019/238423 (Novozymes A/S, Denmark) discloses methods of producing a dough with a reduced amount of added sugar comprising adding a raw starch degrading alpha-amylase and a glucoamylase to the dough ingredients.

SUMMARY OF THE INVENTION

[0007] The inventors found that thermostabilized variants of certain glucoamylases showed greatly improved performance in freshkeeping or anti-staling of a baked or par-baked product. Another improved performance of the thermostabilized variants was that they increased the sweetness or sweet taste of the product, which allowed a reduction in the amount of added sugar in traditional recipes.

[0008] Accordingly in a first aspect, the invention relates to method of producing a baked or par-baked product, said method comprising:

[0009] a) providing a dough comprising a mature thermostable variant of a parent glucoamylase at least 70% identical to SEQ ID NO:1, SEQ ID NO:6, SEQ ID NO:7 or SEQ ID NO:8; and

[0010] b) baking or par-baking the dough to produce a baked or par-baked product.

[0011] A second aspect of the invention, relates to baking compositions comprising a mature thermostable variant of a parent glucoamylase as defined in the first aspect.

[0012] Other aspects of the invention relate to uses of the baking compositions of the second aspect for sugar replacement in a method of producing a baked or par-baked product, for increasing the sweetness of a baked or par-baked product, for reducing the amount of sugar in the dough in a method of producing a baked or par-baked product and/or for extending the shelf-life of a baked or par-baked product in a method of producing a baked or par-baked product, as well as in methods as defined in the first aspect, whereby the baked or par-baked product after final bake-off has a reduced initial firmness and/or an increased initial elasticity, and/or a reduced increase in firmness and/or a higher elasticity after 1, 7 or 14 days, when cooled to room temperature, packed in a sealed container and stored at room temperature until analysis, compared to a control made without any added glucoamylase.

[0013] Preferably, the mature thermostable variant of a parent glucoamylase of the invention is at least 71% identical to SEQ ID NO:1, SEQ ID NO:6, SEQ ID NO:7 or SEQ ID NO:8, e.g. at least 72%, e.g. at least 73%, e.g. at least 74%, e.g. at least 75%, e.g. at least 76%, e.g. at least 77%, e.g. at least 78%, e.g. at least 79%, e.g., at least 80%, e.g. at least 81%, e.g. at least 82%, e.g. at least 83%, e.g. at least 84%, e.g., at least 85%, e.g. at least 86%, e.g. at least 87%, e.g. at least 88%, e.g. at least 89%, e.g., at least 90%, e.g., at least 91%, e.g., at least 92%, e.g., at least 93%, e.g., at least 94%, e.g., at least 95%, e.g. at least 96%, e.g., at least 97%, e.g., at least 98%, e.g., at least 99% identical to SEQ ID NO:1, SEQ ID NO:6, SEQ ID NO:7 or SEQ ID NO:8.

FIGURES

[0014] FIG. 1 shows a multiple alignment of the amino acid sequences of the mature proteins of:

[0015] Wildtype AMG from *Penicillium oxalicum* (PoAMG) of SEQ ID NO:1

[0016] PoAMG variant denoted 'AMG NL' of SEQ ID NO:2

[0017] PoAMG variant denoted 'AMG anPAV498' of SEQ ID NO:3

[0018] PoAMG variant denoted 'AMG JPO001' of SEQ ID NO:4

[0019] PoAMG variant denoted 'AMG JPO124' of SEQ ID NO:5

[0020] PoAMG variant denoted 'AMG JPO172' of SEQ ID NO:6

[0021] Wildtype AMG from *Penicillium miczynskii* (PoAMG) of SEQ ID NO:7

[0022] Wildtype AMG from *Penicillium russellii* (PoAMG) of SEQ ID NO:8

[0023] Wildtype AMG from *Penicillium glabrum* (PoAMG) of SEQ ID NO:9

DETAILED DESCRIPTION OF THE INVENTION

Definitions

[0024] Sequence identity: The relatedness between two amino acid sequences or between two nucleotide sequences is described by the parameter "sequence identity".

[0025] For purposes of the present invention, the sequence identity between two amino acid sequences is determined using the Needleman-Wunsch algorithm (Needleman and Wunsch, 1970, *J. Mol. Biol.* 48:443-453) as implemented in the Needle program of the EMBOSS package (EMBOSS: The European Molecular Biology Open Software Suite, Rice et al., 2000, *Trends Genet.* 16:276-277), preferably version 5.0.0 or later. The parameters used are gap open penalty of 10, gap extension penalty of 0.5, and the EBLOSUM62 (EMBOSS version of BLOSUM62) substitution matrix. The output of Needle labelled “longest identity” (obtained using the—no brief option) is used as the percent identity and is calculated as follows:

$$\frac{(\text{Identical Residues} \times 100)}{(\text{Length of Alignment} - \text{Total Number of Gaps in Alignment})}$$

[0026] Variant: The term “variant” means a polypeptide comprising an alteration, i.e., a substitution, insertion, and/or deletion, at one or more (e.g., several) positions. A substitution means replacement of the amino acid occupying a position with a different amino acid; a deletion means removal of the amino acid occupying a position; and an insertion means adding one or more amino acids adjacent to and immediately following the amino acid occupying a position. The amino acid changes may be of a minor nature, that is conservative amino acid substitutions or insertions that do not significantly affect the folding and/or activity of the protein; small deletions, typically of 1-30 amino acids; small amino- or carboxyl-terminal extensions, such as an amino-terminal methionine residue; a small linker peptide of up to 20-25 residues; or a small extension that facilitates purification by changing net charge or another function, such as a poly-histidine tract, an antigenic epitope, or a binding domain. Examples of conservative substitutions are within the groups of basic amino acids (arginine, lysine and histidine), acidic amino acids (glutamic acid and aspartic acid), polar amino acids (glutamine and asparagine), hydrophobic amino acids (leucine, isoleucine and valine), aromatic amino acids (phenylalanine, tryptophan and tyrosine), and small amino acids (glycine, alanine, serine, threonine and methionine). Amino acid substitutions that do not generally alter specific activity are known in the art and are described, for example, by H. Neurath and R. L. Hill, 1979, *In, The Proteins*, Academic Press, New York. Common substitutions are Ala/Ser, Val/Ile, Asp/Glu, Thr/Ser, Ala/Gly, Ala/Thr, Ser/Asn, Ala/Val, Ser/Gly, Tyr/Phe, Ala/Pro, Lys/Arg, Asp/Asn, Leu/Ile, Leu/Val, Ala/Glu, and Asp/Gly.

[0027] Increased strength: The term “increased strength of the dough” is defined herein as the property of a dough that has generally more elastic properties and/or requires more work input to mould and shape compared to a control.

[0028] Increased elasticity: The term “increased elasticity of the dough” is defined herein as the property of a dough which has a higher tendency to regain its original shape after being subjected to a certain physical strain compared to a control.

[0029] Increased stability of the dough: The term “increased stability of the dough” is defined herein as the property of a dough that is less susceptible to mechanical abuse thus better maintaining its shape and volume and is evaluated by the ratio of height:width of a cross section of a loaf after normal and/or extended proof compared to a control.

[0030] Reduced stickiness of the dough: The term “reduced stickiness of the dough” is defined herein as the

property of a dough that has less tendency to adhere to surfaces compared to a control, e.g., in the dough production machinery, and it is either evaluated empirically by the skilled test baker or measured by, e.g., a texture analyser (e.g. TAXT2) as known in the art.

[0031] Improved extensibility: The term “improved extensibility of the dough” is defined herein as the property of a dough that can be subjected to increased strain or stretching without rupture compared to a control.

[0032] Improved machinability: The term “improved machinability of the dough” is defined herein as the property of a dough that is generally less sticky and/or firmer and/or more elastic compared to a control.

[0033] Increased volume of the baked product: The term “increased volume of the baked product” is measured as the volume of a given loaf of bread compared to a control. The volume may be determined as known in the art.

[0034] Improved crumb structure of the baked product: The term “improved crumb structure of the baked product” is defined herein as the property of a baked product with finer cells and/or thinner cell walls in the crumb and/or more uniform/homogenous distribution of cells in the crumb compared to a control and is usually evaluated visually by the skilled baker or by digital image analysis as known in the art (e. g., C-cell, Calibre Control International Ltd, Appleton, Warrington, UK).

[0035] Improved softness of the baked product: The term “improved softness of the baked product” is the opposite of “firmness” and is defined herein as the property of a baked product that is more easily compressed compared to a control and is evaluated either empirically by the skilled test baker or measured by, e.g., a texture analyser (e.g. TAXT2 or TA-XT Plus from Stable Micro Systems Ltd, Surrey, UK) as known in the art.

[0036] Sensory attributes of the baked products: The sensory attributes may be evaluated using procedures well established in the baking industry, and may include, for example, the use of a panel of trained taste-testers.

[0037] Thermostability improvement: The thermostability improvement (Td) in ° C. is a measure of how much the variants have improved in thermostability over their parent glucoamylase under the same conditions, determined as exemplified herein.

[0038] The first aspect of the invention relates to method of producing a baked or par-baked product, said method comprising:

[0039] a) providing a dough comprising a mature thermostable variant of a parent glucoamylase at least 70% identical to SEQ ID NO:1, SEQ ID NO:6, SEQ ID NO:7 or SEQ ID NO:8; and

[0040] b) baking or par-baking the dough to produce a baked or par-baked product.

[0041] A second aspect of the invention, relates to baking compositions comprising a mature thermostable variant of a parent glucoamylase as defined in the first aspect.

[0042] Other aspects of the invention relate to uses of the baking compositions of the second aspect for sugar replacement in a method of producing a baked or par-baked product, for increasing the sweetness of a baked or par-baked product, for reducing the amount of sugar in the dough in a method of producing a baked or par-baked product and/or for extending the shelf-life of a baked or par-baked product in a method of producing a baked or par-baked product, as well as in methods as defined in the

first aspect, whereby the baked or par-baked product after final bake-off has a reduced initial firmness and/or an increased initial elasticity, and/or a reduced increase in firmness and/or a higher elasticity after 1, 7 or 14 days, when cooled to room temperature, packed in a sealed container and stored at room temperature until analysis, compared to a control made without any added glucoamylase.

[0043] Preferably, the mature thermostable variant of a parent glucoamylase of the invention is at least 71% identical to SEQ ID NO:1, SEQ ID NO:6, SEQ ID NO:7 or SEQ ID NO:8, e.g., at least 72%, e.g., at least 73%, e.g., at least 74%, e.g., at least 75%, e.g., at least 76%, e.g., at least 77%, e.g., at least 78%, e.g., at least 79%, e.g., at least 80%, e.g., at least 81%, e.g., at least 82%, e.g., at least 83%, e.g., at least 84%, e.g., at least 85%, e.g., at least 86%, e.g., at least 87%, e.g., at least 88%, e.g., at least 89%, e.g., at least 90%, e.g., at least 91%, e.g., at least 92%, e.g., at least 93%, e.g., at least 94%, e.g., at least 95%, e.g., at least 96%, e.g., at least 97%, e.g., at least 98%, e.g., at least 99% identical to SEQ ID NO:1, SEQ ID NO:6, SEQ ID NO:7 or SEQ ID NO:8.

The Dough

[0044] As used herein “dough” means any dough used to prepare a baked product, in particular a bread.

[0045] According to the present invention, the dough used to prepare a baked product may be made from any suitable dough ingredients comprising flour.

[0046] The flour may be from any baking grain known in the art, such as, wheat flour, corn flour, rye flour, barley flour, oat flour, rice flour, sorghum flour, potato flour, soy flour, and any combinations thereof (e.g., wheat flour combined with one of the other flour sources; or rice flour combined with one of the other flour sources).

[0047] In a preferred embodiment, the flour is wheat flour.

[0048] In a preferred embodiment, at least 10% (w/w) or more of the total flour content is wheat flour, e.g., at least 15% or more of the total flour content is wheat flour, e.g., at least 20% or more of the total flour content is wheat flour, e.g., at least 25% or more of the total flour content is wheat flour, e.g., at least 30% or more of the total flour content is wheat flour, e.g., at least 35% or more of the total flour content is wheat flour, e.g., at least 40% or more of the total flour content is wheat flour, e.g., at least 45% or more of the total flour content is wheat flour, e.g., at least 50% or more of the total flour content is wheat flour, e.g., at least 55% or more of the total flour content is wheat flour, e.g., at least 60% or more of the total flour content is wheat flour, e.g., at least 65% or more of the total flour content is wheat flour, e.g., at least 70% or more of the total flour content is wheat flour, e.g., at least 75% or more of the total flour content is wheat flour, e.g., at least 80% or more of the total flour content is wheat flour, e.g., at least 85% or more of the total flour content is wheat flour, e.g., at least 90% or more of the total flour content is wheat flour, e.g., at least 95% or more of the total flour content is wheat flour, e.g., 100% of total the flour is wheat flour.

[0049] The dough of the invention is normally a leavened dough or a dough to be subjected to leavening. The dough may be leavened in various ways, such as by adding dough ingredients such as chemical leavening agents, e.g., sodium bicarbonate or by adding a leaven (fermenting dough), but it is preferred to leaven the dough by adding a suitable yeast

culture, such as a culture of *Saccharomyces cerevisiae* (baker's yeast), e.g., a commercially available strain of *S. cerevisiae*.

[0050] The dough of the invention may typically comprise some added sugar as the method according to the invention is able to reduce the amount of added sugar, but normally a partial reduction of sugar is obtained.

[0051] In one embodiment, the amount of added sugar is reduced by at least 10% (w/w) compared to the amount of sugar added to a dough in an original recipe, e.g., the amount of added sugar is reduced by at least 20% (w/w) compared to the amount of sugar added to a dough in an original recipe, e.g., the amount of added sugar is reduced by at least 30% (w/w) compared to the amount of sugar added to a dough in an original recipe, e.g., the amount of added sugar is reduced by at least 40% (w/w) compared to the amount of sugar added to a dough in an original recipe, e.g., the amount of added sugar is reduced by at least 50% (w/w) compared to the amount of sugar added to a dough in an original recipe, e.g., the amount of added sugar is reduced by at least 60% (w/w) compared to the amount of sugar added to a dough in an original recipe, e.g., the amount of added sugar is reduced by at least 70% (w/w) compared to the amount of sugar added to a dough in an original recipe, e.g., the amount of added sugar is reduced by at least 80% (w/w) compared to the amount of sugar added to a dough in an original recipe, e.g., the amount of added sugar is reduced by at least 90% (w/w) compared to the amount of sugar added to a dough in an original recipe, e.g., the amount of added sugar is reduced by 100% (w/w) compared to the amount of sugar added to a dough in an original recipe.

[0052] The dough may also comprise other conventional dough ingredients, e.g., proteins, such as milk powder, gluten, and soy; eggs (either whole eggs, egg yolks or egg whites); an oxidant such as ascorbic acid, potassium bromate, potassium iodate, azodicarbonamide (ADA) or ammonium persulfate; an amino acid such as L-cysteine; a salt such as sodium chloride, calcium acetate, sodium sulphate, calcium sulphate, diluents such as silica dioxide, and starch of different origins. Still other conventional ingredients include hydrocolloids such as CMC, guar gum, xanthan gum, locust bean gum, etc.

[0053] The dough ingredients may typically comprise fat (triglyceride) and/or oil and/or shortenings, in particular oil such as sunflower oil or rapeseed oil.

[0054] The dough may be prepared applying any conventional mixing process, such as the continuous mix process, straight-dough process, or the sponge and dough method.

[0055] The present invention is particularly useful for preparing dough and baked products in industrialized processes in which the dough used to prepare the baked products are prepared mechanically using automated or semi-automated equipment.

[0056] The process of preparing bread generally involves the sequential steps of dough making, sheeting or dividing, shaping or rolling, and proofing the dough, which steps are well known in the art.

[0057] As used herein, “baked product” means any kind of baked product including bread types such as pan bread, toast bread, open bread, pan bread with and without lid, buns, Fino bread, Hammam bread, Samoli bread, baguettes, brioche hamburger buns, rolls, brown bread, whole meal bread, rich bread, bran bread, flat bread, tortilla, biscuits, and any

variety thereof. According to the present invention, the baked product may also be a cake or any patisserie product as known in the art.

Raw Starch Degrading Alpha-Amylase

[0058] As used herein, a “raw starch degrading alpha-amylase” refers to an enzyme that can directly degrade raw starch granules below the gelatinization temperature of starch.

[0059] Examples of raw starch degrading alpha-amylases include the ones disclosed in WO 2005/003311, U.S. Patent

or *Penicillium glabrum*. Preferably, the parent glucoamylase is from a species of *Penicillium*, preferably from *Penicillium oxalicum*, *Penicillium miczynskii*, *Penicillium russellii* or *Penicillium glabrum*.

[0065] Examples of other suitable fungi include *Aspergillus niger*, *Aspergillus awamori*, *Aspergillus oryzae*, *Rhizopus delemar*, *Rhizopus niveus*, *Rhizopus oryzae* and *Talaromyces emersonii*.

[0066] Below is shown the %-identity between the AMG amino acid sequences aligned in FIG. 1, and also provided in the sequence list:

<i>P_oxalicum</i>	100.00	99.83	98.99	98.82	96.64	95.97	77.07	77.12	74.32
AMG_NL	99.83	100.00	99.16	98.99	96.81	96.13	77.07	77.12	74.32
AMG_anPAV498	98.99	99.16	100.00	99.83	97.65	96.97	76.73	76.95	73.82
AMG_JPO001	98.82	98.99	99.83	100.00	97.82	97.14	76.73	76.95	73.82
AMG_JPO124	96.64	96.81	97.65	97.82	100.00	99.33	77.07	77.12	74.32
AMG_JPO172	95.97	96.13	96.97	97.14	99.33	100.00	76.73	76.78	73.99
<i>P_miczynskii</i>	77.07	77.07	76.73	76.73	77.07	76.73	100.00	94.75	80.51
<i>P_russellii</i>	77.12	77.12	76.95	76.95	77.12	76.78	94.75	100.00	79.66
<i>P_glabrum</i>	74.32	74.32	73.82	73.82	74.32	73.99	80.51	79.66	100.00

Publication no. 2005/0054071, and U.S. Pat. No. 7,326,548. Examples also include those enzymes disclosed in Table 1 to 5 of the examples in U.S. Pat. No. 7,326,548, in U.S. Patent Publication no. 2005/0054071 (Table 3 on page 15), as well as the enzymes disclosed in WO 2004/020499 and WO 2006/06929 and WO 2006/066579.

[0060] In one embodiment, the raw starch degrading alpha-amylase is a GH13_1 amylase.

[0061] In one embodiment, the raw starch degrading alpha-amylase enzyme has at least 70%, e.g. at least 71%, e.g. at least 72%, e.g. at least 73%, e.g. at least 74%, e.g. at least 75%, e.g. at least 76%, e.g. at least 77%, e.g. at least 78%, e.g. at least 79%, e.g., at least 80%, e.g. at least 81%, e.g. at least 82%, e.g. at least 83%, e.g. at least 84%, e.g., at least 85%, e.g. at least 86%, e.g. at least 87%, e.g. at least 88%, e.g. at least 89%, e.g., at least 90%, e.g., at least 91%, e.g., at least 92%, e.g., at least 93%, e.g., at least 94%, e.g., at least 95%, e.g. at least 96%, e.g., at least 97%, e.g., at least 98%, e.g., at least 99% identity to the raw starch degrading alpha-amylase shown in EP U.S. Pat. No. 2,981,170 (No-vozymes A/S).

[0062] In one embodiment, the raw starch degrading alpha-amylase according to the invention may be added to flour or dough in an amount of 0.01-10 mg enzyme protein per kg flour, e.g., in an amount of 0.1-5 mg enzyme protein per kg flour.

Glucoamylases

[0063] Glucoamylases are also called amyloglucosidases, and Glucan 1,4-alpha-glucosidase (EC 3.2.1.3), more commonly they are referred to as AMGs.

[0064] According to the present invention, different types of amyloglucosidases may be used as parent for the generation of a thermostable amyloglucosidase variant, e.g. the amyloglucosidase may be a polypeptide that is encoded by a DNA sequence that is found in a fungal strain of *Aspergillus*, *Rhizopus*, *Talaromyces* or *Penicillium*; preferably the DNA sequence that is found in a fungal strain of *Penicillium*, even more preferably the DNA sequence that is found in a fungal strain of *Penicillium oxysporum*, *Penicillium oxalicum*, *Penicillium miczynskii*, *Penicillium russellii*

[0067] In one embodiment, the glucoamylase according to the invention may be added to flour or dough in an amount 0.01-1,000 mg enzyme protein (mgEP) per kg flour, preferably in an amount of 0.01-500 mg enzyme protein (mgEP) per kg flour, even more preferably in an amount of 0.1-100 mg enzyme protein (mgEP) per kg flour.

[0068] Thermostable variants of the PoAMG have been generated (see table 2 below). In a preferred embodiment, the mature thermostable glucoamylase variant of the invention comprises one or more or all of the combinations of amino acid substitutions listed in table 2 below.

[0069] In a preferred embodiment, the mature variant of the invention comprises at least one amino acid modification in one or more or all of the positions corresponding to positions 1, 2, 4, 6, 7, 11, 31, 34, 65, 79, 103, 132, 327, 445, 447, 481, 566, 568, 594 and 595 in SEQ ID NO:1; preferably the at least one amino acid modification comprises a substitution in one or more or all of the positions corresponding to positions 1, 2, 4, 11, 65, 79 and 327 in SEQ ID NO:1, preferably the at least one amino acid modification comprises a substitution in one or more or all of the positions corresponding to R1A, P2N, P4S, P11F, T65A, K79V and Q327F in SEQ ID NO:1; or preferably the at least one amino acid modification comprises a substitution in one or more or all of the positions corresponding to positions 1, 6, 7, 31, 34, 79, 103, 132, 445, 447, 481, 566, 568, 594 and 595 in SEQ ID NO:1, preferably the at least one amino acid modification comprises a substitution in one or more or all of the positions corresponding to R1A, G6S, G7T, R31F, K34Y, K79V, S103N, A132P, D445N, V447S, S481P, D566T, T568V, Q594R and F595S in SEQ ID NO:1; or preferably the at least one amino acid modification comprises a substitution in one or more or all of the positions corresponding to positions 1, 6, 7, 31, 34, 50, 79, 103, 132, 445, 447, 481, 484, 501, 539, 566, 568, 594 and 595 in SEQ ID NO:1, preferably the at least one amino acid modification comprises a substitution in one or more or all of the positions corresponding to R1A, G6S, G7T, R31F, K34Y, E50R, K79V, S103N, A132P, D445N, V447S, S481P, T484P, E501A, N539P, D566T, T568V, Q594R and F595S in SEQ ID NO:1.

[0070] The thermostability improvements (Td) of the variants in table 2 are listed in Table 3, where the Td of the

PoAMG variant denoted “anPAV498” (the parent) was set to zero. In a preferred embodiment, the mature thermostable variant of the invention has a thermostability improvement (Td) over its parent of at least 3° C., preferably at least 4° C., 5° C., 6° C., 7° C. or 8° C., preferably determined as exemplified herein.

[0071] In another preferred embodiment, the mature thermostable variant of the invention has a relative activity at 91° C. of at least 150, preferably at least 200, more preferably at least 250, most preferably at least 300 compared to its parent.

[0072] Preferably, the mature thermostable variant glucoamylase enzyme is comprised in the dough in an amount of 0.01-1,000 mg enzyme protein (mgEP) per kg flour, preferably in an amount of 0.01-500 mg enzyme protein (mgEP) per kg flour, even more preferably in an amount of 0.1-100 mg enzyme protein (mgEP) per kg flour.

Amylases

[0073] Alpha-Amylases (alpha-1,4-glucan-4-glucanohydrolases, EC. 3.2.1.1) constitute a group of enzymes which catalyze hydrolysis of starch and other linear and branched 1,4-glucosidic oligo- and polysaccharides.

[0074] A number of alpha-amylases are referred to as Termamyl™ and “Termamyl™-like alpha-amylases” and are known from, e.g., WO 90/11352, WO 95/10603, WO 95/26397, WO 96/23873 and WO 96/23874.

[0075] Another group of alpha-amylases are referred to as Fungamyl™ and “Fungamyl™-like alpha-amylases”, which are alpha-amylases related to the alpha-amylase derived from *Aspergillus oryzae* disclosed in WO 01/34784.

[0076] Suitable commercial alpha-amylase compositions according to the present invention include, e.g., BAKEZYME P 300 (available from DSM) and FUNGAMYL 2500 SG, FUNGAMYL 4000 BG, FUNGAMYL 4000 SG, FUNGAMYL 800 L, FUNGAMYL ULTRA BG and FUNGAMYL ULTRA SG (available from Novozymes A/S).

[0077] In one embodiment, the alpha-amylase according to the invention may be added to flour or dough in an amount of 0.01-1,000 mg enzyme protein (mgEP) per kg flour, preferably in an amount of 0.01-500 mg enzyme protein (mgEP) per kg flour, even more preferably in an amount of 0.1-100 mg enzyme protein (mgEP) per kg flour.

Additional Enzymes

[0078] Optionally, one or more additional enzymes, such as alpha-amylase, maltogenic amylase, beta amylase, aminopeptidase, carboxypeptidase, catalase, cellulytic enzyme, chitinase, cutinase, cyclodextrin glycosyltransferase, deoxyribonuclease, esterase, glucan 1,4-alpha-maltotetrahydrolase, glucanase, galactanase, alpha-galactosidase, beta-galactosidase, glucose oxidase, alpha-glucosidase, beta-glucosidase, haloperoxidase, hemicellulytic enzyme, invertase, laccase, lipase, mannanase, mannosidase, oxidase, pectinolytic enzymes, peptidoglutaminase, peroxidase, phospholipase, phytase, polyphenoloxidase, proteolytic enzyme, ribonuclease, transglutaminase, and xylanase may be used together with the enzyme composition according to the invention.

[0079] The additional enzyme(s) may be of any origin, including mammalian, plant, and microbial (bacterial, yeast or fungal) origin.

[0080] The maltogenic alpha-amylase (EC 3.2.1.133) may be from *Bacillus*. A maltogenic alpha-amylase from *B. stearothermophilus* strain NCIB 11837 is commercially available from Novozymes A/S under the tradename Novamyl®.

[0081] The maltogenic alpha-amylase may also be a variant of the maltogenic alpha-amylase from *B. stearothermophilus* as disclosed in, e.g., WO1999/043794; WO2006/032281; or WO2008/148845, e.g., Novamyl® 3D.

[0082] An anti-staling amylase for use in the invention may also be an amylase (glucan 1,4-alpha-maltotetrahydrolase (EC 3.2.1.60)) from *Pseudomonas saccharophila* or variants thereof, such as any of the amylases disclosed in WO1999/050399, WO2004/111217 or WO2005/003339.

[0083] The glucose oxidase may be a fungal glucose oxidase, in particular an *Aspergillus niger* glucose oxidase (such as GLUZYME®, available from Novozymes A/S).

[0084] The xylanase which may be of microbial origin, e.g., derived from a bacterium or fungus, such as a strain of *Aspergillus*, in particular of *A. aculeatus*, *A. niger*, *A. awamori*, or *A. tubigensis*, from a strain of *Trichoderma*, e.g. *T. reesei*, or from a strain of *Hemicolola*, e.g., *H. insolens*.

[0085] Suitable commercially available xylanase preparations for use in the present invention include PANZEA BG, PENTOPAN MONO BG and PENTOPAN 500 BG (available from Novozymes A/S), GRINDAMYL POWERBAKE (available from Danisco), and BAKEZYME BXP 5000 and BAKEZYME BXP 5001 (available from DSM).

[0086] The protease may be from *Bacillus*, e.g., *B. amyloliquefaciens*. A suitable protease may be Neutrase® available from Novozymes A/S.

[0087] The phospholipase may have phospholipase A1, A2, B, C, D or lysophospholipase activity; it may or may not have lipase activity. It may be of animal origin, e.g. from pancreas, snake venom or bee venom, or it may be of microbial origin, e.g., from filamentous fungi, yeast or bacteria, such as *Aspergillus* or *Fusarium*, e.g., *A. niger*, *A. oryzae* or *F. oxysporum*. A preferred lipase/phospholipase from *Fusarium oxysporum* is disclosed in WO 98/26057. Also, the variants described in WO 00/32758 may be used.

[0088] Suitable phospholipase compositions are LIPOPAN F, LIPOPAN XTRA, and LIPOPAN MAX (available from Novozymes A/S) or PANAMORE GOLDEN and PANAMORE SPRING (available from DSM).

[0089] Preferably, the one or more additional enzyme is added in an amount of 0.01-1,000 mg enzyme protein (mgEP) per kg flour, preferably in an amount of 0.01-500 mg enzyme protein (mgEP) per kg flour, even more preferably in an amount of 0.1-100 mg enzyme protein (mgEP) per kg flour.

Enzyme Compositions

[0090] The mature thermostable variant glucoamylase of the invention as well as any additional enzyme(s) may be added to flour or dough in any suitable form, such as, e.g., in the form of a liquid, in particular a stabilized liquid, or it may be added to flour or dough as a substantially dry powder or granulate.

[0091] Granulates may be produced, e.g., as disclosed in U.S. Pat. Nos. 4,106,991 and 4,661,452. Liquid enzyme preparations may, for instance, be stabilized by adding a sugar or sugar alcohol or lactic acid according to established procedures. Other enzyme stabilizers are well-known in the art.

[0092] The enzyme(s) may be added to the bread dough ingredients in any suitable manner, such as individual components (separate or sequential addition of the enzymes) or addition of the enzymes together in one step or one composition.

Baking Composition

[0093] The present invention further relates to baking compositions comprising a mature thermostable variant of a parent glucoamylase as defined in the first aspect of the invention.

[0094] The baking composition may contain other dough-improving and/or bread-improving additives, e.g., any of the additives, including enzymes, mentioned above.

[0095] The baking composition may be, e.g., a dough composition, a flour composition, a flour pre-mix, or a bread improver.

[0096] Preferably, the baking compositions of the invention also comprise one or more additional enzyme selected from the group consisting of a alpha-amylase, maltogenic amylase, beta amylase, aminopeptidase, carboxypeptidase, catalase, cellulytic enzyme, chitinase, cutinase, cyclodextrin glycosyltransferase, deoxyribonuclease, esterase, glucan 1,4-alpha-maltotetrahydrolase, glucanase, galactanase, alpha-galactosidase, beta-galactosidase, glucose oxidase, alpha-glucosidase, beta-glucosidase, haloperoxidase, hemi-cellulytic enzyme, invertase, laccase, lipase, mannanase, mannosidase, oxidase, pectinolytic enzymes, peptidoglutaminase, peroxidase, phospholipase, phytase, polyphenoloxidase, proteolytic enzyme, ribonuclease, transglutaminase, and xylanase.

[0097] Preferably, the baking compositions of the invention also comprise flour, sugar, yeast, salt and/or fat.

[0098] It will often be advantageous to provide the enzymes used in the treatment of the present invention in admixture with other ingredients used to improve the properties of baked products. These baking compositions are commonly known in the art as "pre-mixes," which usually comprise flour.

[0099] Hence, in a further aspect, the present invention relates to a bread premix for improving the quality of dough by reducing the amount of added sugar, which premix comprises the enzyme combination of the present invention.

[0100] In one embodiment, the present invention further relates to a bread pre-mix comprising the enzyme combination of the present invention and flour, such as, flour from grains, such as, wheat flour, corn flour, rye flour, barley flour, oat flour, rice flour, or sorghum flour, and combinations thereof.

[0101] In another embodiment, the present invention relates to a bread pre-mix comprising the enzyme combination of the present invention and flour, such as, flour from grains, such as, wheat flour, corn flour, rye flour, barley flour, oat flour, rice flour, sorghum, soy flour, and combinations thereof, and one or more additional enzymes, as previously described.

[0102] The pre-mix may be in the form of a granulate or agglomerated powder, e.g., wherein typically 95% (by weight) of the granulate or agglomerated powder has a particle size between 25 and 500 μm .

[0103] Granulates and agglomerated powders may be prepared by conventional methods, e.g., by spraying the enzymes onto a carrier in a fluid-bed granulator. The carrier may consist of particulate cores having a suitable particle

size. The carrier may be soluble or insoluble, e.g. a salt (such as NaCl or sodium sulfate), a sugar (such as sucrose or lactose), a sugar alcohol (such as sorbitol), starch, rice, corn grits, or soy.

Bread Properties

[0104] Organoleptic qualities or sensory attributes of the bread may be measured as known in the art. The properties of the bread may be referred to herein as sensory attributes, which include anti-staling (bread crumb firmness/hardness), crumb properties and mouth feel, or more precisely, the attributes of bread as detected in the mouth during eating (e.g., bread softness/resistance to first bite, crumb moistness, crumb chewiness and gumminess, and crumb smoothness and melting properties).

[0105] In one embodiment, the sensory attribute of the baked product is an increased sweetness by using the enzyme solution according to the invention.

[0106] In one embodiment, the sensory attribute of the baked product is an increased crumb sweetness by using the enzyme solution according to the invention.

[0107] In a preferred embodiment of the invention, the baked or par-baked product after final bake-off has a reduced initial firmness and/or an increased initial elasticity, and/or a reduced increase in firmness and/or a higher elasticity after 1, 7 or 14 days, when cooled to room temperature, packed in a sealed container and stored at room temperature until analysis, compared to a control made without any added glucoamylase.

[0108] In another preferred embodiment, the baked or par-baked product after final bake-off has at least the same sweetness or sweet taste as a control product made with double the amount of the mature glucoamylase the amino acid sequence of which is shown in SEQ ID NO:10, preferably determined as exemplified herein; preferably the baked or par-baked product after final bake-off has a higher sweetness or more sweet taste than a control product made with double the amount of the mature glucoamylase the amino acid sequence of which is shown in SEQ ID NO:10, preferably determined as exemplified herein.

[0109] The invention described and claimed herein is not to be limited in scope by the specific embodiments herein disclosed, since these embodiments are intended as illustrations of several aspects of the invention. Any equivalent embodiments are intended to be within the scope of this invention as well as combinations of one or more of the embodiments.

[0110] Various references are cited herein, the disclosures of which are incorporated by reference in their entirety. The present invention is further described by the following example which should not be construed as limiting the scope of the invention.

EXAMPLES

Example 1: Construction of PoAMG Libraries

PoAMG Libraries were Constructed as Follows:

[0111] A forward or reverse primer having NNK or desired mutation(s) at target site(s) with 15 bp overlaps each other were designed. Inverse PCR, which means amplification of entire plasmid DNA sequences by inversely directed primers, were carried out with appropriate template plasmid DNA (e.g. plasmid DNA containing JPO-0001 gene) by the

following conditions. The resultant PCR fragments were purified by QIAquick Gel extraction kit [QIAGEN], and then introduced into *Escherichia coli* ECOS Competent *E. coli* DH5 α [NIPPON GENE CO., LTD.]. The plasmid DNAs were extracted from *E. coli* transformants by MagEx-tractor plasmid extraction kit [TOYOBO], and then introduced into *A. niger* competent cells.

PCR Reaction Mix:

- [0112] PrimeSTAR Max DNA polymerase [TaKaRa]
- [0113] Total 25 μ l
- [0114] 1.0 μ l Template DNA (1 ng/ μ l)
- [0115] 9.5 μ l H₂O
- [0116] 12.5 μ l 2 $\times$ PrimeSTAR Max pre-mix
- [0117] 1.0 μ l Forward primer (5 μ M)
- [0118] 1.0 μ l Reverse primer (5 μ M)

PCR Program:

- [0119] 98° C./2 min
- [0120] 25 $\times$ (98° C./10 sec, 60° C./15 sec, 72° C./2 min)
- [0121] 10° C./hold

Example 2: Screening for Better Thermostability

[0122] *B. subtilis* libraries constructed as in EXAMPLE 1 were fermented in either 96-well or 24-well MTP containing COVE liquid medium (2.0 g/L sucrose, 2.0 g/L iso-maltose, 2.0 g/L maltose, 4.9 mg/L, 0.2 ml/L 5N NaOH, 10 ml/L COVE salt, 10 ml/L 1M acetamide), 32° C. for 3 days. Then, AMG activities in culture supernatants were measured at several temperatures by pNPG assay described as follows.

pNPG Thermostability Assay:

[0123] The culture supernatants containing desired enzymes was mixed with same volume of pH 5.0 200 mM NaOAc buffer. Twenty microliter of this mixture was dispensed into either 96-well plate or 8-strip PCR tube, and then heated by thermal cycler at various temperatures for 30 min. Those samples were mixed with 10 μ l of substrate solution containing 0.1% (w/v) pNPG [wako] in pH 5.0 200 mM NaOAc buffer and incubated at 70° C. for 20 min for enzymatic reaction. After the reaction, 60 μ l of 0.1M Borax buffer was added to stop the reaction. Eighty microliter of reaction supernatant was taken out and its OD₄₀₅ value was read by photometer to evaluate the enzyme activity.

TABLE 1a

Lists of the relative activity of PoAMG variants when compared with their parent anPAV498 or JPO-0001 (anPAV498 w. leader-/propeptide)	
Name	Relative activity of 80° C./75° C. (%)
anPAV498	17%
JPO-004	32%
JPO-005	15%
JPO-006	16%
JPO-007	3%
AnPav498	13%
JPO-009	16%

TABLE 1a-continued

Lists of the relative activity of PoAMG variants when compared with their parent anPAV498 or JPO-0001 (anPAV498 w. leader-/propeptide)	
JPO-011	15%
JPO-012	15%
JPO-013	17%
JPO-020	20%
Name	Relative activity of 80° C./70° C. (%)
JPO-001	10%
JPO-004	29%
JPO-009	13%
JPO-014	21%
JPO-020	16%
JPO-021	30%
JPO-052	33%
Name	Relative activity of 79° C./70° C. (%)
JPO-001	23%
JPO-021	46%
JPO-022	39%
JPO-023	44%
JPO-025	51%
JPO-027	49%
JPO-029	37%
Name	Relative activity of 77° C./70° C. (%)
JPO-001	72%
JPO-029	82%
JPO-047	80%
JPO-048	90%
JPO-049	84%
JPO-050	86%
JPO-064	87%
name	Relative activity of 79° C./77° C. (%)
JPO-001	36%
JPO-029	51%
JPO-047	45%
JPO-048	81%
JPO-049	53%
JPO-050	58%
JPO-064	65%
Name	Relative activity of 79° C./77° C. (%)
JPO-001	41%
JPO-021	60%
JPO-022	48%
JPO-023	57%
JPO-025	56%
JPO-027	64%
JPO-029	66%
JPO-047	50%
JPO-048	72%
JPO-051	82%
JPO-058	73%
JPO-062	72%

TABLE 1a-continued

Lists of the relative activity of PoAMG variants when compared with their parent anPAV498 or JPO-0001 (anPAV498 w. leader-/propeptide)	
JPO-063	85%
JPO-064	83%

TABLE 1b

Lists of the relative activity of PoAMG variants when compared with their parent JPO-022	
Name	Relative activity of 77° C./70° C. (%)
JPO-022	60%
JPO-027	67%
JPO-042	8%
JPO-044	86%
JPO-045	67%
JPO-046	48%
JPO-022	76%
JPO-023	75%
JPO-025	80%
JPO-027	84%
JPO-058	92%
JPO-059	88%
JPO-060	86%
JPO-061	83%
JPO-062	87%

Name	Relative activity of 79° C./77° C. (%)
JPO-022	49%
JPO-023	51%
JPO-025	52%
JPO-027	58%
JPO-058	69%
JPO-059	36%
JPO-060	41%
JPO-061	44%
JPO-062	57%

TABLE 1c

List of the relative activity of PoAMG variants when compared with their parent JPO-063	
Name	Relative activity of 79° C./77° C. (%)
JPO-063	91%
JPO-066	96%
JPO-071	89%
JPO-072	84%
JPO-074	103%
JPO-075	86%
JPO-076	92%
JPO-077	95%
JPO-078	88%
JPO-079	100%

Name	Relative activity of 84° C./80° C. (%)
JPO-063	16%
JPO-065	26%
JPO-067	21%
JPO-070	12%
JPO-071	13%
JPO-074	32%

TABLE 1c-continued

List of the relative activity of PoAMG variants when compared with their parent JPO-063	
JPO-081	17%
JPO-082	24%
JPO-083	46%
JPO-084	26%
JPO-044	37%

Name	Relative activity of 82° C./70° C. (%)
JPO-063	21%
JPO-093	43%
JPO-081	25%
JPO-088	39%
JPO-094	38%
JPO-096	38%
JPO-106	53%

Name	Relative activity of 83° C./80° C. (%)
JPO-063	46%
JPO-051	44%
JPO-096	64%
JPO-106	88%
JPO-110	81%
JPO-111	100%
JPO-112	86%
JPO-113	83%
JPO-114	47%
JPO-115	90%

TABLE 1d

List of the relative activity of PoAMG variants when compared with their parent JPO-096	
Name	Relative activity of 83° C./70° C. (%)
JPO-082	53%
JPO-088	70%
JPO-091	69%
JPO-092	65%
JPO-093	62%
JPO-094	74%
JPO-095	69%
JPO-096	67%
JPO-097	65%
JPO-098	65%

Name	Relative activity of 83° C./80° C. (%)
JPO-051	20%
JPO-096	43%
JPO-109	51%
JPO-126	33%
JPO-129	48%
JPO-130	18%
JPO-131	51%
JPO-132	34%

TABLE 1e

List of the relative activity of PoAMG variants when compared with their parents JPO-129	
Name	Relative activity of 84° C./80° C. (%)
JPO-129	62%
JPO-156	51%
JPO-160	34%
JPO-161	41%

TABLE 1e-continued

List of the relative activity of PoAMG variants when compared with their parents JPO-129	
Name	Relative activity of 84° C./80° C. (%)
JPO-162	49%
JPO-163	21%
JPO-164	57%
JPO-165	77%

TABLE 1f

List of the relative activity of PoAMG variants when compared with their parent JPO-166	
Name	Relative activity of 84° C./75° C. (%)
JPO-166	19%
JPO-167	66%
JPO-168	58%
JPO-169	53%
JPO-171	47%
JPO-172	98%

TABLE 2

Amino acid substitutions in the variants of the PoAMG mature sequence	
Name	Amino acid substitutions
PoAMG	The wildtype mature AMG from <i>Penicillium</i> (SEQ ID NO: 1)
AMG NL	K79V
anPAV498	P2N P4S P11F T65A K79V Q327F
JPO-001	R1A P2N P4S P11F T65A K79V Q327F
JPO-018	D75N R77D A78Q
JPO-019	D75S R77G A78W V79D F80Y
JPO-023	R1A K34Y S103N
JPO-024	R1A K34Y D445N V447S
JPO-025	R1A K34Y Y504T
JPO-026	R1A S103N D445N V447S
JPO-027	R1A S103N Y504T
JPO-028	R1A D445N V447S Y504T
JPO-029	R1A K34Y S103N D445N V447S
JPO-044	R1A K34Y S103N D445N V447S E501V Y504T
JPO-047	R1A K34Y S103N Y504T
JPO-048	R1A K34Y S103N D445N V447S D566T
JPO-049	R1A K34Y S103N Q594R F595S
JPO-050	R1A K34Y S103N Y504T Q594R F595S
JPO-051	R1A K34Y S103N D445N V447S Y504T Q594R F595S
JPO-052	R1A S105L
JPO-053	R1A S105E
JPO-055	R1A A132R
JPO-058	R1A K34Y S105L Y504T Q594R F595S
JPO-059	R1A K34Y S103N S105L Y504T Q594R F595S
JPO-060	R1A K34Y S103N S105L Y504T Q594R F595S
JPO-061	R1A K34Y S103N S105L Y504T D566T Q594R F595S
JPO-062	R1A K34Y S103N S105L D445N V447S Y504T D566T Q594R F595S
JPO-063	R1A K34Y S103N S105L A132R D445N V447S Y504T D566T Q594R F595S
JPO-064	R1A K34Y S103N S105L D445N V447S D566T Q594R F595S
JPO-065	R1AK34Y S103N S105LA132R D445N V447S E501VY504T D566T Q594R R1A F595S
JPO-066	R1A K34Y S103N A132R D445N V447S Y504T D566T Q594R F595S
JPO-069	R1A K34Y S103N S105L A132R D445N V447S Y504T D566T V592T
JPO-071	R1A G6S G7T K34Y S103N S105L A132R D445N V447S Y504T D566T Q594R R1A F595S
JPO-074	R1A K34Y S103N P107L A132R D445N V447S Y504T D566T Q594R F595S
JPO-083	R1A G6S G7T K34Y S103N P107L A132R D445N V447S Y504T D566T Q594R R1A F595S
JPO-084	R1A G6S G7T K34Y S103N P107L A132R D445N V447S Y504T D566T V592T R1A Q594R F595S
JPO-091	R1A G6S R7T K34Y S103N P107L A132P D445N V447S Y504T D566T Q594R F595S
JPO-092	R1A G6S G7T K34Y S103N P107L A132R D445N V447S Y504T D566T T568V Q594R F595S
JPO-093	R1A G6S G7T K34Y S103N P107L A132P D445N V447S Y504T D566T T568V Q594R F595S
JPO-094	R1A G6S G7T K34Y S103N P107L A132R D445N V447S S481P Y504T D566T Q594R F595S
JPO-095	R1A G6S G7T K34Y S103N P107L A132R D445N V447S S481P Y504T D566T T568V Q594R F595S
JPO-096	R1A G6S G7T K34Y S103N P107L A132P D445N V447S D566T T568V Q594R F595S
JPO-097	R1A G6S G7T K34Y S103N P107L T110W A132P D445N V447S Y504T D566T T568V Q594R F595S
JPO-098	R1A G6S G7T K34Y E50R S103N P107L A132P D445N V447S Y504T D566T T568V Q594R F595S

TABLE 2-continued

Amino acid substitutions in the variants of the PoAMG mature sequence	
Name	Amino acid substitutions
JPO-105	R1A G6S G7T K34Y S103N P107L A132P D445N V447S E501V Y504T
JPO-106	R1A G6S G7T R31F K34Y S103N P107L A132P D445N V447S Y504T D566T T568V Q594R F595S
JPO-108	R1A G6S G7T R31F K34Y S103N P107L A132P D445N V447S S481P Y504T D566T T568V Q594R F595S
JPO-109	R1A G6S G7T K34Y E50R S103N P107L A132P D445N V447S S481P Y504T D566T T568V Q594R F595S
JPO-111	R1A G6S G7T R31F K34Y S103N P107L A132P D445N V447S S481P E501V Y504T D566T T568V Q594R F595S
JPO-112	R1A G6S G7T R31F K34Y S103N P107L A132P D445N V447S S481P D566T T568V Q594R F595S
JPO-114	R1A K34Y D75N R77D A78Q S103N R138L D445N V447S Y504T Q594R F595S
JPO-115	R1A G6S G7T R31F K34Y D75N R77D A78Q S103N P107L A132P D445N V447S S481P Y504T D566T T568V Q594R F595S
JPO-124	R1A G6S G7T R31F K34Y S103N A132P D445N V447S S481P D566T T568V Q594R F595S
JPO-125	R1A G6S G7T K34Y E50R S103N A132P D445N V447S S481P D566T T568V Q594R F595S
JPO-126	R1A R31F K34Y D75N R77D A78Q S103N R138L D445N V447S Y504T Q594R F595S
JPO-127	R1A K34Y D75N R77D A78Q S103N R138L D445N V447S Q594R F595S
JPO-128	R1A G6S G7T R31F K34Y S103N A132P D445N V447S
JPO-129	R1A G6S G7T R31F K34Y E50R S103N A132P D445N V447S S481P D566T T568V Q594R F595S
JPO-130	R1A K34Y E50R D75N R77D A78Q S103N R138L D445N V447S Q594R F595S
JPO-131	R1A G6S G7T R31F K34Y E50R D75N R77D A78Q S103N A132P D445N V447S S481P D566T Q594R F595S
JPO-132	R1A R31F K34Y E50R D75N R77D A78Q S103N R138L D445N V447S Q594R F595S
JPO-133	R1A G6S G7T R31F K34Y E50R D75N R77D A78Q S103N A132P R138L D445N V447S S481P D566T Q594R F595S
JPO-138	R1A R135S
JPO-143	R1A G6S G7T R31F K34Y E50R S103N A132P D445N V447S S481P E501L D566T T568V Q594R F595S
JPO-154	R1A G6S G7T R31F K34Y S103N A132P R138G D445N V447S S481P D566T T568V Q594R F595S
JPO-155	R1A G6S G7T R31F K34Y S103N A132P R138L D445N V447S S481P D566T T568V Q594R F595S
JPO-156	R1A G6S G7T R31F K34Y S103N A132P R138P D445N V447S S481P D566T T568V Q594R F595S
JPO-167	R1A G6S G7T R31F K34Y E50R S103N A132P S379P D445N V447S S481P E501A D566T T568V Q594R F595S
JPO-168	R1A G6S G7T R31F K34Y E50R S103N A132P D445N V447S S481P T484P E501A D566T T568V Q594R F595S
JPO-169	R1A G6S G7T R31F K34Y E50R S103N A132P D445N V447S S481P E501A N539P D566T T568V Q594R F595S
JPO-171	R1A G6S G7T R31F K34Y E50R S103N A132P S379P D445N V447S S481P T484P E501A D566T T568V Q594R F595S
JPO-172	R1A G6S G7T R31F K34Y E50R S103N A132P D445N V447S S481P T484P E501A N539P D566T T568V Q594R F595S

Example 3: Fermentation of the *Aspergillus niger*

[0124] *Aspergillus niger* strains were fermented on a rotary shaking table in 500 ml baffled flasks containing 100 ml MU1 with 4 ml 50% urea at 220 rpm, 30° C. The culture broth was centrifuged (10,000×g, 20 min) and the supernatant was carefully decanted from the precipitates.

Example 4: Purification of PoAMG (JPO-001)
Variants

[0125] PoAMG variants were purified by cation exchange chromatography. The peak fractions of each were pooled individually and dialyzed against 20 mM sodium acetate buffer pH 5.0, and then the samples were concentrated using a centrifugal filter unit (Vivaspin Turbo 15, Sartorius). Enzyme concentrations were determined by A280 value.

Example 5: Thermostability Determination (TSA)

[0126] Purified enzyme was diluted with 50 mM sodium acetate buffer pH 5.0 to 0.5 mg/ml and mixed with equal volume of SYPRO Orange (Invitrogen) diluted with Milli-Q water. Eighteen ul of mixture solution were transfer to LightCycler 480 Multiwell Plate 384 (Roche Diagnostics) and the plate was sealed.

Equipment Parameters of TSA:

- [0127] Apparatus: LightCycler 480 Real-Time PCR System (Roche Applied Science)
- [0128] Scan rate: 0.02° C./sec
- [0129] Scan range: 37-96° C.
- [0130] Integration time: 1.0 sec
- [0131] Excitation wave length 465 nm
- [0132] Emission wave length 580 nm

[0133] The obtained fluorescence signal was normalized into a range of 0 and 1. The Td was defined as the temperature at which the signal intensity was 0.5. The thermostability improvements are listed in Table 3 with Td of the PoAMG variant denoted anPAV498 as 0.

Example 6: PoAMG Activity Assay

[0134] Maltodextrin (DE11) assay by GOD-POD method

Substrate Solution

[0135] 30 g maltodextrin (pindex #2 from MAT-SUTANI chemical industry Co., Ltd.)

[0136] 100 ml 120 mM sodium acetate buffer, pH 5.0
Glucose CII test kit (Wako Pure Chemical Industries, Ltd.)

[0137] Twenty ul of enzyme samples were mixed with 100 ul of substrate solution and incubated at set temperatures for 2 hours. The samples were cooled down on the aluminum block for 3 min then 10 ul of the reaction solution was mixed with 590 ul of 1 M Tris-HCl PH 8.0 to stop reaction. Ten ul of the solution was mixed with 200 ul of the working solution of the test kit then stand at room temperature for 15 min. The absorbance at A505 was read. The activities are listed in Table 3 as relative activity of the PoAMG variant denoted anPAV498.

TABLE 3

Variant	Td improvement [° C.] (pH5.0, anPAV498 as 0)	Activity at 91° C. (anPAV498 as 100)
anPAV498	—	100
JPO-001	1.0	94
JPO-004	2.2	—
JPO-009	0.7	—
JPO-013	1.5	—
JPO-014	2.3	—
JPO-020	1.4	74
JPO-021	2.5	113
JPO-052	2.6	85
JPO-053	0.2	71
JPO-055	1.6	85
JPO-023	3.6	—
JPO-024	2.5	—
JPO-025	3.4	—
JPO-027	2.9	—
JPO-029	3.7	191
JPO-048	4.3	163
JPO-051	5.7	222
JPO-058	4.2	157
JPO-062	4.2	159
JPO-063	5.4	107
JPO-064	4.9	178
JPO-065	7.0	127
JPO-066	6.5	178
JPO-069	4.8	95
JPO-071	6.1	128
JPO-074	6.3	108
JPO-081	5.5	213
JPO-082	5.6	215
JPO-089	6.0	171
JPO-090	5.5	155
JPO-018	0.6	84
JPO-019	0.5	86
JPO-044	6.3	225
JPO-083	6.1	103
JPO-084	4.4	66
JPO-099	6.8	156
JPO-091	6.6	130
JPO-092	6.7	113
JPO-093	6.8	132

TABLE 3-continued

Variant	Td improvement [° C.] (pH5.0, anPAV498 as 0)	Activity at 91° C. (anPAV498 as 100)
JPO-094	6.6	126
JPO-095	6.9	—
JPO-096	5.9	—
JPO-097	5.2	—
JPO-098	5.6	—
JPO-112	8.2	—
JPO-114	5.2	218
JPO-115	8.0	—
JPO-108	8.5	—
JPO-109	7.2	—
JPO-111	8.4	—
JPO-124	8.0	385
JPO-125	6.8	324
JPO-126	6.6	268
JPO-127	4.9	246
JPO-129	8.2	399
JPO-130	5.3	278
JPO-131	7.9	367
JPO-132	6.6	336
JPO-138	6.4	125
JPO-133	6.1	143
JPO-143	8.8	280
JPO-154	7.6	252
JPO-155	8.3	282
JPO-156	8.3	290
JPO-145	8.2	—
JPO-147	8.2	—
JPO-150	8.2	—
JPO-152	8.4	—
JPO-153	9.0	399
JPO-161	6.0	200
JPO-165	8.9	403
JPO-166	7.0	237
JPO-167	9.1	387
JPO-168	9.3	332
JPO-169	9.6	269
JPO-171	9.4	255
JPO-172	9.9	432

Example 7: Freshness Effect of AMG in Bread (Part 1)

[0138] Bread was baked in a straight dough process with a recipe according to Table 4. The bread was baked in lidded tins in order to have the same volume of all bread. The ingredients were mixed in a spiral mixer into a dough for 3+7 min at 17 respectively 35 rpm. The doughs were divided into 450 g pieces, rounded, sheeted and place in baking tins. The tins with the doughs were proofed for 55 min at 32° C. and 86% relative humidity. The proofed doughs were baked in a deck oven for 35 min at 230° C.

TABLE 4

	Bakers %
Flour (Kolibri, Meneba, NL)	100
Water	55.5
Yeast	4.5
Sucrose	1.5
Salt	1.5
Ascorbic acid	0.04
Calcium Propionate	0.25
Fungamyl® 4000SG	6 ppm
Panzea® BG	25 ppm

TABLE 5

Seven dough treatments were prepared with different enzymatic additions according to Table 4; AMG Goldcrust® 3300 BG (Goldcrust®) is a commercially available AMG for baking (Novozymes A/S, Denmark); AMG NL and AMG anPAV498 are artificial variants of PoAMG (see table 2).							
	Dough						
	1	2	3	4	5	6	7
AMG Goldcrust®, mgEP/kg flour				25	50		
AMG NL, mgEP/kg flour			25			50	
AMG anPAV498, mgEP/kg flour		25					50

[0139] The doughs were baked and the resulting breads were packed 2 hours after baking in sealed plastic bags and stored at room temperature until analysis.

[0140] The texture of each bread was evaluated with a texture analyzer (TA-XT plus, Stable microsystems, Godalmine, UK). Bread crumb texture properties were characterized by firmness (the same as “hardness” and the opposite of “softness”) and the elasticity of the baked product. A standard method for measuring firmness and elasticity is based on force-deformation of the baked product. A force-deformation of the baked products may be performed with a 40 mm diameter cylindrical probe. The force on the cylindrical probe is recorded as it is pressed down 40% strain a 25 mm thick bread slice at a deformation speed of 1 mm/second. The probe is then kept in this position for 30 seconds while the force is recorded and then probe returns to its original position.

[0141] Firmness (in grams) is defined as the force needed to compress a probe to a 25% strain (corresponding to 6.25 mm compression into a bread crumb slice of 25 mm thickness).

[0142] Elasticity (in %) is defined as the force recoded after 30 seconds compression at 40% strain (corresponding to force at time=40 s for a bread slice of 25 mm thickness) divided by the force needed to press the probe 10 mm into the crumb (corresponding to force at time=10 s for a bread slice of 25 mm thickness) times 100.

[0143] The results from the texture analysis can be found in Table 6 (firmness) and Table 7 (elasticity).

[0144] Fresh bread without enzyme (control) has low firmness and high elasticity, as the bread is stored the firmness increase over time and the elasticity decrease. Traditional AMGs used in baking applications (for example Goldcrust) does not impact the Firmness or elasticity.

[0145] AMG anPAV498 dosed at 25 or 50 mgEP/kg as well as AMG NL dosed at 50 mgEP/kg flour improves (decrease) the initial firmness and reduces the increase in firmness over time. AMG anPAV498 dosed at 25 or 50 mgEP/kg as well as AMG NL dosed at 50 mgEP/kg flour improves (increase) the initial elasticity and prevents the loss of elasticity over time.

TABLE 6

Firmness (g) on day 1, 3 and 7 of bread with enzyme treatments according Table 5			
Day	1	3	7
Control	411	690	1025
25 mgEP/kg AMG anPAV498	332	446	746
25 mgEP/kg AMG NL	413	689	1019
25 mgEP/kg Goldcrust®	406	709	1094
50 mgEP/kg Goldcrust®	417	736	1071
50 mgEP/kg AMG NL	388	579	875
50 mgEP/kg AMG anPAV498	314	426	662

TABLE 7

Elasticity (%) on day 1, 3 and 7 of bread with enzyme treatments according Table 5			
Day	1	3	7
Control	61.0	54.0	48.0
25 mgEP/kg AMG anPAV498	63.3	59.6	53.9
25 mgEP/kg AMG NL	60.7	54.9	49.3
25 mgEP/kg Goldcrust®	60.4	54.7	48.3
50 mgEP/kg Goldcrust®	60.6	53.4	48.4
50 mgEP/kg AMG NL	61.9	56.9	51.1
50 mgEP/kg AMG anPAV498	66.1	63.2	57.1

[0146] Sugars were extracted from the bread crumb using 0.1 M Phosphate buffer pH 8.0 in 70% EtOH. Bread crumb (180 mg) were added to the extraction buffer (1.8 ml) and was incubated for 20 minutes at 70° C. during mixing. The bread crumb was spun down at 12,000 rpm for 5 minutes in a centrifuge and 500 µl of the supernatant was taken and diluted 200× using a 20 mM Phosphate buffer pH 8.0+10 mg/L cellobiose as internal standard. The extracted sugars (glucose, fructose, maltose and maltotriose) were quantified on an ICS-5000 HPLC system with a CarboPac PA1 column. A theoretical sweetness was calculated based on the levels of glucose, fructose and maltose was calculated using sweetness intensity factors. The sweetness factors in Table 8 was based on the determinations in Portmann M O, Birch G. J Sci Food Agric 69(3):275-81, 1995.

TABLE 8

Sugar	Sweetness intensity factor
Glucose	0.5
Fructose	1
Maltose	0.2

[0147] The amount of simple sugars (glucose fructose, maltose and maltotriose) can be found in Table 9 along with a theoretical sweetness calculated on the amount of the individual sugars. All three AMGs increase the amount of simple sugars. Both AMG NL and AMG anPAV498 are more efficient in generating glucose compared to Goldcrust® resulting in a higher theoretical sweetness.

TABLE 9

Amount of sugars (g/kg bread crumb) extracted from dough treated with enzymes according to Table 5					
	glucose	fructose	maltose	maltotriose	Sweetness
Control	0.7	3.4	17.0	1.0	7.2
25 mgEP AMG anPAV498	15.1	4.2	14.8	0.6	14.7
25 mgEP AMG NL	7.6	4.7	14.0	0.5	11.3
25 mgEP Goldcrust®	3.4	4.6	13.4	0.5	9.0
50 mgEP Goldcrust®	7.2	5.3	13.8	0.5	11.6
50 mgEP AMG NL	14.5	5.2	13.7	0.4	15.2
50 mgEP AMG anPAV498	31.1	5.1	14.1	0.5	23.4

Example 8. Freshness Effect of AMG (part 2)

[0148] Bread was baked in a straight dough process with a recipe according to

[0149] Table 10. The bread was baked in lidded tins in order to have the same volume of all bread. The ingredients were mixed in a spiral mixer into a dough for 3+7 min at 17 respectively 35 rpm. The doughs were divided into 450 g pieces, rounded, sheeted and placed in baking tins. The tins with the doughs were proofed for 55 min at 32° C. and 86% relative humidity. The proofed doughs were baked in a deck oven for 35 min at 230° C.

TABLE 10

	Bakers %
Flour (Kolibri, Meneba, NL)	100
Water	55.5
Yeast	4.5
Sucrose	1.5
Salt	1.5
Ascorbic acid	0.04
Calcium Propionate	0.25
Fungamyl® 4000SG	6 ppm
Panzea® BG	25 ppm

TABLE 11

Seven dough treatments were prepared with different enzymatic additions							
Dough							
	1	2	3	4	5	6	7
AMG anPAV498, mgEP/kg flour				25	50		
JPO124, mgEP/kg flour			25			50	
JPO172, mgEP/kg flour		25					50

[0150] The doughs were baked and the resulting breads were packed 2 hours after baking in sealed plastic bags and stored at room temperature until analysis.

[0151] The texture of each bread was evaluated with a texture analyzer (TA-XT plus, Stable microsystems, Godalmine, UK). Bread crumb texture properties were characterized by firmness (the same as “hardness” and the opposite of “softness”) and the elasticity of the baked product.

[0152] A standard method for measuring firmness and elasticity is based on force-deformation of the baked product. A force-deformation of the baked products may be performed with a 40 mm diameter cylindrical probe. The force on the cylindrical probe is recorded as it is pressed down 40% strain on a 25 mm thick bread slice at a deformation speed of 1 mm/second. The probe is then kept in this position for 30 seconds while the force is recorded and then probe returns to its original position.

[0153] Firmness (in grams) is defined as the force needed to compress a probe to a 25% strain (corresponding to 6.25 mm compression into a bread crumb slice of 25 mm thickness).

[0154] Elasticity (in %) is defined as the force recorded after 30 seconds compression at 40% strain (corresponding to force at time=40 s for a bread slice of 25 mm thickness) divided by the force needed to press the probe 10 mm into the crumb (corresponding to force at time=10 s for a bread slice of 25 mm thickness) times 100.

[0155] The results from the texture analysis can be found in Table 12 (firmness) and Table 13 (elasticity).

[0156] Fresh bread without enzyme (control) has low firmness and high elasticity, as the bread is stored the firmness increase over time and the elasticity decrease. Traditional AMGs used in baking applications (for example Goldcrust) does not impact the firmness or elasticity (Example 7).

[0157] All three AMGs (AMG anPAV498, JPO124 and JPO172) dosed at 25 or 50 mgEP/kg improved (decreased) the initial firmness and reduced the increase in firmness over time. All three AMGs (AMG anPAV498, JPO124 and JPO172) dosed at 25 or 50 mgEP/kg improved (increased) the initial elasticity and prevented the loss of elasticity over time.

TABLE 12

Firmness (g) on day 1, 3 and 7 of bread with enzyme treatments according Table 11.			
Treatment	Day 1	Day 3	Day 7
Control	519	687	883
25 mgEP/kg AMG anPAV498	377	513	716
50 mgEP/kg AMG anPAV498	337	434	651
25 mgEP/kg JPO124	311	400	612
50 mgEP/kg JPO124	288	313	516
25 mgEP/kg JPO172	313	351	502
50 mgEP/kg JPO172	240	289	433

TABLE 13

Elasticity (%) on day 1, 3 and 7 of bread with enzyme treatments according Table 11.			
Treatment	Day 1	Day 3	Day 7
Control	59.5	51.6	47.1
25 mgEP/kg AMG anPAV498	63.5	58.5	51.3
50 mgEP/kg AMG anPAV498	66.5	62.2	55.9
25 mgEP/kg JPO124	65.9	62.8	56.5
50 mgEP/kg JPO124	66.9	66.7	61.2
25 mgEP/kg JPO172	66.4	64.0	58.7
50 mgEP/kg JPO172	69.1	67.8	64.4

[0158] Sugars were extracted from the bread crumb using 0.1 M Phosphate buffer pH 8.0 in 70% EtOH. Bread crumb (180 mg) were added to the extraction buffer (1.8 ml) and

was incubated for 20 minutes at 70° C. during mixing. The bread crumb was spun down at 12,000 rpm for 5 minutes in a centrifuge and 500 µl of the supernatant was taken and diluted 200× using a 20 mM Phosphate buffer pH 8.0+10 mg/L cellobiose as internal standard. The extracted sugars (glucose, fructose, maltose and maltotriose) were quantified on an ICS-5000 HPLC system with a CarboPac PA1 column. A theoretical sweetness was calculated based on the levels of glucose, fructose and maltose was calculated using sweetness intensity factors. The sweetness factors in

[0159] Table 13 were based on the determinations in Portmann M O, Birch G. J Sci Food Agric 69(3):275-81, 1995.

TABLE 14

Sugar	Sweetness intensity factor
Glucose	0.5
Fructose	1
Maltose	0.2

The amount of simple sugars (glucose fructose, maltose and maltotriose) can be found in Table 1 along with a theoretical sweetness calculated on the amount of the individual sugars. All three AMGs increase the amount of simple sugars and increased the calculated sweetness. The higher dosage of the AMGs the more glucose was generated and higher theoretical sweetness. JPO0172 and JPO124 were more efficient than AMG anPAV498 in increasing the glucose and the theoretical sweetness.

TABLE 15

Simple sugars (g/kg bread crumb) extracted from bread treated with enzymes according to Table 11.					
	Glucose	Fructose	Maltose	Maltotriose	Sweetness
Control	1.5	6.1	27.5	1.4	12
25 mgEP/kg JPO172	39.3	7.7	24.6	0.7	32
25 mgEP/kg JPO124	38.6	9.7	28.1	0.8	35
25 mgEP/kg AMG anPAV498	24.0	8.3	24.8	0.8	25
50 mgEP/kg AMG anPAV498	48.0	10.2	23.2	0.6	39
50 mgEP/kg JPO124	62.3	10.6	22.1	0.5	46
50 mgEP/kg JPO172	68.6	10.8	21.7	0.5	49

The change in sugar levels in the bread crumb as a function of bread storage time at ambient temperature can be found in Tables 16-20. The glucose level table 16 which is the product of the AMG is stable over bread storage time. The same picture is seen for the other sugars extracted from the bread crumb fructose (table 17), maltose (table 18), maltotriose (table 19) and Maltotetraose (table 20)

TABLE 16

Glucose levels (g/kg bread crumb) in bread crumb over time as a function of enzyme treatment.				
Glucose	Day 0	Day 3	Day 7	
AMG anPAV498	0	1.5	1.4	1.5
	25	23.9	24.9	24.3
	50	48.0	45.3	46.9

TABLE 16-continued

Glucose levels (g/kg bread crumb) in bread crumb over time as a function of enzyme treatment.				
Glucose		Day 0	Day 3	Day 7
JP0124	0	1.5	1.4	1.5
	25	38.6	37.0	36.9
	50	62.3	62.4	68.0
JP0172	0	1.5	1.4	1.5
	25	39.3	38.6	40.7
	50	68.6	72.0	74.0

TABLE 17

Maltose levels (g/kg bread crumb) in bread crumb over time as a function of enzyme treatment.				
Maltose		Day 0	Day 3	Day 7
AMG ANPAV498anPAV498	0	27.5	28.0	24.4
	25	24.8	25.4	25.5
	50	23.2	22.5	23.0
JP0124	0	27.5	28.0	24.4
	25	28.1	25.1	24.8
	50	22.1	19.6	21.5
JP0172	0	27.5	28.0	24.4
	25	24.6	24.8	24.7
	50	21.7	21.2	21.2

TABLE 18

Fructose levels (g/kg bread crumb) in bread crumb over time as a function of enzyme treatment.				
Fructose		Day 0	Day 3	Day 7
AMG ANPAV498	0	6.1	6.5	6.7
	25	8.3	8.7	9.0
	50	10.2	10.1	10.4
JP0124	0	6.1	6.5	6.7
	25	9.7	8.9	9.2
	50	10.6	10.6	10.4
JP0172	0	6.1	6.5	6.7
	25	7.7	8.1	8.5
	50	10.8	10.4	11.3

TABLE 19

Maltotriose levels (g/kg bread crumb) in bread crumb over time as a function of enzyme treatment.				
Maltotriose		Day 0	Day 3	Day 7
AMG ANPAV498	0	1.4	1.3	1.2
	25	0.8	0.7	0.8
	50	0.6	0.6	0.6
JP0124	0	1.4	1.3	1.2
	25	0.8	0.7	0.7
	50	0.5	0.5	0.6
JP0172	0	1.4	1.3	1.2
	25	0.7	0.7	0.7
	50	0.5	0.5	0.5

TABLE 20

Maltotetraose levels (g/kg bread crumb) in bread crumb over time as a function of enzyme treatment.				
Maltotetraose		Day 0	Day 3	Day 7
AMG anPAV498	0	0.5	0.4	0.4
	25	0.4	0.3	0.4
	50	0.3	0.3	0.2
JP0124	0	0.5	0.4	0.4
	25	0.3	0.3	0.3
	50	0.2	0.2	0.2
JP0172	0	0.5	0.4	0.4
	25	0.3	0.3	0.3
	50	0.2	0.2	0.2

Example 9. Freshness Effect of AMG in Combination with Novamyl

[0160] Bread was baked in a straight dough process with a recipe according to table 21. The bread was baked in lidded tins in order to have the same volume of all bread. The ingredients were mixed in a spiral mixer into a dough for 3+7 min at 17 respectively 35 rpm. The doughs were divided into 450 g pieces, rounded, sheeted and place in baking tins. The tins with the doughs were proofed for 55 min at 32° C. and 86% relative humidity. The proofed doughs were baked in a deck oven for 35 min at 230° C.

TABLE 21

	Bakers %
Flour (Kolibri, Meneba, NL)	100
Water	55.5
Yeast	4.5
Sucrose	1.5
Salt	1.5
Ascorbic acid	0.04
Calcium Propionate	0.25
Fungamyl 4000SG	6 ppm
Panzea BG	25 ppm

TABLE 22

Seven treatments were prepared with different enzymatic additions							
	Dough						
	1	2	3	4	5	6	7
AMG anPAV498, mgEP/kg flour	25	50	25	—	25	50	—
Novamyl® 3D, ppm	—	30	15	—	30	—	30

[0161] The bread was packed 2 hours after baking in sealed plastic bags and stored at room temperature until analysis.

[0162] The texture of each bread was evaluated with a texture analyzer (TA-XT plus, Stable microsystems, Godalmine, UK). Bread crumb texture properties were characterized by firmness (the same as “hardness” and the opposite of “softness”) and the elasticity of the baked product. A standard method for measuring firmness and elasticity is based on force-deformation of the baked product. A force-deformation of the baked products may be performed with a 40 mm diameter cylindrical probe. The force on the cylindrical probe is recorded as it is pressed

down 40% strain on a 25 mm thick bread slice at a deformation speed of 1 mm/second. The probe is then kept in this position for 30 seconds while the force is recorded and then probe returns to its original position.

[0163] Firmness (in grams) is defined as the force needed to compress a probe to a 25% strain (corresponding to 6.25 mm compression into a bread crumb slice of 25 mm thickness). Elasticity (in %) is defined as the force recoded after 30 seconds compression at 40% strain (corresponding to force at time=40 s for a bread slice of 25 mm thickness) divided by the force needed to press the probe 10 mm into the crumb (corresponding to force at time=10 s for a bread slice of 25 mm thickness) times 100.

[0164] The results from the texture analysis can be found in table 23 (firmness) and table 24 (elasticity). Fresh bread without enzyme (Control) has low firmness and high elasticity, as the bread is stored the firmness increase over time and the elasticity decrease.

[0165] AMG anPAV498 improves the initial firmness and elasticity as well as reduce the changes in firmness and elasticity over time.

[0166] Novamyl® 3D does not impact the initial firmness or elasticity. However, Novamyl® 3D reduces the change in firmness and elasticity over time.

[0167] The combination of AMG anPAV498 and Novamyl® 3D both improves the initial firmness and elasticity compared to no enzyme or Novamyl® 3D alone, as well as the change in firmness and elasticity over time. The combination results in a bread with the best firmness and elasticity after 7 days of storage.

TABLE 23

Firmness (g) on day 1, 3 and 7 of bread with enzyme treatments according Table 22.			
Treatment	Day 1	Day 7	Day 14
25 mgEP/kg flour AMG anPAV498	321	446	741
50 mgEP/kg flour AMG anPAV498 + 30 ppm Novamyl® 3D	506	552	704
25 mgEP/kg flour AMG anPAV498 + 15 ppm Novamyl® 3D	344	544	668
Control	499	733	1099
25 mgEP/kg flour AMG anPAV498 + 30 ppm Novamyl® 3D	407	512	579
50 mgEP/kg flour AMG anPAV498	317	401	769
30 ppm Novamyl® 3D	480	607	769

TABLE 24

Elasticity (%) on day 1, 3 and 7 of bread with enzyme treatments according Table 22.			
Treatment	Day 1	Day 7	Day 14
25 mgEP/kg flour AMG anPAV498	65.9	60.8	50.5
50 mgEP/kg flour AMG anPAV498 + 30 ppm Novamyl® 3D	61.8	61.5	58.4
25 mgEP/kg flour AMG anPAV498 + 15 ppm Novamyl® 3D	64.1	61.4	58.2
Control	60.6	55.2	46.4
25 mgEP/kg flour AMG anPAV498 + 30 ppm Novamyl® 3D	61.2	59.5	58.0
50 mgEP/kg flour AMG anPAV498	67.3	64.4	56.1
30 ppm Novamyl® 3D	60.2	58.4	55.0

[0168] Sugars were extracted from the bread crumb using 0.1 M Phosphate buffer pH 8.0 in 70% EtOH. Bread crumb

(180 mg) were added to the extraction buffer (1.8 ml) and was incubated for 20 minutes at 70° C. during mixing. The bread crumb was spun down at 12,000 rpm for 5 minutes in a centrifuge and 500 µl of the supernatant was taken and diluted 200× using a 20 mM Phosphate buffer pH 8.0+10 mg/L cellobiose as internal standard. The extracted sugars (glucose, fructose, maltose and maltotriose) were quantified on an ICS-5000 HPLC system with a CarboPac PA1 column. A theoretical sweetness was calculated based on the levels of glucose, fructose and maltose was calculated using sweetness intensity factors. The sweetness factors in [0169] Table 25 were based on the determinations in Portmann M O, Birch G. J Sci Food Agric 69(3):275-81, 1995.

TABLE 25

Sugar	Sweetness intensity factor
Glucose	0.5
Fructose	1
Maltose	0.2

[0170] The amounts of different sugars extracted from the bread and the theoretically calculated sweetness based on the sugar amounts can be found in Table 26. The higher dosage of AMG anPAV498 the more glucose in the bread. The higher dosage of Novamyl® 3D the more maltose and maltotriose in the dough. The combination of AMG anPAV498 and Novamyl® 3D increase both glucose, maltose and maltotriose. The main contributor to the calculated sweetness is the dose of AMG anPAV498 since glucose impacts sweetness more than maltose and maltotriose.

TABLE 26

Simple sugars (g/kg bread crumb) extracted from bread treated with enzymes according to Table 22.					
	Glucose	Fructose	Maltose	Maltotriose	Sweetness
25 mgEP/kg AMG anPAV498	17.4	4.6	17.2	0.6	17
50 mgEP/kg AMG anPAV498 + 30 ppm Novamyl® 3D	33.4	7.0	24.5	12.7	29
25 mgEP/kg AMG anPAV498 + 15 ppm flour Novamyl® 3D	18.0	7.1	21.9	6.8	20
Control	1.0	4.0	20.3	1.0	9
25 mgEP/kg AMG anPAV498 + 30 ppm Novamyl® 3D	15.6	5.4	29.1	9.6	19
50 mgEP/kg AMG anPAV498	34.9	6.3	17.1	0.5	27
30 ppm Novamyl® 3D	2.3	3.5	32.8	5.2	11

Example 10. Dosage Response of AMG NL (Partial Sugar Replacement)

[0171] Bread was baked in a straight dough process with a recipe according to Table 27. The ingredients were mixed in a spiral mixer into a dough for 3+7 min at 17 respectively 35 rpm. The doughs were divided into 450 g pieces, rounded, sheeted and place in baking tins. The tins with the doughs were proofed for 55 min at 32° C. and 86% relative humidity. The proofed doughs were baked in a deck oven for 25 min at 230° C.

TABLE 27

Fungamyl® 4000SG is a commercially available fungal amylase for baking (Novozymes A/S, Denmark), Panzea® BG is a commercially available bacterial xylanase for baking (Novozymes A/S, Denmark).	
Bakers %	
Flour (Kolibri, Meneba, NL)	100
Water	55.5
Yeast	3
Sucrose	5
Salt	0.5
Ascorbic acid	0.04
Calcium Propionate	0.25
Fungamyl® 4000SG	7 ppm
Panzea® BG	25 ppm

TABLE 28

Eight treatments were prepared with different enzymatic additions, AMG Goldcrust® is a commercially available AMG for baking (Novozymes A/S, Denmark) and JA126 is a raw-starch degrading amylase (Novozymes A/S, Denmark).	
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	Dough							
	1	2	3	4	5	6	7	8
AMG Goldcrust® mgEP/kg flour	112.5	—	—	—	—	—	—	—
AMG NL, mgEP/kg flour		17.9	26.8	35.7	44.6	53.6	62.5	71.4

TABLE 28-continued

Eight treatments were prepared with different enzymatic additions, AMG Goldcrust® is a commercially available AMG for baking (Novozymes A/S, Denmark) and JA126 is a raw-starch degrading amylase (Novozymes A/S, Denmark).								
	Dough							
	1	2	3	4	5	6	7	8
JA126, mgEP/kg flour	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35

The dough properties were evaluated by a trained baker and the volume of the bread was determined using Volscan

profiler (Stable microsystems, Godalming, UK). The results from the evaluation can be found in

[0172] Table 29. The doughs with 44.6 and 53.6 mgEP/kg flour of AMG NL (Doughs 5 & 6) had similar volume, the same extensibility and elasticity as the dough with 112.5 mgEP/kg flour of AMG Goldcrust®, but the doughs were slightly less sticky and soft.

TABLE 29

	Dough							
	1	2	3	4	5	6	7	8
Stickiness	5	4	4	4	4	4	4	4
Softness	5	4	4	4	4	4	4	4
Extensibility	5	3	3	3	5	5	5	5
Elasticity	5	6	6	6	5	5	5	5
Spec vol ml/g	4.4	4.3	4.2	4.2	4.3	4.2	4.1	4.1

[0173] Sugars were extracted from the bread crumb using 0.1 M Phosphate buffer pH 8.0 in 70% EtOH. Bread crumb (180 mg) were added to the extraction buffer (1.8 ml) and was incubated for 20 minutes at 70° C. during mixing. The bread crumb was spun down at 12,000 rpm for 5 minutes in a centrifuge and 500 µl of the supernatant was taken and diluted 200× using a 20 mM Phosphate buffer pH 8.0+10 mg/L cellobiose as internal standard. The extracted sugars (glucose, fructose, maltose and maltotriose) were quantified on an ICS-5000 HPLC system with a CarboPac PA1 column. A theoretical sweetness was calculated based on the levels of glucose, fructose and maltose was calculated using sweetness intensity factors. The sweetness factors in Table 30 were based on the determinations in Portmann M O, Birch G. J Sci Food Agric 69(3):275-81, 1995.

TABLE 30

Sugar	Sweetness intensity factor
Glucose	0.5
Fructose	1
Maltose	0.2

[0174] The amounts of sugars (g/kg bread crumb) and theoretical sweetnesses can be found in Table 31. Based on these sugar levels it can be calculated that a dough with 44.6 mg enzyme protein (mgEP) per kg flour of AMG NL have a higher theoretical sweetness than a dough with 112.5 mgEP/kg flour of AMG Goldcrust® and a dough with 53.6 mgEP/kg flour AMG NL generates more glucose than a dough with 112.5 mgEP/kg flour of AMG Goldcrust®.

TABLE 31

	Dough							
	1	2	3	4	5	6	7	8
AMG Goldcrust®, mgEP/kg	112.5	—	—	—	—	—	—	—
AMG NL, mgEP/kg	—	17.9	26.8	35.7	44.6	53.6	62.5	71.4
Glucose, mg/g	17.6	11.4	13.5	15.3	17.0	18.9	19.7	21.4
Fructose, mg/g	10.7	10.3	10.7	10.8	10.6	10.7	10.7	10.7
Maltose, mg/g	12.1	16.1	16.0	16.0	15.6	15.7	14.2	13.8
Maltotriose, mg/g	0.3	0.5	0.5	0.5	0.4	0.4	0.4	0.4

TABLE 31-continued

	Dough							
	1	2	3	4	5	6	7	8
Sweetness calculation*	21.9	19.3	20.7	21.6	22.2	23.3	23.4	24.2

Example 11. Dosage Response of AMG anPAV498 (Partial Sugar Replacement)

[0175] Bread was baked in a straight dough process with a recipe according to Table 32. The ingredients were mixed in a spiral mixer into a dough for 3+7 min at 17 respectively 35 rpm. The doughs were divided into 450 g pieces, rounded, sheeted and place in baking tins. The tins with the doughs were proofed for 55 min at 32° C. and 86% relative humidity. The proofed doughs were baked in a deck oven for 25 min at 230° C.

TABLE 32

	Bakers %
Flour (Kolibri, Meneba, NL)	100
Water	55.5
Yeast	3
Sucrose	5
Salt	0.5
Ascorbic acid	0.04
Calcium Propionate	0.25
Fungamyl 4000SG	7 ppm
Panzea BG	25 ppm

TABLE 33

Eight treatments were prepared with different enzymatic additions								
	Dough							
	1	2	3	4	5	6	7	8
Goldcrust®, mgEP/kg	112.5	—	—	—	—	—	—	—
AMG anPAV498 mgEP/kg	—	12.0	15.1	18.1	21.1	24.1	27.1	30.1
Ja126 mgEP/kg	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35

[0176] The dough properties were evaluated by a trained baker and the volume of the bread was determined using Volscan profiler (Stable microsystems, Godalming, UK). The results from the evaluation can be found in

[0177] Table 34. All the doughs had similar dough properties and produced bread with similar volume.

TABLE 34

	Dough							
	1	2	3	4	5	6	7	8
Stickiness	5	5	5	5	5	5	5	5
Softness	5	5	5	5	6	6	6	6

TABLE 34-continued

	Dough							
	1	2	3	4	5	6	7	8
Extensibility	5	5	5	6	6	6	6	6
Elasticity	5	6	6	5	5	5	5	5
Average spec	4.3	4.4	4.3	4.3	4.3	4.2	4.2	4.2
vol ml/g								
Spec vol	100	101	100	99	100	97	98	96
index %								

[0178] Sugars were extracted from the bread crumb using 0.1 M Phosphate buffer pH 8.0 in 70% EtOH. Bread crumb (180 mg) were added to the extraction buffer (1.8 ml) and was incubated for 20 minutes at 70° C. during mixing. The bread crumb was spun down at 12 000 rpm for 5 minutes in a centrifuge and 500 µl of the supernatant was taken and diluted 200× using a 20 mM Phosphate buffer pH 8.0+10 mg/L cellobiose as internal standard. The extracted sugars (glucose, fructose, maltose and maltotriose) were quantified on an ICS-5000 HPLC system with a CarboPac PA1 column. A theoretical sweetness was calculated based on the levels of glucose, fructose and maltose was calculated using sweetness intensity factors. The sweetness factors in Table 35 were based on the determinations in Portmann M O, Birch G. J Sci Food Agric 69(3):275-81, 1995.

TABLE 35

Sugar	Sweetness intensity factor
Glucose	0.5
Fructose	1
Maltose	0.2

[0179] The amount of sugars (g/kg bread crumb) and is theoretical sweetness can be found in Table 36. Based on these sugar levels it can be calculated that a dough with 24.1 mgEP/kg flour of AMG anPAV498 generates a higher theoretical sweetness than a dough with 112.5 mgEP/kg flour of Goldcrust® and a dough with 27.1 mgEP/kg flour AMG anPAV498 generates more glucose than a dough with 112.5 mgEP/kg flour of Goldcrust®.

TABLE 36

	Dough							
	1	2	3	4	5	6	7	8
Glucose, mg/g	18.0	12.1	14.2	14.8	16.2	17.3	18.4	19.0
Fructose, mg/g	10.6	10.2	10.6	10.5	10.4	10.4	10.3	10.2
Maltose, mg/g	11.3	16.3	16.9	16.5	16.3	16.1	15.8	15.3
Maltotriose, mg/g	0.3	0.6	0.7	0.6	0.6	0.6	0.5	0.5
Sweetness calculation*	21.9	19.5	21.1	21.2	21.8	22.3	22.7	22.7

Example 12. Sensory Comparison of Sweetness of Goldcrust® to AMG NL and AMG anPAV498 (Partial Sugar Replacement)

[0180] Bread was baked in a straight dough process with a recipe according to Table 37. The ingredients were mixed in a spiral mixer into a dough for 3+8 min at 17 respectively 35 rpm. The doughs were divided into 350 g pieces,

rounded, sheeted and place in baking tins. The tins with the doughs were proofed for 85 and 115 min at 35° C. and 85% relative humidity. The proofed doughs were baked in a deck oven for 25 min at 230° C.

TABLE 37

	Bakers %
Flour	100
Water	57
Fresh Yeast	3
Salt	1
Sugar	5
Ascorbic acid	0.06
Fungamyl 4000SG	7 ppm
Panzea BG	25 ppm

TABLE 38

Three treatments were prepared with different enzymatic additions			
Dough	1	2	3
Goldcrust®, mgEP/kg	124.3		
AMG NL, mgEP/kg		52.2	
AMG anPAV498, mgEP/kg			23.6
Ja126 mgEP/kg	0.34	0.34	0.34

TABLE 39

Dough properties			
Dough	1	2	3
Average spec vol ml/g	7.1	7.0	7.2
Spec vol index %	100	99	102

[0181] Sugars were extracted from the bread crumb using 0.1 M Phosphate buffer pH 8.0 in 70% EtOH. Bread crumb (180 mg) were added to the extraction buffer (1.8 ml) and was incubated for 20 minutes at 70° C. during mixing. The bread crumb was spun down at 12 000 rpm for 5 minutes in a centrifuge and 500 µl of the supernatant was taken and diluted 200× using a 20 mM Phosphate buffer pH 8.0+10 mg/L cellobiose as internal standard. The extracted sugars (glucose, fructose, maltose and maltotriose) were quantified on an ICS-5000 HPLC system with a CarboPac PA1 column. A theoretical sweetness was calculated based on the levels of glucose, fructose and maltose was calculated using sweetness intensity factors. The sweetness factors in Table 40 were based on the determinations in Portmann M O, Birch G. J Sci Food Agric 69(3):275-81, 1995.

TABLE 40

Sugar	Sweetness intensity factor
Glucose	0.5
Fructose	1
Maltose	0.2

[0182] The amount of sugars (mg/g bread crumb) and its theoretical sweetness can be found in Table 41 Table 316. AMG Goldcrust® generates the more glucose, while the level of maltose is higher for AMG NL and AMG anPAV498. However, the calculated sweetness of AMG NL

at 52.2 mgEP/kg flour and AMG anPAV498 at 23.6 mgEP/kg flour is actually similar to that of Goldcrust® at a far higher dosing of 124.3 mgEP/kg flour (table 32).

TABLE 41

Dough	1	2	3
Glucose, mg/g	18.9	17.7	15.9
Fructose, mg/g	8.8	8.3	7.9
Maltose, mg/g	4.8	8.6	11.5
Maltotriose, mg/g	M	M	M
Sweetness calculation*	19.3	18.9	18.2

Sensory Evaluation Method

[0183] Each sensory assessor was served 2 slices of each bread type (day 1). Samples were served blind, 3-digit coded, and in random order. 7 assessors participated in the evaluation. Intensity of the bread crumb sweet taste was evaluated on a 1-9 point intensity scale ranging from little to very intense.

[0184] The sweetness did not differ significantly between samples and no other significant differences were noted between samples.

TABLE 42

Dough	1	2	3
Sweet taste	5.6	6.0	5.6

Example 13. Sensory Evaluation, Full Sugar Replacement

[0185] Toast bread (panned bread, open top) —no sucrose added to doughs

TABLE 43

Recipe, % (w/w):	
Ingredients:	Baker's %
Wheat flour, Kolibri %	100
Fresh yeast %	4
Salt %	0.5
Water %	57.5
Enzyme solution	*
Fungamyl ® 4000 SG ppm	7
Panza ® BG ppm	25
Ascorbic acid ppm	40

* Enzyme solutions:

Control (=no starch degrading enzyme and no glucoamylase)

Enzyme solution A: 0.35 mg raw starch degrading alpha-amylase (JA126) protein per kg flour and 112.5 mg Golderust ® glucoamylase protein per kg flour

Enzyme solution B: 0.35 mg raw starch degrading alpha-amylase (JA126) protein per kg flour, and 53.6 mg AMG NL glucoamylase protein per kg flour

Enzyme solution C: 0.35 mg raw starch degrading alpha-amylase (JA126) protein per kg flour, and 21.1 mg AMG anPAV498 glucoamylase protein per kg flour

TABLE 44

Baking procedure	
Procedure	Time, min
Mixing at speed low/high speed (17 rpm/35 rpm)	3/7
Temperature after mixing, ° C.	25.3-26.2
Floor time	20

TABLE 44-continued

Baking procedure	
Procedure	Time, min
Scaling 320 g, in baking tins	10
Table resting/bench time	15
Fermentation time at 32° C., min	80
Baking at temperature 230° C.	25

Sensory Evaluation Method

[0186] Each assessor was served 2 slices of each bread type (day 1). Samples were served blind, 3-digit coded, and in random order. Moist and Soft were evaluated by hand, and sweet by tasting the breadcrumb. The sensory attributes were evaluated on a 1-9 point intensity scale, ranging from little to very. 4 trained assessors participated in the evaluation. Two sensory replicates were performed.

Results:

The doughs had same stickiness and softness. AMG-NL gave more extensible and less elastic dough (Table 45).

TABLE 45

Dough parameters				
	Control	A	B	C
Stickiness	5	6	6	6
Softness	5	5	5	5
Extensibility	5	5	6	5
Elasticity	5	5	4	5

[0187] The data shown in Table 45 demonstrate that solution C gave most moist and soft bread, whereas for sweet taste there was no significant difference. No other differences were noted between samples. There was no significant difference in bread specific volume (Table 46).

TABLE 46

Specific volume index, %				
Control	A	B	C	
100	101	102	103	

TABLE 47

Mean sensory scores of the enzyme bread, 1 day after baking.				
Sensory attribute	A	B	C	p-value
Moist	5.1AB	5.0B	5.8B	0.0338
Soft	5.1B	5.1B	6.6A	0.002
Sweet taste	4.0	3.6	4.0	0.7998

Tukey HSD: Means followed by different letters within sensory attribute were significantly ($P < 0.05$) different between samples

[0188] Sugars were extracted from the bread crumb using 0.1 M Phosphate buffer pH 8.0 in 70% EtOH. Bread crumb (180 mg) were added to the extraction buffer (1.8 ml) and was incubated for 20 minutes at 70° C. during mixing. The bread crumb was spun down at 12 000 rpm for 5 minutes in a centrifuge and 500 µl of the supernatant was taken and

diluted 200× using a 20 mM Phosphate buffer pH 8.0+10 mg/L cellobiose as internal standard. The extracted sugars (glucose, fructose, maltose and maltotriose) were quantified on an ICS-5000 HPLC system with a CarboPac PA1 column.

[0189] The glucose levels in the breads were slightly higher with B and C than A (Table 48), which means improved sweetness for B and C compared to the control A. Maltose was higher with B and C than with A.

TABLE 48

Sugar levels (mg/g bread crumb) in the bread			
Enzyme solution	Glucose mg/g bread	Fructose mg/g bread	Maltose mg/g bread
Control	0.5	0.1	11.2
A	9.6	1.4	6.6
B	10.9	0.3	10.6
C	11.3	0.1	10.7

Example 14. Freshness Effect of AMGs in US Sponge and Dough Recipe

[0190] Bread was baked in a sponge and dough process with a recipe according to Table 49. The bread was baked in lidded tins in order to have the same volume of all bread. The ingredients of the sponge were mixed in a pin mixer into a dough for 2+1 min at 50 respectively 150 rpm. The sponge was proofed for 2 hours at 27° C. and 75% rH. The sponge was placed in the pin mixer with the rest of the ingredients of the dough and mixed into a dough for 1+3 minutes at 50 and 150 rpm respectively.

[0191] The doughs were divided into 400 gram pieces, rounded, sheeted and place in baking tins with lid. The tins with the doughs were proofed for 60 min at 43° C. and 80% relative humidity. The proofed doughs were baked in a revolving oven for 20 min at 215° C.

TABLE 49

Recipe	
Ingredient	Amount
Sponge	
Flour, %	70
Water, %	40.6
Dry yeast, %	2.3
SSL, %	0.5
Soybean oil, %	3
Dimodan HP75, %	0.5
Calcium propionate, %	0.1
Dough	
Flour, %	30
Water, %	16.4
Sugar, %	8
Salt, %	2.3
Calcium Propionate, %	0.4
Ascorbic acid, ppm	60
Fungamyl® 4000 SG, ppm	6
Panzea® BG, ppm	25

TABLE 50

Seven treatments were prepared with different enzymatic additions							
	1	2	3	4	5	6	7
AMG anPAV498, mgEP/kg flour				25	50		
JPO124, mgEP/kg flour			25			50	
JPO172, mgEP/kg flour		25					50

[0192] The bread was packed 2 hours after baking in sealed plastic bags and stored at room temperature until analysis. The texture of the bread was evaluated with a texture analyzer (TA-XT plus, Stable microsystems, Godalmine, UK). Bread crumb texture properties were characterized by firmness (the same as “hardness” and the opposite of “softness”) and the elasticity of the baked product. A standard method for measuring firmness and elasticity is based on force-deformation of the baked product. A force-deformation of the baked products may be performed with a 40 mm diameter cylindrical probe. The force on the cylindrical probe is recorded as it is pressed down 40% strain on a 25 mm thick bread slice at a deformation speed of 1 mm/second. The probe is then kept in this position for 30 seconds while the force is recorded and then probe returns to its original position.

[0193] Firmness (in grams) is defined as the force needed to compress a probe to a 25% strain (corresponding to 6.25 mm compression into a bread crumb slice of 25 mm thickness).

[0194] Elasticity (in %) is defined as the force recoded after 30 seconds compression at 40% strain (corresponding to force at time=40 s for a bread slice of 25 mm thickness) divided by the force needed to press the probe 10 mm into the crumb (corresponding to force at time=10 s for a bread slice of 25 mm thickness) times 100. The results from the texture evaluation can be found in Tables 46 and 47, respectively.

[0195] The control bread increased in firmness (table 51) and lost elasticity (table 52) over storage time, also known as bread staling.

[0196] Surprisingly, all three AMGs tested in this study had an anti-staling effect seen as less increase in firmness over storage time—table 46. The AMGs also had a positive effect on elasticity that starts at a higher level, and after 14 days of storage the AMG breads had a higher elasticity than the control—table 47.

TABLE 51

Firmness over storage time			
Treatment	Day 1	Day 7	Day 14
Control	168	490	750
AMG anPAV498, 25 mgEP/kg flour	191	423	667
AMG anPAV498, 50 mgEP/kg flour	182	387	561
JPO124, 25 mgEP/kg flour	146	374	572
JPO124, 50 mgEP/kg flour	143	319	465
JPO172, 25 mgEP/kg flour	124	318	447
JPO172, 50 mgEP/kg flour	143	301	434

TABLE 52

Elasticity over storage time			
Treatment	Day 1	Day 7	Day 14
Control	52.8	49.9	44.7
anPAV498, 25 mgEP/kg flour	53.9	50.1	46.8
anPAV498, 50 mgEP/kg flour	55.9	51.6	48.1
JPO124, 25 mgEP/kg flour	55.7	52.2	48.4
JPO124, 50 mgEP/kg flour	58.0	54.1	51.7
JPO172, 25 mgEP/kg flour	55.6	53.3	50.2
JPO172, 50 mgEP/kg flour	59.8	56.3	52.7

Example 15. Use of Thermostable AMG in Cake Applications

Muffins were baked using a commercial cake mix (Tegral Satin Creame Cake Neutral SG, Puratos, UK) using the recipe on the bag.

TABLE 53

Muffin recipe.	
Ingredient	Amount, %
Tegral satin cream cake mix	100
Egg	40
Oil (rapeseed oil)	30
Water	22.5

TABLE 54

Nine muffin treatments were prepared with different enzyme additions:	
1. Blank	
2. JPO124 100 mgEP/kg cake mix	
3. JPO124 900 mgEP/kg cake mix	
4. JPO172 100 mgEP/kg cake mix	
5. JPO172 900 mgEP/kg cake mix	
6. OC50 1250 MANU/kg cake mix (6.25 mgEP/kg cake mix)	
7. OC50 2500 MANU/kg cake mix (12.5 mgEP/kg cake mix)	
8. OC50 3750 MANU/kg cake mix (18.75 mgEP/kg cake mix)	
9. OC50 5000 MANU/kg cake mix (25 mgEP/kg cake mix)	

TABLE 55

Muffin making process:	
1. Eggs, oil and water according to table 48 were added to the mixing bowl.	
2. The individual treatments, according to table 49, were added to each dough	
3. The cakemix was added to the mixing bowl and mixed for 1 min with a hand mixer on speed 1 into a cake batter.	
4. The cake batter was placed in muffin tin (50 g batter to each tin) using a piping bag.	
5. The muffins were baked for 28 minutes in a deck oven with a top heat at 200° C. and bottom heat at 180° C. and with a tray upside down in the bottom of the oven.	
6. The muffins were allowed to cool down for 1 h and placed in sealed plastic bag with modified atmosphere and stored at room temperature until analysis.	

[0197] The textural properties of the muffin were analyzed using a texture analyzer performing a texture profile analysis (TPA). In the analysis of the muffin, the top of the muffin was

cut off at the same level as the muffin tin leaving a 3 cm muffin. The muffin was placed on the texture analyzer and a 25 mm diameter cylindrical probe was pressed down into the muffin twice to a 7 mm depth at a constant upward and downward speed of 1 mm/s with 5 seconds between the two compressions. The force (gram) as a function of time (seconds) and distance (mm) was recorded.

[0198] The peak force of the first compression corresponds to the hardness (gram) of the muffin.

[0199] The area below the force-distance curve of the second compression divided by the area below the force-distance curve of the first compression corresponds to cohesiveness and were expressed in %.

[0200] The area below the force-distance curve of the first upward move divided by the area below the first downward move corresponds to resilience and were expressed in %.

Table 56 below illustrates the benefits of using JPO172 and JPO124 in muffins. The muffins treated with JPO124 and JPO172 have a surprisingly improved (higher) resilience and Cohesiveness; even higher than other known solutions for improving cake freshness.

TABLE 56

Textural properties of muffins with different enzymatic treatments.			
Treatment	Hardness	Resilience, %	Cohesiveness, %
Blank	174	30.2	66.4
JPO124 100 me/kg cake mix	175	30.2	66.4
JPO124 900 mgEP/kg cake mix	184	31.5	67.2
JPO172 100 mgEP/kg cake mix	174	32.8	68.7
JPO172 900 mgEP/kg cake mix	177	33.7	69.7
OC50 1250 MANU/kg cake mix (6.25 mgEP)	160	30.3	66.9
OC50 2500 MANU/kg cake mix (12.5 mgEP)	159	30.7	67.3
OC50 3750 MANU/kg cake mix (18.75 mgEP)	157	31.6	68.2
OC50 5000 MANU/kg cake mix (25 mgEP)	155	31.9	68.1

Example 16. Sensory Comparison of Freshness of AMG Goldcrust®, AMG NL, AMG AnPAV498, JP172 and Novamyl 3D®

[0201] Bread was baked in a straight dough process with a recipe according to Table 57. The ingredients were mixed in a spiral mixer into a dough for 3+6 min at 17 respectively 35 rpm. The doughs were divided into 450 g pieces, rounded, sheeted and placed in baking tins. The tins with the doughs were proofed for 55 mins at 32° C. and 86% relative humidity. The proofed doughs were baked in a deck oven for 35 min at 230° C.

TABLE 57

Bakers %	
Flour	100
Water	57
Fresh Yeast	4.5
Salt	1.5
Sugar	1.5
Calcium propionate	0.25
Ascorbic acid	0.04
Fungamyl 4000SG	7 ppm
Panzea BG	25 ppm

TABLE 58

Treatments were prepared with different enzymatic additions						
	Dough					
	1	2	3	4	5	6
Control (no glucoamylase)						
Goldcrust ®		50				
3300 BG, mgEP/kg						
AMG NL mgEP/kg			50			
AMG				500		
AnPAV498 mgEP/kg*)						
JPO172 mgEP/kg					50	
Novamyl						25
3D ppm						

*)Note:
AMG AnPAV498 was mistakenly overdosed ten times in this experiment; it should have been 50 mgEP/kg, but was 500 mgEP/kg.

Sensory Evaluation Method

[0202] Sensory evaluation was performed on day 1 and day 8. A training session was held prior to evaluation, identifying the relevant attributes and procedures (Table 59). Texture was evaluated by hand. 4-5 trained assessors participated in the evaluation. Each assessor was served 2 slices without crust of each bread type. Samples were served blind, 3-digit coded, and in random order. Intensity of the sensory attributes were evaluated on a 1-9 point intensity scale ranging from little to very intense. Two sensory replicates were performed on each evaluation day.

TABLE 59

Description of sensory attributes, procedures and evaluation		
Attribute	Procedure	Evaluate
Moist	Touch the surface of the bread, hold for 3 sec	Degree of moistness/cooling perceived
Soft	Gently compress the bread	Ease of compressing the crumb
Resilient/springy	Compress the slice completely	Degree to which the crumb recovers to original shape
	Use the other bread slice	
Foldable	Fold the bread slice	Extent of coherency at the fold

Sensory Results

JPO172 and AMG AnPAV498 scored highest on all the evaluated freshness attributes on day 1, and on Moist, Soft and foldable Day 8. AMG Goldcrust® did not differ from Control.

TABLE 60

Mean values of sensory scores of the bread day 1.				
Day 1	Moist	Soft	Resilient/Springy	Foldable
Control	5.8	5.6	6.3	5.5
50 mgEP/kg Goldcrust ®	5.8	5.1	6.6	6.0
50 mgEP/kg JPO172	8.0	8.0	7.6	9.0
50 mgEP/kg AMG NL	6.1	6.1	6.6	6.7
500 mgEP/kg AMG AnPAV498	8.2	8.1	7.9	8.8
25 ppm Novamyl 3D	5.8	6.4	6.0	7.1

TABLE 61

Mean values of sensory scores of the bread day 8.				
Day 8	Moist	Soft	Resilient/Springy	Foldable
Control	2.3	2.0	7.3	1.2
50 mgEP/kg Goldcrust ®	2.4	1.7	7.7	1.7
50 mgEP/kg JPO172	6.1	6.5	7.8	7.5
50 mgEP/kg flour AMG NL	2.5	2.6	7.7	1.8
500 mgEP/kg AMG AnPAV498	6.3	6.9	7.8	8.1
25 ppm Novamyl 3D	4.2	4.7	6.6	4.6

Example 17. Freshness Effect of AMG the First 24 Hours

[0203] Bread was baked in a straight dough mini baking process with a recipe according to

[0204] Table 62. The bread was baked in lidded tins in order to have the same volume of all bread. The ingredients were mixed in a spiral mixer into a dough for 4 minutes at 90 rpm. The doughs were divided into 20 g pieces, rounded and place in baking tins. The tins with the doughs were proofed on a conveyor belt for 55 min at 36° C. and 80% relative humidity. The proofed doughs were baked in mini tunnel oven for 12 min at 210° C.

TABLE 62

Bakers %	
Flour (Kolibri, Meneba, NL)	100
Water	58
Yeast	4.5
Sucrose	1.5
Salt	1.5
Ascorbic acid	0.04
Calcium Propionate	0.3
Fungamyl ® 4000SG	8 ppm
Panzea ® BG	25 ppm

TABLE 63

Ten dough treatments were prepared with different enzymatic additions
1. Control (Blank)
2. Datem 0.5%
3. JPO172 50 mgEP/kg flour
4. Opticake 50 BG 200 MANU/kg
5. JPO124 50 mgEP/kg flour
6. Novamyl 3D 440 MANU/kg flour
7. SSL 0.5%
8. Distilled monoglycerides 0.5%
9. Novamyl 10 000 BG 750 MANU/kg
10. Lipopan Extra 200 LU/kg

[0205] The doughs were baked and the resulting breads were packed 0.5 hours after baking in sealed plastic bags and stored at room temperature until analysis.

[0206] The texture of each bread was evaluated with a texture analyzer (TA-XT plus, Stable microsystems, Godalmine, UK). Bread crumb texture properties were characterized by firmness (the same as “hardness” and the opposite of “softness”) and the elasticity of the baked product.

[0207] A standard method for measuring firmness and elasticity is based on force-deformation of the baked product. A force-deformation of the baked products may be performed with a 20 mm diameter spherical probe. The force on the probe is recorded as it is pressed down 40% strain on a 25 mm thick bread slice at a deformation speed of 1 mm/second. The probe is then kept in this position for 30 seconds while the force is recorded and then probe returns to its original position.

[0208] Firmness (in grams) is defined as the force needed to compress a probe to a 25% strain (corresponding to 6.25 mm compression into a bread crumb slice of 25 mm thickness).

[0209] Elasticity (in %) is defined as the force recoded after 30 seconds compression at 40% strain (corresponding to force at time=40 s for a bread slice of 25 mm thickness) divided by the force needed to press the probe 10 mm into the crumb (corresponding to force at time=10 s for a bread slice of 25 mm thickness) times 100.

[0210] The results from the texture analysis can be found in Table 64 (firmness) and Table 65 (elasticity).

[0211] Fresh bread without enzyme (control) has low firmness and high elasticity, as the bread is stored the firmness increase over time and the elasticity decrease. Traditional AMGs used in baking applications (for example AMG Goldcrust®) do not impact the firmness or elasticity (see Example 7).

[0212] Two of the AMGs herein (JPO124 and JPO172) dosed at 50 mgEP/kg improved (reduced) the increase in firmness over time. Both AMGs (JPO124 and JPO172) dosed at 50 mgEP/kg improved (increased) the initial elasticity and prevented the loss of elasticity over time.

TABLE 64

Firmness (g) 2, 5 and 24 hours after baking of bread with enzyme treatments according Table 63.			
Treatment	2 Hour	5 hours	24 hour
Control	166	189	463
JPO124 50 mgEP/kg	165	175	311
JPO172 50 mgEP/kg	163	166	236
Novamyl 10 000 BG	166	188	358
750 MANU/kg			
Novamyl 3D 440 MANU/kg	155	181	329
OC50 200 MANU/kg	175	180	325
Lipopan Xtra 200 LU/kg	145	156	334
DMG 0.5%	188	240	482
DATEM 0.5%	163	173	403
SSL 0.5%	194	193	407

TABLE 65

Elasticity (%) 2, 5 and 24 hours after baking of bread with enzyme treatments according Table 63.			
Treatment	2 hours	5 hours	24 hours
Control	64.5	63.8	55.9
JPO124 50 mgEP/kg	66.2	66.6	63.2
JPO172 50 mgEP/kg	66.2	66.7	65.2
Novamyl 10 000 BG	61.8	61.5	59.0
750 MANU/kg			
Novamyl 3D 440 MANU/kg	63.3	61.9	58.1
OC50 200 MANU/kg	62.2	63.5	59.4
Lipopan Xtra 200 LU/kg	62.0	61.4	54.6
DMG 0.5%	62.2	61.2	54.8
DATAM 0.5%	63.4	62.9	56.3
SSL 0.5%	58.2	57.8	51.8

Example 18. Freshness Effect of AMG at High Dosages

[0213] Bread was baked in a straight dough mini baking process with a recipe according to

[0214] Table 66. The bread was baked in lidded tins in order to have the same volume of all bread. The ingredients were mixed in a spiral mixer into a dough for 4 minutes at 90 rpm. The doughs were divided into 20 g pieces, rounded and place in baking tins. The tins with the doughs were proofed on a conveyor belt for 55 min at 36° C. and 80% relative humidity. The proofed doughs were baked in mini tunnel oven for 12 min at 210° C.

TABLE 66

	Bakers %
Flour (Kolibri, Meneba, NL)	100
Water	58
Yeast	4.5
Sucrose	1.5
Salt	1.5
Ascorbic acid	0.04
Calcium Propionate	0.3
Fungamyl ® 4000SG	8 ppm
Panzea ® BG	25 ppm

TABLE 67

Ten treatments were prepared with different enzymatic additions	
1. Control	
2. Opticake 50BG 200 MANU/kg	
3. JPO124 50 mgEP/kg flour	
4. JPO124 100 mgEP/kg flour	
5. JPO124 300 mgEP/kg flour	
6. JPO124 500 mgEP/kg flour	
7. JPO172 50 mgEP/kg flour	
8. JPO172 100 mgEP/kg flour	
9. JPO172 300 mgEP/kg flour	
10. JPO172 500mgEP/kg flour	

[0215] The doughs were baked and the resulting breads were packed 0.5 hours after baking in sealed plastic bags and stored at room temperature until analysis.

[0216] The texture of each bread was evaluated with a texture analyzer (TA-XT plus, Stable microsystems, Godalmine, UK). Bread crumb texture properties were characterized by firmness (the same as “hardness” and the opposite of “softness”) and the elasticity of the baked product.

[0217] A standard method for measuring firmness and elasticity is based on force-deformation of the baked product. A force-deformation of the baked products may be performed with a 20 mm diameter spherical probe. The force on the probe is recorded as it is pressed down 40% strain on a 25 mm thick bread slice at a deformation speed of 1 mm/second. The probe is then kept in this position for 30 seconds while the force is recorded and then probe returns to its original position.

[0218] Firmness (in grams) is defined as the force needed to compress a probe to a 25% strain (corresponding to 6.25 mm compression into a bread crumb slice of 25 mm thickness).

[0219] Elasticity (in %) is defined as the force recoded after 30 seconds compression at 40% strain (corresponding to force at time=40 s for a bread slice of 25 mm thickness) divided by the force needed to press the probe 10 mm into the crumb (corresponding to force at time=10 s for a bread slice of 25 mm thickness) times 100.

[0220] The results from the texture analysis can be found in Table 68 (firmness) and Table 69 (elasticity).

[0221] Fresh bread without enzyme (control) has low firmness and high elasticity, as the bread is stored the firmness increase over time and the elasticity decrease. Traditional AMGs used in baking applications (for example AMG Goldcrust®) do not impact the firmness or elasticity (see Example 7).

[0222] The two new AMGs (JPO124 and JPO172) improved (reduced) initial firmness and the increase in firmness over time. The higher dosage the lower increase in firmness over time. Both AMGs (JPO124 and JPO172) improved (increased) the initial elasticity and prevented the loss of elasticity over time. The higher dosage of the AMG the higher initial elasticity and the lower loss of elasticity over time.

TABLE 68

Firmness (g) on day 1 and 7 of bread with enzyme treatments according Table 67.		
Treatment	Day 1	Day 7
Control	259	1083
Opticake 50BG 200 MANU/kg flour	217	451
JPO124 50 mgEP/kg flour	185	639
JPO124 100 mgEP/kg flour	159	416
JPO124 300 mgEP/kg flour	146	230
JPO124 500 mgEP/kg flour	201	237
JPO172 50 mgEP/kg flour	174	463
JPO172 100 mgEP/kg flour	145	369
JPO172 300 mgEP/kg flour	162	210
JPO172 500 mgEP/kg flour	209	280

TABLE 69

Elasticity (%) on day 1 and 7 of bread with enzyme treatments according Table 67.		
Treatment	Day 1	Day 7
ontrol	57.2	45.3
Opticake 50BG 200 MANU/kg flour	60.2	52.9
JPO124 50 mgEP/kg flour	63.9	53.1
JPO124 100 mgEP/kg flour	65.9	59.2
JPO124 300 mgEP/kg flour	68.4	67.3
JPO124 500 mgEP/kg flour	69.0	68.7
JPO172 50 mgEP/kg flour	64.5	57.1
JPO172 100 mgEP/kg flour	66.3	62.5
JPO172 300 mgEP/kg flour	67.6	67.2
JPO172 500 mgEP/kg flour	68.4	68.6

Example 19. Freshness Effect of AMG Combined with Lip182

[0223] Bread was baked in a straight dough mini baking process with a recipe according to

[0224] Table 10. The bread was baked in lidded tins in order to have the same volume of all bread. The ingredients were mixed in a spiral mixer into a dough for 4 minutes at 90 rpm. The doughs were divided into 20 g pieces, rounded and place in baking tins. The tins with the doughs were proofed on a conveyor belt for 55 min at 36° C. and 80% relative humidity. The proofed doughs were baked in mini tunnel oven for 12 min at 210° C.

TABLE 70

Bakers %	
Flour (Kolibri, Meneba, NL)	100
Water	58
Yeast	4.5
Sucrose	1.5
Salt	1.5
Ascorbic acid	0.04
Calcium Propionate	0.3
Fungamyl® 4000SG	8 ppm
Panzea® BG	25 ppm

TABLE 71

Various treatments were prepared with different enzymatic additions			
Treatment	Opticake 50BG, MANU/kg flour	JPO172, mgEP/kg flour	Lip182, PLA(B)/kg flour
1	200		
2			
3			2.5
4			5
5		25	
6		50	
7		25	2.5
8		25	5
9		50	2.5
10		50	5

[0225] The doughs were baked and the resulting breads were packed 0.5 hours after baking in sealed plastic bags and stored at room temperature until analysis.

[0226] The texture of each bread was evaluated with a texture analyzer (TA-XT plus, Stable microsystems, Godalmine, UK). Bread crumb texture properties were characterized by firmness (the same as “hardness” and the opposite of “softness”) and the elasticity of the baked product.

[0227] A standard method for measuring firmness and elasticity is based on force-deformation of the baked product. A force-deformation of the baked products may be performed with a 20 mm diameter spherical probe. The force on the probe is recorded as it is pressed down 40% strain on a 25 mm thick bread slice at a deformation speed of 1 mm/second. The probe is then kept in this position for 30 seconds while the force is recorded and then probe returns to its original position.

[0228] Firmness (in grams) is defined as the force needed to compress a probe to a 25% strain (corresponding to 6.25 mm compression into a bread crumb slice of 25 mm thickness).

[0229] Elasticity (in %) is defined as the force recoded after 30 seconds compression at 40% strain (corresponding to force at time=40 s for a bread slice of 25 mm thickness) divided by the force needed to press the probe 10 mm into the crumb (corresponding to force at time=10 s for a bread slice of 25 mm thickness) times 100.

[0230] The results from the texture analysis can be found in Table 72 (firmness) and Table 73 (elasticity).

[0231] Fresh bread without enzyme (control) has low firmness and high elasticity, as the bread is stored the firmness increase over time and the elasticity decrease. Traditional AMGs used in baking applications (for example, AMG Goldcrust®) does not impact the firmness or elasticity (see Example 7).

[0232] The AMG JPO172 improved (reduced) initial firmness and the increase in firmness over time. The lipase Lip182 had no effect on firmness alone compared to a control bread. The combination of Lip182 and JPO172 resulted in the bread with lowest firmness on both day 1 and 7.

[0233] The AMG JPO172 improved (increased) the initial elasticity and prevented the loss of elasticity over time. The lipase Lip182 had similar elasticity as the control and the combination of JPO172 and Lip182 was similar to JPO172 alone.

TABLE 72

Firmness (g) on day 1 and 7 of bread with enzyme treatments according Table 71.		
Treatment	Day 1	Day 7
Control	291	936
Opticake 50 BG, 200 MANU/kg flour	170	333
Lip182, 2.5 PLA(B)/kg flour	340	875
Lip182, 5 PLA(B)/kg flour	326	1031
JPO172, 25 mgEP/kg flour	198	481
JPO172, 50 mgEP/kg flour	172	495
JPO172, 25 mgEP/kg flour – Lip182, 2.5 PLA(B)/kg flour	217	728
JPO172 25 mgEP/kg flour – Lip182, 5 PLA(B)/kg flour	193	578
JPO172, 50 mgEP/kg flour – Lip182, 2.5 PLA(B)/kg flour	171	565
JPO172, 50 mgEP/kg flour – Lip182, 5 PLA(B)/kg flour	158	377

TABLE 73

Elasticity (%) on day 1 and 7 of bread with enzyme treatments according Table 71.		
Treatment	Day 1	Day 7
Control	58.2	45.8
Opticake 50 BG, 200 MANU/kg flour	60.8	55.8
Lip182, 2.5 PLA(B)/kg flour	57.6	46.4
Lip182, 5 PLA(B)/kg flour	57.2	46.7
JPO172, 25 mgEP/kg flour	63.4	51.3
JPO172, 50 mgEP/kg flour	66.0	57.7
JPO172, 25 mgEP/kg flour – Lip182, 2.5 PLA(B)/kg flour	61.7	49.9
JPO172 25 mgEP/kg flour – Lip182, 5 PLA(B)/kg flour	61.9	51.0
JPO172, 50 mgEP/kg flour – Lip182, 2.5 PLA(B)/kg flour	64.8	55.2
JPO172, 50 mgEP/kg flour – Lip182, 5 PLA(B)/kg flour	64.7	57.6

Example 20. Freshness Effect of AMG Combined with Gluzyne Fortis

[0234] Bread was baked in a straight dough mini baking process with a recipe according to

[0235] Table 74. The bread was baked in lidded tins in order to have the same volume of all bread. The ingredients were mixed in a spiral mixer into a dough for 4 minutes at 90 rpm. The doughs were divided into 20 g pieces, rounded and place in baking tins. The tins with the doughs were proofed on a conveyor belt for 55 min at 36° C. and 80% relative humidity. The proofed doughs were baked in mini tunnel oven for 12 min at 210° C.

TABLE 74

Bakers %	
Flour (Kolibri, Meneba, NL)	100
Water	58
Yeast	4.5
Sucrose	1.5
Salt	1.5
Ascorbic acid	0.04
Calcium Propionate	0.3
Fungamyl® 4000SG	8 ppm
Panzea® BG	25 ppm

TABLE 75

Various treatments were prepared with different enzymatic additions. Additional water was added to achieve similar dough rheology.				
Treatment	Opticake 50BG, MANU/kg flour	JPO172, mgEP/kg flour	Gluzyme Fortis, GODU/kg of flour	Additional water, %
1				
2	200			
3			150	0.5
4			300	1.5
5		25		
6		50		
7		25	150	0.5
8		25	300	1.5
9		50	150	0.5
10		50	300	1.5

[0236] The doughs were baked and the resulting breads were packed 0.5 hours after baking in sealed plastic bags and stored at room temperature until analysis.

[0237] The texture of each bread was evaluated with a texture analyzer (TA-XT plus, Stable microsystems, Godalmine, UK). Bread crumb texture properties were characterized by firmness (the same as “hardness” and the opposite of “softness”) and the elasticity of the baked product.

[0238] A standard method for measuring firmness and elasticity is based on force-deformation of the baked product. A force-deformation of the baked products may be performed with a 20 mm diameter spherical probe. The force on the probe is recorded as it is pressed down 40% strain on a 25 mm thick bread slice at a deformation speed of 1 mm/second. The probe is then kept in this position for 30 seconds while the force is recorded and then probe returns to its original position.

[0239] Firmness (in grams) is defined as the force needed to compress a probe to a 25% strain (corresponding to 6.25 mm compression into a bread crumb slice of 25 mm thickness).

[0240] Elasticity (in %) is defined as the force recoded after 30 seconds compression at 40% strain (corresponding to force at time=40 s for a bread slice of 25 mm thickness) divided by the force needed to press the probe 10 mm into the crumb (corresponding to force at time=10 s for a bread slice of 25 mm thickness) times 100.

[0241] The results from the texture analysis can be found in Table 76 (firmness) and Table 77 (elasticity).

[0242] Fresh bread without enzyme (control) has low firmness and high elasticity, as the bread is stored the firmness increase over time and the elasticity decrease. Traditional AMGs used in baking applications (for example Goldcrust) does not impact the firmness or elasticity (Example 7).

[0243] The AMG JPO172 improved (reduced) initial firmness and the increase in firmness over time. The glucose oxidase (Gluzyme Fortis) alone reduced firmness compared to a control bread to some degree. The combination of glucose oxidase and JPO172 resulted in the bread with lowest firmness on both day 1 and 7.

[0244] The AMGs JPO172 improved (increased) the initial elasticity and prevented the loss of elasticity over time. The glucose oxidase (Gluzyme Fortis) alone had similar elasticity as the control and the combination of JPO172 and the glucose oxidase was similar to JPO172 alone.

TABLE 76

Firmness (g) on day 1 and 7 of bread with enzyme treatments according Table 75.		
	Day 1	Day 7
Blank	281	944
Opticake ® 50 BG, 200 MANU/kg flour	169	423
Gluzyme Fortis ®, 150 GODU/kg flour	216	825
Gluzyme Fortis ®, 300 GODU/kg flour	239	851
JPO172, 25 mgEP/kg flour	157	521
JPO172, 50 mgEP/kg flour	154	390
Gluzyme Fortis ®, 150 GODU/kg flour + JPO172, 25 mgEP/kg flour	189	572
Gluzyme Fortis ®, 300 GODU/kg flour + JPO172, 25 mgEP/kg flour	159	456
Gluzyme Fortis ®, 150 GODU/kg flour + JPO172, 50 mgEP/kg flour	155	445
Gluzyme Fortis ®, 300 GODU/kg flour + JPO172, 50 mgEP/kg flour	128	354

TABLE 77

Elasticity (%) on day 1 and 7 of bread with enzyme treatments according Table 75.		
	Day 1	Day 7
Blank	55.8	44.1
Opticake ® 50 BG, 200 MANU/kg flour	57.8	52.0
Gluzyme Fortis ®, 150 GODU/kg flour	57.1	44.3
Gluzyme Fortis ®, 300 GODU/kg flour	57.4	44.1
JPO172, 25 mgEP/kg flour	61.8	52.1
JPO172, 50 mgEP/kg flour	64.4	57.0
Gluzyme Fortis ®, 150 GODU/kg flour + JPO172, 25 mgEP/kg flour	60.8	48.4
Gluzyme Fortis ®, 300 GODU/kg flour + JPO172, 25 mgEP/kg flour	61.9	49.5
Gluzyme Fortis ®, 150 GODU/kg flour + JPO172, 50 mgEP/kg flour	64.7	56.0
Gluzyme Fortis ®, 300 GODU/kg flour + JPO172, 50 mgEP/kg flour	63.4	55.4

Example 21. Sensory Comparison of Freshness Effect of AMG Goldcrust®, AMG NL, AMG AnPAV498 and JPO124 in Sponge and Dough Recipe

[0245] Bread was baked in a sponge and dough process with a recipe according to table 78. The bread was baked in lidded tins in order to have the same volume of all bread. The ingredients of the sponge were mixed in a pin mixer into a dough for 2+1 min at 50 respectively 150 rpm. The sponge was proofed for 2 hours at 27° C. and 75% rH. The sponge was placed in the pin mixer with the rest of the ingredients of the dough and mixed into a dough for 1+3 minutes at 50 and 150 rpm respectively.

[0246] The doughs were divided into 400 gram pieces, rounded, sheeted and place in baking tins with lid. The tins with the doughs were proofed for 60 min at 43° C. and 80% relative humidity. The proofed doughs were baked in a revolving oven for 20 min at 215° C.

TABLE 78

Recipe	
Ingredient	Amount
Sponge	
Flour, %	70
Water, %	40.6
Dry yeast, %	2.0
SSL, %	0.5
Soybean oil, %	3
Calcium propionate, %	0.1
Dough	
Flour, %	30
Water, %	16.4
Sugar, %	8
Salt, %	2.0
Calcium Propionate, %	0.4
Ascorbic acid, ppm	60
Fungamyl ® 4000 SG, ppm	7
Panzea ® BG, ppm	25

TABLE 79

Treatments were prepared with different enzymatic additions					
Dough	1	2	3	4	5
Control (no glucoamylase)					
AMG		50			
Goldcrust ®, mgEP/kg					
AMG NL mg EP/kg			50		
AMG				50	
AnPAV498 mgEP/kg					
JPO124 mgEP/kg					50

Sensory Evaluation Method

[0247] Sensory evaluation was performed on day 1 and day 7. A training session was held prior to evaluation, identifying the relevant attributes and procedures (Table 80). Texture was evaluated by hand. 5 trained assessors participated in the evaluation. Each assessor was served 2 slices of each bread type. Samples were served blind, 3-digit coded, and in random order. Intensity of the sensory attributes were evaluated on a 1-9 point intensity scale ranging from little to very intense. Two sensory replicates were performed on each evaluation day.

TABLE 80

Description of sensory attributes, procedures and evaluation		
Attribute	Procedure	Evaluate
Moist	Touch the surface of the bread, hold for 3 sec	Degree of moistness/cooling perceived
Soft	Gently compress the bread	Ease of compressing the crumb
Resilient/springy	Compress the slice completely	Degree to which the crumb recovers to original shape
Use the other bread slice		
Foldable	Fold the bread slice	Extend of coherency at the fold

Sensory Results

JPO124 scored the highest on Moist, Soft and Foldable day 7, followed by AMG AnPAV498.

TABLE 81

Mean values of sensory scores of the bread day 1.				
Day 1	Moist	Soft	Resilient/springy	Foldable
Control	6.7	6.6	5.0	5.3
50 mgEP/kg Goldcrust ®	6.8	6.8	5.1	6.3
50 mgEP/kg JPO124	7.4	7.1	5.3	6.5
50 mgEP/kg AMG NL	7.0	6.9	4.4	5.7
50 mgEP/kg AMG	6.8	6.8	5.4	5.9
AnPAV498				

TABLE 82

Mean values of sensory scores of the bread day 7.				
Day 7	Moist	Soft	Resilient/springy	Foldable
Control	3.2	3.5	4.8	1.2
50 mgEP/kg Goldcrust ®	3.7	3.9	5.3	1.3
50 mgEP/kg JPO124	4.6	5.2	4.5	2.7
50 mgEP/kg AMG NL	3.7	3.6	4.5	1.6
50 mgEP/kg AMG	4.0	4.3	4.3	2.3
AnPAV498				

Example 22. Freshness Effect of JPO172 in Low pH Mixed Rye/Wheat Sour Dough Bread

[0248] Bread was baked in a straight dough process with a recipe according to Table 83. Nine different treatments were done according to table 84. The ingredients were mixed in a spiral mixer into a dough for 6+4 min at 17 respectively 35 rpm. The doughs were divided into 650 g pieces, rounded, sheeted and place in baking tins. The pH of the final dough was of 4.3. The bread was baked in lidded tins in order to have the same volume of all bread. The tins with the doughs were proofed for 60 min at 32° C. and 85% relative humidity. The proofed doughs were baked in a deck oven for 20 min at 225° C. 10

TABLE 83

Recipe	
Ingredients	Amount, %
Flour 1 (Rye flour)	50
Flour 2 (Wheat flour)	50
Water	65.5
Yeast 70% water	3
Rye sour	2.24
Salt	2
Ascorbic acid	0.06
Calcium Propionate	0.25
Vital Wheat Gluten	1
Vinegar	1
Citric acid	0.3

TABLE 84

Treatments.			
	Pentopan 500, ppm (flour)	JPO172 mg EP/kg flour	Sample name
1	50		Control
2	50	12.5	12.5 mgEP/kg flour JPO172
3	50	25	25 mgEP/kg flour JPO172
4	50	50	50 mgEP/kg flour JPO172
5	50	100	100 mgEP/kg flour JPO172

[0249] After baking, the bread was allowed to cool down for 2 hours and placed in sealed plastic bags. The bread was stored at room temperature until analysis.

[0250] The texture of each bread was evaluated with a texture analyzer (TA-XT plus, Stable microsystems, Godalmine, UK). Bread crumb texture properties were characterized by firmness (the same as “hardness” and the opposite of “softness”) and the elasticity of the baked product. A standard method for measuring firmness and elasticity is based on force-deformation of the baked product. A force-deformation of the baked products may be performed with a 40 mm diameter cylindrical probe. The force on the cylindrical probe is recorded as it is pressed down at a deformation speed of 1 mm/second. The probe is then kept in this position for 30 seconds while the force is recorded and then the probe returns to its original position.

[0251] Firmness (in grams) is defined as the force needed to compress a probe to a 25% strain (corresponding to 6.25 mm compression into a bread crumb slice of 25 mm thickness).

[0252] Elasticity (in %) is defined as the force recoded after 30 seconds compression at 40% strain (corresponding to force at time=40 s for a bread slice of 25 mm thickness) divided by the force needed to press the probe 10 mm into the crumb (corresponding to force at time=10 s for a bread slice of 25 mm thickness) times 100.

[0253] The results from the texture analysis can be found in Table 85 (firmness) and Table 86 (elasticity).

[0254] Fresh bread without any treatment (Control) have a low firmness and high elasticity, as the bread is stored the bread becomes more firm and loses elasticity. Bread with JPO172 was less firm and had a higher elasticity after baking. The change in firmness and elasticity over time was also reduced compared to a control bread, making the bread with JPO172 less firm and more elastic on day 7 compared to a control bread on day 1.

TABLE 85

Effect on various treatments on Firmness				
Treatment	Day 1	Day 3	Day 7	
Control	1316	2331	3040	
12.5 mgEP/kg flour JPO172	1012	1592	2072	
25 mgEP/kg flour JPO172	872	1217	1668	
50 mgEP/kg flour JPO172	858	1129	1312	
100 mgEP/kg flour JPO172	879	992	1097	

TABLE 86

Effect on various treatments on Elasticity.			
Treatment	Day 1	Day 3	Day 7
Control	55.4	44.9	43.6
12.5 mgEP/kg flour JPO172	61.9	54.0	48.2
25 mgEP/kg flour JPO172	64.6	60.6	54.6
50 mgEP/kg flour JPO172	65.4	63.1	61.1
100 mgEP/kg flour JPO172	66.2	65.6	65.0

Example 23. Freshness Effect of JPO172 in Tortillas

[0255] Tortillas were made using the recipe in Table 87, different enzymatic solutions were added according to table 88. The ingredients were mixed in a pin mixer for 1+6 minutes at low and high speed respectively. The doughs were allowed to rest for 2 minutes. The dough was divided into 30 g pieces and shaped into rolls. The tortillas were baked in a two-step process where the dough pieces first went through a hot press at 160° C. for 6 seconds, secondly the tortilla was baked for 20 seconds and flipped over and baked for another 20 seconds.

TABLE 87

Recipe.	
Ingredient	Amount, %
Flour	100.0
Water	54
Baking powder	3.0
Glycerol	4.5
Salt	2.0
Sugar	2.00
Citric acid	0.40
Calcium propionate	0.50
DMG	0.50
SSL	0.25
Guar Gum	0.30
Sunflower oil	6.00

TABLE 88

Various treatments were prepared with different enzymatic additions.			
Enzyme Dough#	JPO172 mg EP/ kg flour	Sensea ® wrap ppm (flour)	Sample name
1			control
2	400		400 mgEP/kg JPO172
3	2000		2000 mgEP/kg JPO172
4		200	200 ppm Sensea Wrap
5		400	400 ppm Sensea Wrap

[0256] The tortillas were allowed to cool down for 30 minutes after baking and the placed in a sealed plastic bag that were stored at room temperature until analysis.

[0257] The texture properties of tortillas were evaluated with a texture analyzer (Stable Microsystems, Godalming, UK) using the Tortilla/Pastry Burst Rig (HDP/TPB). In the test procedure, the sample is held between two plates and the 1" spherical probe is driven through the center. The force

and distance to extend the sample are measured and used as an indication of ‘deformation resistance’ and ‘extensibility’, respectively.

[0258] The tortillas are typically used as wraps where the tortilla is wrapped around different types of fillings. An important parameter is the extensibility which describes the resistance to rupture. A fresh tortilla is extensible. However, it loses this extensibility quite rapidly upon storage, as can be seen in table 90. The addition of JPO172 results in a tortilla that has an extensibility similar to a freshly baked tortilla after 28 days.

TABLE 89

Deformation resistance, g of tortilla				
Treatment	Dose	Day 1	Day 14	Day 28
JPO172, mgEP/kg flour	0	663	482	333
	400	727	634	492
Sensea Wrap, ppm	2000	868	616	498
	0	663	482	333
	200	649	543	420
	400	689	542	445

TABLE 90

Extensibility in mm of tortilla				
Treatment	Dose	Day 1	Day 14	Day 28
JPO172, mgEP/kg flour	0	19	7	5
	400	16	12	9
	2000	19	15	14
Sensea ®	0	19	7	5
Wrap, ppm	200	18	13	9
	400	21	14	13

Example 24. Freshness Effect of JPO172 in Brioche

[0259] Bread was baked in a straight dough process with a recipe according to Table 91. Eight different treatments were done according to table 92. The ingredients were mixed in a spiral mixer into a dough for 4+8 min at 17 respectively 35 rpm. The doughs were divided into 420 g pieces, rounded, sheeted and place in baking tins. The doughs were proofed for 2.5 hours at 30° C. and 75% rH. The bread was baked for 34 minutes at 175° C.

TABLE 91

Recipe.	
Recipe:	%
Water	33
Flour	97
Vital gluten	3
Salt (0.3% less due to salted butter)	2
Whole Egg (without preservative)	20
Dried yeast (sugar tolerant - GOLD)	2
Sugar	25
Butter (Salted Lurpak ®)	20
Ascorbic-Acid (ppm)	0.006

TABLE 92

Treatments			
	Fungamyl 4000 SG, ppm (flour)	Panzea BG, ppm (flour)	JPO172 mg EP/kg flour
1	6	20	
2	6	20	50
3	6	20	75
4	6	20	100
5	6	20	150
6	6	20	200

[0260] After baking the bread was allowed to cool down for 2 hours and placed in sealed plastic bags. The bread was stored at room temperature until analysis.

[0261] The texture of each bread was evaluated with a texture analyzer (TA-XT plus, Stable microsystems, Godalmine, UK). Bread crumb texture properties were characterized by firmness (the same as “hardness” and the opposite of “softness”) and the elasticity of the baked product. A standard method for measuring firmness and elasticity is based on force-deformation of the baked product. A force-deformation of the baked products may be performed with a 34 mm diameter cylindrical probe. The force on the cylindrical probe is recorded as it is pressed down 28% strain a 25 mm thick bread slice at a deformation speed of 1 mm/second. The probe is then kept in this position for 30 seconds while the force is recorded and then probe returns to its original position.

[0262] Firmness (in grams) is defined as the force needed to compress a probe to a 25% strain (corresponding to 6.25 mm compression into a bread crumb slice of 25 mm thickness).

[0263] Elasticity (in %) is defined as the force recoded after 30 seconds compression at 28% strain (corresponding to force at time=40 s for a bread slice of 25 mm thickness) divided by the force needed to press the probe 10 mm into the crumb (corresponding to force at time=10 s for a bread slice of 25 mm thickness) times 100.

[0264] The results from the texture analysis can be found in Table 93 (firmness) and Table 94 (elasticity).

[0265] Fresh bread without any treatment (Control) have a low firmness and high elasticity after baking, as the bread is stored the bread becomes more firm and loses elasticity. Bread with JPO172 was less firm and had a higher elasticity after baking compared to the control. As the bread with JPO172 was stored the firmness and elasticity changed only slightly, resulting in a Brioche on day 60 with JPO172 having similar Firmness and better Elasticity as a control on day 1.

TABLE 93

Effect on various treatments on Firmness.				
Treatment	Day 1	Day 21	Day 39	Day 60
Control	244	918	1111	1151
50 mgEP JPO172/Kg flour	213	407	502	566
75 mgEP JPO172/Kg flour	157	324	307	414
100 mgEP JPO172/Kg flour	126	306	269	348
150 mgEP JPO172/Kg flour	144	300	316	299
200 mgEP JPO172/Kg flour	159	321	317	334

TABLE 94

Effect on various treatments on Elasticity.				
Treatment	Day 1	Day 21	Day 39	Day 60
Control	52.3	42.2	40.4	40.9
50 mgEP JPO172\Kg flour	58.0	51.9	51.0	50.0
75 mgEP JPO172\Kg flour	59.9	55.5	54.9	53.1
100 mgEP JPO172\Kg flour	60.0	56.7	55.5	54.8
150 mgEP JPO172\Kg flour	60.3	57.2	56.5	55.9
200 mgEP JPO172\Kg flour	60.5	56.9	56.6	55.9

Example 25. JPO124 and JPO172 in Lebanese Double Layer Flat Bread

[0266] Lebanese double layer flat bread was baked in a straight dough process with ingredients according to table 95. Seven different treatments were done according to table 96. The ingredients were mixed in a spiral mixer into a dough for 2.5 minutes at 35 rpm. The dough were proofed for 40 minutes at 32° C. and 82% rH. The dough was rolled out rolled out to a thickness of 2 mm, and a 20 cm circular dough piece was cut out from the sheet. The circular dough pieces were proofed at room temperature for 20 minutes. The dough was placed in an oven at 750° C. and baked for 9 seconds.

TABLE 95

Recipe.	
Ingredient	Amount, %
Flour	100.0
Water	51
Dry instant Yeast	0.7
Sucrose	4.0
Salt	0.4
Calcium propionate	0.20

TABLE 96

Treatments.		
JPO 124, mg EP/kg flour	JPO 172, mg EP/kg flour	Sample name
1		control
2	400	400 mgEP/kg JPO172
3	2000	2000 mgEP/kg JPO172
4	4000	4000 mgEP/kg JPO172
5	400	400 mgEP/kg JPO124
6	4000	4000 mgEP/kg JPO124

[0267] The flat breads were allowed to cool down for 30 minutes after baking and then placed in a sealed plastic bag that was stored at room temperature until analysis.

[0268] The texture properties of Lebanese flat bread were evaluated with a texture analyzer (Stable Microsystems, Godalming, UK) on day 3 using the Tortilla/Pastry Burst Rig (HDP/TPB). In the test procedure, the sample is held between two plates and the 4 mm spherical probe is driven through the center. The force and distance to extend the sample are measured and used as an indication of ‘deformation resistance’ and ‘extensibility’, respectively.

[0269] Sensory evaluation was performed on day 3. A training session was held prior to evaluation, identifying the relevant attributes and procedures (Table ZZ). Texture was evaluated by hand. 4-5 trained assessors participated in the evaluation. Each assessor was served 2 slices without crust of each bread type. Samples were served blind, 3-digit coded, and in random order. The intensities of the sensory attributes were evaluated on a 1-9 point intensity scale ranging from little to very intense. Two sensory replicates were performed on each evaluation day.

TABLE 97

Sensory evaluations.		
Attribute	Procedure	Evaluate
Soft	Gently compress the bread	Ease of compressing the flat bread
Flexible	Place the flat bread on the tip of your fingers	How much the flat bread bends down
Foldable	Roll the flat bread around a 20 mm round stick	Degree of cracking
Elastic	Grab the flat bread at the edges and stretch the flat bread	Resistance to extension
Layer separation	Open the flat bread at an edge and separate the layers of the flat bread	Ease of layer separation

[0270] The results for the sensory evaluation can be found in table 98 and the results from the texture evaluation can be found in table 99. The bread without any enzymes added (control) was scored low (2-3) on all sensory parameters. The flat bread with JPO172 and JPO124 scored higher on all parameters, the higher the dosage the higher the score. The improvement detected in the sensory evaluation was also seen in the texture analysis, where the bread with JPO124 or JPO172 had higher extensibility compared to flat bread without any enzyme (Control).

TABLE 98

Sensory evaluation of Lebanese flat bread on day 3.					
	Soft	Flex-ible	Foldable	Elastic	Layer separation
Control	2.8	2.6	2.0	2.2	2.6
400 mgEP/kg JPO172	3.3	2.4	3.5	2.9	4.9
2000 mgEP/kg JPO172	6.6	4.2	6.9	5.4	6.8
4000 mgEP/kg JPO172	6.6	4.0	6.9	4.9	6.7
400 mgEP/kg JPO124	4.5	3.7	3.5	3.3	5.9
4000 mgEP/kg JPO124	7.1	5.5	7.1	5.3	6.9

TABLE 99

Texture evaluation of Lebanese flat bread on day 3.		
	Toughness	Extensibility
Control	271	3.8
400 mgEP/kg JPO172	266	5.1
2000 mgEP/kg JPO172	204	5.0
4000 mgEP/kg JPO172	217	5.5
400 mgEP/kg JPO124	323	4.9
4000 mgEP/kg JPO124	268	6.4

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Phe	Ser	Gly	Ala	Trp	Gly	Arg	Pro	Gln	Arg	Asp	Gly	Pro	Ala	Leu	Arg	130	135	140
Ala	Thr	Ala	Met	Ile	Thr	Tyr	Ala	Asn	Tyr	Leu	Ile	Ser	His	Gly	Gln	145	150	155
Lys	Ser	Asp	Val	Ser	Gln	Val	Met	Trp	Pro	Ile	Ile	Ala	Asn	Asp	Leu	165	170	175
Ala	Tyr	Val	Gly	Gln	Tyr	Trp	Asn	Asn	Thr	Gly	Phe	Asp	Leu	Trp	Glu	180	185	190
Glu	Val	Asp	Gly	Ser	Ser	Phe	Phe	Thr	Ile	Ala	Val	Gln	His	Arg	Ala	195	200	205
Leu	Val	Glu	Gly	Ser	Gln	Leu	Ala	Lys	Lys	Leu	Gly	Lys	Ser	Cys	Asp	210	215	220
Ala	Cys	Asp	Ser	Gln	Pro	Pro	Gln	Ile	Leu	Cys	Phe	Leu	Gln	Ser	Phe	225	230	235
Trp	Asn	Gly	Lys	Tyr	Ile	Thr	Ser	Asn	Ile	Asn	Thr	Gln	Ala	Ser	Arg	245	250	255
Ser	Gly	Ile	Asp	Leu	Asp	Ser	Val	Leu	Gly	Ser	Ile	His	Thr	Phe	Asp	260	265	270
Pro	Glu	Ala	Ala	Cys	Asp	Asp	Ala	Thr	Phe	Gln	Pro	Cys	Ser	Ala	Arg	275	280	285
Ala	Leu	Ala	Asn	His	Lys	Val	Tyr	Val	Asp	Ser	Phe	Arg	Ser	Ile	Tyr	290	295	300
Lys	Ile	Asn	Ala	Gly	Leu	Ala	Glu	Gly	Ser	Ala	Ala	Asn	Val	Gly	Arg	305	310	315
Tyr	Pro	Glu	Asp	Val	Tyr	Gln	Gly	Gly	Asn	Pro	Trp	Tyr	Leu	Ala	Thr	325	330	335
Leu	Gly	Ala	Ser	Glu	Leu	Leu	Tyr	Asp	Ala	Leu	Tyr	Gln	Trp	Asp	Arg	340	345	350
Leu	Gly	Lys	Leu	Glu	Val	Ser	Glu	Thr	Ser	Leu	Ser	Phe	Phe	Lys	Asp	355	360	365
Phe	Asp	Ala	Thr	Val	Lys	Ile	Gly	Ser	Tyr	Ser	Arg	Asn	Ser	Lys	Thr	370	375	380
Tyr	Lys	Lys	Leu	Thr	Gln	Ser	Ile	Lys	Ser	Tyr	Ala	Asp	Gly	Phe	Ile	385	390	395
Gln	Leu	Val	Gln	Gln	Tyr	Thr	Pro	Ser	Asn	Gly	Ser	Leu	Ala	Glu	Gln	405	410	415
Tyr	Asp	Arg	Asn	Thr	Ala	Ala	Pro	Leu	Ser	Ala	Asn	Asp	Leu	Thr	Trp	420	425	430
Ser	Phe	Ala	Ser	Phe	Leu	Thr	Ala	Thr	Gln	Arg	Arg	Asp	Ala	Val	Val	435	440	445
Pro	Pro	Ser	Trp	Gly	Ala	Lys	Ser	Ala	Asn	Lys	Val	Pro	Thr	Thr	Cys	450	455	460
Ser	Ala	Ser	Pro	Val	Val	Gly	Thr	Tyr	Lys	Ala	Pro	Thr	Ala	Thr	Phe	465	470	475
Ser	Ser	Lys	Thr	Lys	Cys	Val	Pro	Ala	Lys	Asp	Ile	Val	Pro	Ile	Thr	485	490	495
Phe	Tyr	Leu	Ile	Glu	Asn	Thr	Tyr	Tyr	Gly	Glu	Asn	Val	Phe	Met	Ser			

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500	505	510
Gly Asn Ile Thr Ala Leu Gly	Asn Trp Asp Ala Lys Lys Gly Phe Pro	
515	520	525
Leu Thr Ala Asn Leu Tyr Thr	Gln Asp Gln Asn Leu Trp Phe Ala Ser	
530	535	540
Val Glu Phe Ile Pro Ala Gly	Thr Pro Phe Glu Tyr Lys Tyr Tyr Lys	
545	550	555
Val Glu Pro Asn Gly Asp Ile	Thr Trp Glu Lys Gly Pro Asn Arg Val	
565	570	575
Phe Val Ala Pro Thr Gly Cys	Pro Val Gln Pro His Ser Asn Asp Val	
580	585	590
Trp Gln Phe		
595		

<210> SEQ ID NO 3
 <211> LENGTH: 595
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: AMG anPAV498

<400> SEQUENCE: 3

Arg Asn Asp Ser Lys Gly Gly	Asn Leu Thr Phe Phe Ile His Lys Glu
1	15
Gly Glu Arg Ser Leu Gln Gly	Ile Leu Asp Asn Leu Gly Gly Arg Gly
20	30
Lys Lys Thr Pro Gly Thr Ala	Ala Gly Leu Phe Ile Ala Ser Pro Asn
35	45
Thr Glu Asn Pro Asn Tyr Tyr	Tyr Thr Trp Thr Arg Asp Ser Ala Leu
50	60
Ala Ala Lys Cys Leu Ile Asp	Leu Phe Glu Asp Ser Arg Ala Val Phe
65	80
Pro Ile Asp Arg Lys Tyr Leu	Glu Thr Gly Ile Arg Asp Tyr Val Ser
85	95
Ser Gln Ala Ile Leu Gln Ser	Val Ser Asn Pro Ser Gly Thr Leu Lys
100	110
Asp Gly Ser Gly Leu Gly Glu	Pro Lys Phe Glu Ile Asp Leu Asn Pro
115	125
Phe Ser Gly Ala Trp Gly Arg	Pro Gln Arg Asp Gly Pro Ala Leu Arg
130	140
Ala Thr Ala Met Ile Thr Tyr	Ala Asn Tyr Leu Ile Ser His Gly Gln
145	160
Lys Ser Asp Val Ser Gln Val	Met Trp Pro Ile Ile Ala Asn Asp Leu
165	175
Ala Tyr Val Gly Gln Tyr Trp	Asn Asn Thr Gly Phe Asp Leu Trp Glu
180	190
Glu Val Asp Gly Ser Ser Phe	Phe Thr Ile Ala Val Gln His Arg Ala
195	205
Leu Val Glu Gly Ser Gln Leu	Ala Lys Lys Leu Gly Lys Ser Cys Asp
210	220
Ala Cys Asp Ser Gln Pro Pro	Gln Ile Leu Cys Phe Leu Gln Ser Phe
225	240
Trp Asn Gly Lys Tyr Ile Thr	Ser Asn Ile Asn Thr Gln Ala Ser Arg

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245				250				255							
Ser	Gly	Ile	Asp	Leu	Asp	Ser	Val	Leu	Gly	Ser	Ile	His	Thr	Phe	Asp
			260												270
Pro	Glu	Ala	Ala	Cys	Asp	Asp	Ala	Thr	Phe	Gln	Pro	Cys	Ser	Ala	Arg
			275												285
Ala	Leu	Ala	Asn	His	Lys	Val	Tyr	Val	Asp	Ser	Phe	Arg	Ser	Ile	Tyr
			290												300
Lys	Ile	Asn	Ala	Gly	Leu	Ala	Glu	Gly	Ser	Ala	Ala	Asn	Val	Gly	Arg
			305												320
Tyr	Pro	Glu	Asp	Val	Tyr	Phe	Gly	Gly	Asn	Pro	Trp	Tyr	Leu	Ala	Thr
			325												335
Leu	Gly	Ala	Ser	Glu	Leu	Leu	Tyr	Asp	Ala	Leu	Tyr	Gln	Trp	Asp	Arg
			340												350
Leu	Gly	Lys	Leu	Glu	Val	Ser	Glu	Thr	Ser	Leu	Ser	Phe	Phe	Lys	Asp
			355												365
Phe	Asp	Ala	Thr	Val	Lys	Ile	Gly	Ser	Tyr	Ser	Arg	Asn	Ser	Lys	Thr
			370												380
Tyr	Lys	Lys	Leu	Thr	Gln	Ser	Ile	Lys	Ser	Tyr	Ala	Asp	Gly	Phe	Ile
			385												400
Gln	Leu	Val	Gln	Gln	Tyr	Thr	Pro	Ser	Asn	Gly	Ser	Leu	Ala	Glu	Gln
			405												415
Tyr	Asp	Arg	Asn	Thr	Ala	Ala	Pro	Leu	Ser	Ala	Asn	Asp	Leu	Thr	Trp
			420												430
Ser	Phe	Ala	Ser	Phe	Leu	Thr	Ala	Thr	Gln	Arg	Arg	Asp	Ala	Val	Val
			435												445
Pro	Pro	Ser	Trp	Gly	Ala	Lys	Ser	Ala	Asn	Lys	Val	Pro	Thr	Thr	Cys
			450												460
Ser	Ala	Ser	Pro	Val	Val	Gly	Thr	Tyr	Lys	Ala	Pro	Thr	Ala	Thr	Phe
			465												480
Ser	Ser	Lys	Thr	Lys	Cys	Val	Pro	Ala	Lys	Asp	Ile	Val	Pro	Ile	Thr
			485												495
Phe	Tyr	Leu	Ile	Glu	Asn	Thr	Tyr	Tyr	Gly	Glu	Asn	Val	Phe	Met	Ser
			500												510
Gly	Asn	Ile	Thr	Ala	Leu	Gly	Asn	Trp	Asp	Ala	Lys	Lys	Gly	Phe	Pro
			515												525
Leu	Thr	Ala	Asn	Leu	Tyr	Thr	Gln	Asp	Gln	Asn	Leu	Trp	Phe	Ala	Ser
			530												540
Val	Glu	Phe	Ile	Pro	Ala	Gly	Thr	Pro	Phe	Glu	Tyr	Lys	Tyr	Tyr	Lys
			545												560
Val	Glu	Pro	Asn	Gly	Asp	Ile	Thr	Trp	Glu	Lys	Gly	Pro	Asn	Arg	Val
			565												575
Phe	Val	Ala	Pro	Thr	Gly	Cys	Pro	Val	Gln	Pro	His	Ser	Asn	Asp	Val
			580												590
Trp	Gln	Phe													
			595												

<210> SEQ ID NO 4

<211> LENGTH: 595

<212> TYPE: PRT

<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: AMG JPO-001

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<400> SEQUENCE: 4

Ala	Asn	Asp	Ser	Lys	Gly	Gly	Asn	Leu	Thr	Phe	Phe	Ile	His	Lys	Glu
1				5					10					15	
Gly	Glu	Arg	Ser	Leu	Gln	Gly	Ile	Leu	Asp	Asn	Leu	Gly	Gly	Arg	Gly
			20					25					30		
Lys	Lys	Thr	Pro	Gly	Thr	Ala	Ala	Gly	Leu	Phe	Ile	Ala	Ser	Pro	Asn
		35					40					45			
Thr	Glu	Asn	Pro	Asn	Tyr	Tyr	Tyr	Thr	Trp	Thr	Arg	Asp	Ser	Ala	Leu
	50					55					60				
Ala	Ala	Lys	Cys	Leu	Ile	Asp	Leu	Phe	Glu	Asp	Ser	Arg	Ala	Val	Phe
65				70					75						80
Pro	Ile	Asp	Arg	Lys	Tyr	Leu	Glu	Thr	Gly	Ile	Arg	Asp	Tyr	Val	Ser
				85					90					95	
Ser	Gln	Ala	Ile	Leu	Gln	Ser	Val	Ser	Asn	Pro	Ser	Gly	Thr	Leu	Lys
			100						105				110		
Asp	Gly	Ser	Gly	Leu	Gly	Glu	Pro	Lys	Phe	Glu	Ile	Asp	Leu	Asn	Pro
		115					120					125			
Phe	Ser	Gly	Ala	Trp	Gly	Arg	Pro	Gln	Arg	Asp	Gly	Pro	Ala	Leu	Arg
	130					135					140				
Ala	Thr	Ala	Met	Ile	Thr	Tyr	Ala	Asn	Tyr	Leu	Ile	Ser	His	Gly	Gln
145					150					155					160
Lys	Ser	Asp	Val	Ser	Gln	Val	Met	Trp	Pro	Ile	Ile	Ala	Asn	Asp	Leu
			165						170					175	
Ala	Tyr	Val	Gly	Gln	Tyr	Trp	Asn	Asn	Thr	Gly	Phe	Asp	Leu	Trp	Glu
			180					185					190		
Glu	Val	Asp	Gly	Ser	Ser	Phe	Phe	Thr	Ile	Ala	Val	Gln	His	Arg	Ala
		195					200					205			
Leu	Val	Glu	Gly	Ser	Gln	Leu	Ala	Lys	Lys	Leu	Gly	Lys	Ser	Cys	Asp
	210					215					220				
Ala	Cys	Asp	Ser	Gln	Pro	Pro	Gln	Ile	Leu	Cys	Phe	Leu	Gln	Ser	Phe
225				230						235					240
Trp	Asn	Gly	Lys	Tyr	Ile	Thr	Ser	Asn	Ile	Asn	Thr	Gln	Ala	Ser	Arg
			245						250					255	
Ser	Gly	Ile	Asp	Leu	Asp	Ser	Val	Leu	Gly	Ser	Ile	His	Thr	Phe	Asp
		260					265						270		
Pro	Glu	Ala	Ala	Cys	Asp	Asp	Ala	Thr	Phe	Gln	Pro	Cys	Ser	Ala	Arg
		275					280					285			
Ala	Leu	Ala	Asn	His	Lys	Val	Tyr	Val	Asp	Ser	Phe	Arg	Ser	Ile	Tyr
	290					295					300				
Lys	Ile	Asn	Ala	Gly	Leu	Ala	Glu	Gly	Ser	Ala	Ala	Asn	Val	Gly	Arg
305				310						315				320	
Tyr	Pro	Glu	Asp	Val	Tyr	Phe	Gly	Gly	Asn	Pro	Trp	Tyr	Leu	Ala	Thr
			325						330					335	
Leu	Gly	Ala	Ser	Glu	Leu	Leu	Tyr	Asp	Ala	Leu	Tyr	Gln	Trp	Asp	Arg
		340						345					350		
Leu	Gly	Lys	Leu	Glu	Val	Ser	Glu	Thr	Ser	Leu	Ser	Phe	Phe	Lys	Asp
		355					360					365			
Phe	Asp	Ala	Thr	Val	Lys	Ile	Gly	Ser	Tyr	Ser	Arg	Asn	Ser	Lys	Thr
	370					375					380				
Tyr	Lys	Lys	Leu	Thr	Gln	Ser	Ile	Lys	Ser	Tyr	Ala	Asp	Gly	Phe	Ile
385					390					395					400

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<210> SEQ ID NO 5
<211> LENGTH: 595
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: AMG JPO-124

<400> SEQUENCE: 5

Ala Asn Asp Ser Lys Ser Thr Asn Leu Thr Phe Phe Ile His Lys Glu
1          5          10          15
Gly Glu Arg Ser Leu Gln Gly Ile Leu Asp Asn Leu Gly Gly Phe Gly
          20          25          30
Lys Tyr Thr Pro Gly Thr Ala Ala Gly Leu Phe Ile Ala Ser Pro Asn
          35          40          45
Thr Glu Asn Pro Asn Tyr Tyr Tyr Thr Trp Thr Arg Asp Ser Ala Leu
          50          55          60
Ala Ala Lys Cys Leu Ile Asp Leu Phe Glu Asp Ser Arg Ala Val Phe
65          70          75          80
Pro Ile Asp Arg Lys Tyr Leu Glu Thr Gly Ile Arg Asp Tyr Val Ser
          85          90          95
Ser Gln Ala Ile Leu Gln Asn Val Ser Asn Pro Ser Gly Thr Leu Lys
          100          105          110
Asp Gly Ser Gly Leu Gly Glu Pro Lys Phe Glu Ile Asp Leu Asn Pro
          115          120          125
Phe Ser Gly Pro Trp Gly Arg Pro Gln Arg Asp Gly Pro Ala Leu Arg
          130          135          140

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Ala	Thr	Ala	Met	Ile	Thr	Tyr	Ala	Asn	Tyr	Leu	Ile	Ser	His	Gly	Gln
145					150					155					160
Lys	Ser	Asp	Val	Ser	Gln	Val	Met	Trp	Pro	Ile	Ile	Ala	Asn	Asp	Leu
				165					170					175	
Ala	Tyr	Val	Gly	Gln	Tyr	Trp	Asn	Asn	Thr	Gly	Phe	Asp	Leu	Trp	Glu
			180					185					190		
Glu	Val	Asp	Gly	Ser	Ser	Phe	Phe	Thr	Ile	Ala	Val	Gln	His	Arg	Ala
		195					200					205			
Leu	Val	Glu	Gly	Ser	Gln	Leu	Ala	Lys	Lys	Leu	Gly	Lys	Ser	Cys	Asp
		210				215					220				
Ala	Cys	Asp	Ser	Gln	Pro	Pro	Gln	Ile	Leu	Cys	Phe	Leu	Gln	Ser	Phe
225					230					235					240
Trp	Asn	Gly	Lys	Tyr	Ile	Thr	Ser	Asn	Ile	Asn	Thr	Gln	Ala	Ser	Arg
				245					250					255	
Ser	Gly	Ile	Asp	Leu	Asp	Ser	Val	Leu	Gly	Ser	Ile	His	Thr	Phe	Asp
			260					265					270		
Pro	Glu	Ala	Ala	Cys	Asp	Asp	Ala	Thr	Phe	Gln	Pro	Cys	Ser	Ala	Arg
		275					280					285			
Ala	Leu	Ala	Asn	His	Lys	Val	Tyr	Val	Asp	Ser	Phe	Arg	Ser	Ile	Tyr
		290				295					300				
Lys	Ile	Asn	Ala	Gly	Leu	Ala	Glu	Gly	Ser	Ala	Ala	Asn	Val	Gly	Arg
305					310					315					320
Tyr	Pro	Glu	Asp	Val	Tyr	Phe	Gly	Gly	Asn	Pro	Trp	Tyr	Leu	Ala	Thr
				325					330					335	
Leu	Gly	Ala	Ser	Glu	Leu	Leu	Tyr	Asp	Ala	Leu	Tyr	Gln	Trp	Asp	Arg
			340					345					350		
Leu	Gly	Lys	Leu	Glu	Val	Ser	Glu	Thr	Ser	Leu	Ser	Phe	Phe	Lys	Asp
		355					360					365			
Phe	Asp	Ala	Thr	Val	Lys	Ile	Gly	Ser	Tyr	Ser	Arg	Asn	Ser	Lys	Thr
	370					375					380				
Tyr	Lys	Lys	Leu	Thr	Gln	Ser	Ile	Lys	Ser	Tyr	Ala	Asp	Gly	Phe	Ile
385					390					395					400
Gln	Leu	Val	Gln	Gln	Tyr	Thr	Pro	Ser	Asn	Gly	Ser	Leu	Ala	Glu	Gln
				405					410					415	
Tyr	Asp	Arg	Asn	Thr	Ala	Ala	Pro	Leu	Ser	Ala	Asn	Asp	Leu	Thr	Trp
			420					425					430		
Ser	Phe	Ala	Ser	Phe	Leu	Thr	Ala	Thr	Gln	Arg	Arg	Asn	Ala	Ser	Val
		435					440					445			
Pro	Pro	Ser	Trp	Gly	Ala	Lys	Ser	Ala	Asn	Lys	Val	Pro	Thr	Thr	Cys
		450				455					460				
Ser	Ala	Ser	Pro	Val	Val	Gly	Thr	Tyr	Lys	Ala	Pro	Thr	Ala	Thr	Phe
465				470						475					480
Pro	Ser	Lys	Thr	Lys	Cys	Val	Pro	Ala	Lys	Asp	Ile	Val	Pro	Ile	Thr
				485					490					495	
P															

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Val	Glu	Phe	Ile	Pro	Ala	Gly	Thr	Pro	Phe	Glu	Tyr	Lys	Tyr	Tyr	Lys
545					550					555					560
Val	Glu	Pro	Asn	Gly	Thr	Ile	Val	Trp	Glu	Lys	Gly	Pro	Asn	Arg	Val
			565						570					575	
Phe	Val	Ala	Pro	Thr	Gly	Cys	Pro	Val	Gln	Pro	His	Ser	Asn	Asp	Val
			580					585					590		
Trp	Arg	Ser													
		595													
<210> SEQ ID NO 6															
<211> LENGTH: 595															
<212> TYPE: PRT															
<213> ORGANISM: Artificial sequence															
<220> FEATURE:															
<223> OTHER INFORMATION: AMG JPO-172															
<400> SEQUENCE: 6															
Ala	Asn	Asp	Ser	Lys	Ser	Thr	Asn	Leu	Thr	Phe	Phe	Ile	His	Lys	Glu
1				5					10					15	
Gly	Glu	Arg	Ser	Leu	Gln	Gly	Ile	Leu	Asp	Asn	Leu	Gly	Gly	Phe	Gly
			20					25					30		
Lys	Tyr	Thr	Pro	Gly	Thr	Ala	Ala	Gly	Leu	Phe	Ile	Ala	Ser	Pro	Asn
		35					40					45			
Thr	Arg	Asn	Pro	Asn	Tyr	Tyr	Tyr	Thr	Trp	Thr	Arg	Asp	Ser	Ala	Leu
	50					55					60				
Ala	Ala	Lys	Cys	Leu	Ile	Asp	Leu	Phe	Glu	Asp	Ser	Arg	Ala	Val	Phe
65					70				75						80
Pro	Ile	Asp	Arg	Lys	Tyr	Leu	Glu	Thr	Gly	Ile	Arg	Asp	Tyr	Val	Ser
				85					90					95	
Ser	Gln	Ala	Ile	Leu	Gln	Asn	Val	Ser	Asn	Pro	Ser	Gly	Thr	Leu	Lys
		100						105					110		
Asp	Gly	Ser	Gly	Leu	Gly	Glu	Pro	Lys	Phe	Glu	Ile	Asp	Leu	Asn	Pro
		115					120					125			
Phe	Ser	Gly	Pro	Trp	Gly	Arg	Pro	Gln	Arg	Asp	Gly	Pro	Ala	Leu	Arg
	130					135					140				
Ala	Thr	Ala	Met	Ile	Thr	Tyr	Ala	Asn	Tyr	Leu	Ile	Ser	His	Gly	Gln
145					150					155					160
Lys	Ser	Asp	Val	Ser	Gln	Val	Met	Trp	Pro	Ile	Ile	Ala	Asn	Asp	Leu
			165						170					175	
Ala	Tyr	Val	Gly	Gln	Tyr	Trp	Asn	Asn	Thr	Gly	Phe	Asp	Leu	Trp	Glu
		180					185						190		
Glu	Val	Asp	Gly	Ser	Ser	Phe	Phe	Thr	Ile	Ala	Val	Gln	His	Arg	Ala
	195						200					205			
Leu	Val	Glu	Gly	Ser	Gln	Leu	Ala	Lys	Lys	Leu	Gly	Lys	Ser	Cys	Asp
	210					215					220				
Ala	Cys	Asp	Ser	Gln	Pro	Pro	Gln	Ile	Leu	Cys	Phe	Leu	Gln	Ser	Phe
225					230					235					240
Trp	Asn	Gly	Lys	Tyr	Ile	Thr	Ser	Asn	Ile	Asn	Thr	Gln	Ala	Ser	Arg
			245						250					255	
Ser	Gly	Ile	Asp	Leu	Asp	Ser	Val	Leu	Gly	Ser	Ile	His	Thr	Phe	Asp
		260						265					270		
Pro	Glu	Ala	Ala	Cys	Asp	Asp	Ala	Thr	Phe	Gln	Pro	Cys	Ser	Ala	Arg
	275						280						285		

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<210> SEQ ID NO 7
<211> LENGTH: 593
<212> TYPE: PRT
<213> ORGANISM: Penicillium miczynskii

<400> SEQUENCE: 7

Asn Pro Thr Ser Lys Asp Ala Lys Met Ala Ser Tyr Ile Ser Lys Glu
1          5          10          15

Gly Gln Arg Ser Ile Val Gly Ile Thr Glu Asn Leu Gly Gly Lys Gly
          20          25          30

Ser Lys Thr Pro Gly Thr Ala Ala Gly Leu Phe Ile Ala Ser Pro Asn
          35          40          45

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Met	Ala	Asn	Pro	Asn	Tyr	Tyr	Tyr	Thr	Trp	Thr	Arg	Asp	Ser	Ala	Leu
50						55					60				
Thr	Phe	Lys	Cys	Leu	Ile	Asp	Leu	Phe	Glu	Thr	Ser	Asp	Gln	Asp	Tyr
65					70					75					80
Ile	Ser	Arg	Lys	Glu	Leu	Glu	Thr	Asp	Ile	Arg	Asn	Tyr	Val	Ser	Ser
			85						90					95	
Gln	Ala	Val	Leu	Gln	Asn	Val	Ser	Asn	Pro	Ser	Gly	Thr	Leu	Lys	Asp
			100						105					110	
Gly	Ser	Gly	Leu	Gly	Glu	Pro	Lys	Phe	Glu	Ile	Asp	Leu	Asn	Pro	Phe
		115					120					125			
Ser	Gly	Ser	Trp	Gly	Arg	Pro	Gln	Arg	Asp	Gly	Pro	Ala	Leu	Arg	Ala
	130					135					140				
Thr	Ala	Met	Ile	Thr	Tyr	Ala	Asp	Trp	Leu	Ile	Ser	His	Gly	Ser	Lys
145					150						155				160
Ser	Glu	Ala	Ala	Asn	Ile	Met	Trp	Pro	Ile	Ile	Ala	Asn	Asp	Leu	Ala
				165					170					175	
Tyr	Val	Gly	Gln	Tyr	Trp	Asn	Asn	Thr	Gly	Phe	Asp	Leu	Trp	Glu	Glu
			180					185					190		
Val	Asp	Gly	Ser	Ser	Phe	Phe	Thr	Leu	Ala	Val	Gln	His	Arg	Ser	Leu
		195					200					205			
Val	Gln	Gly	Ala	Ser	Leu	Ala	Lys	Lys	Leu	Gly	Lys	Ser	Cys	Pro	Ala
	210					215					220				
Cys	Val	Ser	Gln	Ala	Pro	Gln	Ile	Leu	Cys	Phe	Leu	Gln	Ser	Phe	Trp
225					230					235					240
Asn	Gly	Asn	Tyr	Ile	Thr	Ala	Asn	Ile	Asn	Leu	Asp	Thr	Ser	Arg	Ser
			245						250					255	
Gly	Ile	Asp	Leu	Asn	Ser	Ile	Leu	Gly	Ser	Ile	His	Thr	Phe	Asp	Pro
		260						265					270		
Glu	Ala	Ser	Cys	Asp	Asp	Ser	Thr	Phe	Gln	Pro	Cys	Ser	Ala	Arg	Ala
		275					280					285			
Leu	Ala	Asn	His	Lys	Val	Tyr	Val	Asp	Ala	Phe	Arg	Ser	Ile	Tyr	Lys
	290					295					300				
Val	Asn	Ala	Gly	Leu	Ser	Asn	Gly	Ser	Ala	Ala	Asn	Val	Gly	Arg	Tyr
305					310					315					320
Pro	Glu	Asp	Val	Tyr	Gln	Gly	Gly	Asn	Pro	Trp	Tyr	Leu	Ala	Thr	Leu
			325					330						335	
Ala	Ser	Ala	Glu	Leu	Leu	Tyr	Asp	Ala	Leu	Tyr	Gln	Trp	Ser	Lys	Ile
			340					345					350		
Gly	Lys	Leu	Asp	Val	Thr	Lys	Thr	Ser	Leu	Ala	Phe	Phe	Lys	Asp	Phe
		355					360					365			
Asp	Met	Ala	Val	Lys	Thr	Gly	Thr	Tyr	Ser	Ala	His	Ser	Ser	Thr	Tyr
	370					375					380				
Lys	Ser	Leu	Thr	Ser	Ala	Ile	Arg	Thr	Tyr	Ala	Asp	Asp	Phe	Ile	Gly
385					390					395					400
Leu	Val	Gln	Asp	Tyr	Thr	Pro	Ser	Asn	Gly	Ser	Leu	Ala	Glu	Gln	Tyr
			405						410					415	
Asp	Arg	Asn	Thr	Gly	Ile	Pro	Leu	Ser	Ala	Asn	Asp	Leu	Thr	Trp	Ser
			420					425				430			
Tyr	Ala	Ser	Phe	Ile	Thr	Ala	Ile	Gln	Arg	Arg	Ala	Ser	Val	Val	Pro
	435						440					445			

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Ala	Ser	Trp	Gly	Glu	Lys	Ser	Ala	Asn	Val	Val	Pro	Thr	Ile	Cys	Ser
450						455					460				
Ala	Ser	Pro	Val	Thr	Gly	Thr	Tyr	Gln	Ala	Ala	Pro	Ser	Val	Phe	Pro
465					470					475					480
Thr	Thr	Thr	Gly	Cys	Val	Pro	Ala	Thr	Ser	Ile	Val	Pro	Ile	Thr	Phe
			485						490					495	
Tyr	Leu	Thr	Glu	Thr	Thr	Phe	Tyr	Gly	Glu	Asn	Val	Tyr	Met	Thr	Gly
			500					505					510		
Asn	Ile	Ser	Ala	Leu	Gly	Asn	Trp	Asp	Thr	Asn	Asn	Gly	Phe	Pro	Leu
		515					520					525			
Thr	Ala	Asn	Leu	Tyr	Thr	Asp	Ser	Asp	His	Leu	Trp	Phe	Ala	Ser	Val
	530					535					540				
Glu	Leu	Val	Pro	Ala	Gly	Thr	Pro	Phe	Glu	Tyr	Lys	Tyr	Tyr	Lys	Val
545					550					555					560
Glu	Pro	Asn	Gly	Thr	Val	Ile	Trp	Glu	Asn	Gly	Glu	Asn	Arg	Val	Tyr
			565					570						575	
Val	Ala	Pro	Thr	Gly	Cys	Pro	Ile	Gln	Pro	Ser	Gln	Thr	Asp	Val	Trp
			580					585					590		

Arg

<210> SEQ ID NO 8

<211> LENGTH: 590

<212> TYPE: PRT

<213> ORGANISM: *Penicillium russellii*

<400> SEQUENCE: 8

Ser	Lys	Asp	Gly	Asn	Leu	Ala	Ser	Tyr	Ile	Ala	Lys	Glu	Gly	Gln	Arg
1				5					10					15	
Ser	Ile	Val	Gly	Ile	Thr	Glu	Asn	Leu	Gly	Gly	Lys	Gly	Ser	Lys	Thr
		20						25					30		
Pro	Gly	Thr	Ala	Ala	Gly	Leu	Phe	Ile	Ala	Ser	Pro	Asn	Met	Ala	Asn
		35				40						45			
Pro	Asn	Tyr	Tyr	Tyr	Thr	Trp	Thr	Arg	Asp	Ser	Ala	Leu	Thr	Phe	Lys
	50					55				60					
Cys	Leu	Ile	Asp	Leu	Phe	Glu	Thr	Ser	Asp	Gln	Asp	Tyr	Ile	Ser	Arg
65				70					75					80	
Lys	Glu	Leu	Glu	Thr	Asp	Ile	Arg	Asn	Tyr	Val	Ser	Ser	Gln	Ala	Val
			85					90						95	
Leu	Gln	Asn	Val	Ser	Asn	Pro	Ser	Gly	Thr	Leu	Lys	Asp	Gly	Ser	Gly
		100						105					110		
Leu	Gly	Glu	Pro	Lys	Phe	Glu	Ile	Asp	Leu	Asn	Pro	Phe	Ser	Gly	Ser
	115					120					125				
Trp	Gly	Arg	Pro	Gln	Arg	Asp	Gly	Pro	Ala	Leu	Arg	Ala	Thr	Ala	Met
	130					135					140				
Ile	Thr	Tyr	Ala	Asp	Trp	Leu	Val	Ser	His	Gly	Gln	Lys	Ser	Glu	Ala
145				150						155				160	
Thr	Asn	Ile	Met	Trp	Pro	Ile	Ile	Ala	Asn	Asp	Leu	Ala	Tyr	Val	Gly
			165					170						175	
Gln	Tyr	Trp	Asn	Lys	Thr	Gly	Phe	Asp	Leu	Trp	Glu	Glu	Val	Asp	Gly
		180					185						190		
Ser	Ser	Phe	Tyr	Thr	Leu	Ala	Val	Gln	His	Arg	Ala	Leu	Val	Gln	Gly
	195					200						205			

-continued

Ala	Ser	Leu	Ala	Lys	Lys	Leu	Gly	Lys	Ser	Cys	Thr	Ala	Cys	Val	Ser
210						215					220				
Gln	Ala	Pro	Gln	Ile	Leu	Cys	Phe	Leu	Gln	Ser	Phe	Trp	Asn	Gly	Asn
225					230					235					240
Tyr	Ile	Thr	Ala	Asn	Ile	Asn	Leu	Asp	Thr	Ser	Arg	Ser	Gly	Ile	Asp
				245					250					255	
Leu	Asn	Ser	Ile	Leu	Gly	Ser	Ile	His	Thr	Phe	Asp	Pro	Glu	Ala	Ser
			260					265					270		
Cys	Asp	Asp	Ser	Thr	Phe	Gln	Pro	Cys	Ser	Ala	Arg	Ala	Leu	Ala	Asn
	275						280					285			
His	Lys	Val	Tyr	Val	Asp	Ala	Phe	Arg	Ser	Ile	Tyr	Gly	Val	Asn	Ala
	290					295					300				
Gly	Leu	Ser	Asn	Gly	Thr	Ala	Ala	Asn	Val	Gly	Arg	Tyr	Pro	Glu	Asp
305					310					315					320
Val	Tyr	Gln	Gly	Gly	Asn	Pro	Trp	Tyr	Leu	Ala	Thr	Leu	Ala	Ala	Ala
				325					330					335	
Glu	Leu	Leu	Tyr	Asp	Ala	Leu	Tyr	Gln	Trp	Asn	Gln	Ile	Gly	Lys	Leu
			340					345					350		
Asp	Val	Thr	Lys	Thr	Ser	Leu	Ala	Phe	Phe	Lys	Asp	Phe	Asp	Ala	Ala
		355					360					365			
Val	Lys	Thr	Gly	Thr	Tyr	Ser	Ala	His	Ser	Ser	Ala	Tyr	Arg	Thr	Leu
	370					375					380				
Thr	Ser	Ala	Ile	Arg	Thr	Tyr	Ala	Asp	Asp	Phe	Ile	Ser	Leu	Val	Gln
385					390					395					400
His	Tyr	Thr	Pro	Ser	Asn	Gly	Ser	Leu	Ala	Glu	Gln	Tyr	Asp	Arg	Asp
				405					410					415	
Thr	Gly	Ile	Pro	Leu	Ser	Ala	Asn	Asp	Leu	Thr	Trp	Ser	Tyr	Ala	Ser
			420					425					430		
Phe	Ile	Thr	Ala	Ile	Glu	Arg	Arg	Ala	Ser	Val	Val	Pro	Ala	Ser	Trp
		435					440					445			
Gly	Glu	Lys	Ser	Ala	Asn	Val	Val	Pro	Thr	Thr	Cys	Ser	Ala	Ser	Pro
	450					455					460				
Val	Thr	Gly	Thr	Tyr	Val	Ala	Ala	Thr	Ser	Val	Phe	Pro	Thr	Thr	Thr
465					470					475					480
Gly	Cys	Val	Pro	Ala	Thr	Ser	Ile	Val	Pro	Ile	Thr	Phe	Tyr	Leu	Thr
				485					490					495	
Glu	Ser	Thr	Phe	Tyr	Gly	Glu	Asn	Val	Tyr	Met	Thr	Gly	Asn	Ile	Ser
			500					505					510		
Ala	Leu	Gly	Asn	Trp	Asp	Thr	Ser	Ser	Gly	Phe	Pro	Leu	Thr	Ala	Asn
		515					520					525			
Leu	Tyr	Thr	Asp	Ser	Asp	His	Leu	Trp	Phe	Ala	Ser	Val	Glu	Leu	Val
	530					535					540				
Pro	Ala	Gly	Thr	Pro	Phe	Glu	Tyr	Lys	Tyr	Tyr	Lys	Val	Glu	Pro	Asn
545					550					555					560
Gly	Thr	Val	Ile	Trp	Glu	Asn	Gly	Glu	Asn	Arg	Val	Tyr	Val	Ala	Pro
				565					570					575	
Thr	Gly	Cys	Pro	Ile	Gln	Pro	Ser	Gln	Thr	Asp	Ile	Trp	Arg		
			580					585					590		

<210> SEQ ID NO 9

<211> LENGTH: 594

<212> TYPE: PRT

-continued

<213> ORGANISM: *Penicillium glabrum*

<400> SEQUENCE: 9

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Asn Pro Phe Asn Ser Leu Asp Ser Phe Ile Leu Lys Glu Gly Ala Arg
1           5           10           15
Ser Tyr Gln Gly Ile Ile Asp Asn Leu Gly Asn Lys Gly Val Lys Ala
20           25           30
Pro Gly Thr Ala Ala Gly Leu Phe Val Ala Ser Pro Asn Thr Ala Asn
35           40           45
Pro Asp Tyr Phe Tyr Thr Trp Thr Arg Asp Ser Ala Leu Thr Phe Lys
50           55           60
Cys Leu Ile Asp Leu Phe Asp Gly Gly Ser Thr Glu Phe Gly Leu Lys
65           70           75           80
Asn Ser Glu Leu Glu Thr Asp Ile Arg Asn Tyr Val Ser Ser Gln Ala
85           90           95
Val Leu Gln Asn Val Ser Asn Pro Ser Gly Thr Leu Glu Asp Gly Thr
100          105          110
Gly Leu Gly Glu Pro Lys Phe Glu Val Asp Leu Asn Pro Phe Thr Gly
115          120          125
Ser Trp Gly Arg Pro Gln Arg Asp Gly Pro Ala Leu Arg Ala Thr Ala
130          135          140
Leu Ile Thr Tyr Thr Asn Tyr Leu Leu Ser Gln Gly Gln Lys Ser Glu
145          150          155          160
Ala Val Asn Ile Met Trp Pro Ile Ile Ser Asn Asp Leu Ala Tyr Val
165          170          175
Gly Gln Tyr Trp Asn Asp Thr Gly Phe Asp Leu Trp Glu Glu Thr Asp
180          185          190
Gly Ser Ser Phe Phe Thr Leu Ala Val Gln His Arg Ala Leu Val Gln
195          200          205
Gly Ala Thr Leu Ala Gln Lys Leu Gly Lys Ser Cys Ala Ala Cys Ser
210          215          220
Ser Gln Ala Pro Gln Ile Leu Cys Phe Leu Gln Ser Phe Trp Asn Gly
225          230          235          240
Lys Tyr Ile Thr Ala Asn Ile Asn Leu Asp Thr Ser Arg Thr Gly Ile
245          250          255
Asp Ala Asn Thr Leu Leu Gly Ser Ile His Thr Phe Asp Pro Glu Ala
260          265          270
Ala Cys Asp Asp Ser Thr Phe Gln Pro Cys Ser Ala Arg Ala Leu Ala
275          280          285
Asn His Lys Val Tyr Val Asp Ala Phe Arg Ser Ile Tyr Lys Ile Asn
290          295          300
Ser Gly Ile Ala Glu Gly Ser Pro Ala Asn Val Gly Arg Tyr Pro Glu
305          310          315          320
Asp Val Tyr Gln Gly Gly Asn Pro Trp Tyr Leu Thr Thr Leu Ala Ser
325          330          335
Ala Glu Leu Leu Tyr Asp Ala Leu Tyr Gln Trp Asn Lys Ile Gly Gly
340          345          350
Leu Asp Val Thr Glu Thr Ser Leu Ala Phe Phe Lys Asp Phe His Ser
355          360          365
Ser Val Lys Thr Gly Ser Tyr Ser Ala His Ser Gln Thr Tyr Lys Thr
370          375          380

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Leu Thr Ser Ala Ile Arg Thr Tyr Ala Asp Gly Phe Val Gly Leu Val
385                      390                      395                      400

Gln Lys Tyr Thr Pro Ala Asn Gly Ser Leu Ala Glu Gln Tyr Asn Arg
                      405                      410                      415

Asn Thr Ser Val Pro Leu Ser Ala Asn Asp Leu Thr Trp Ser Phe Ala
                      420                      425                      430

Ser Phe Leu Thr Ala Ile Gln Arg Arg Glu Ser Ile Val Pro Gly Ser
                      435                      440                      445

Trp Gly Glu Lys Ser Ala Asn Thr Val Pro Thr Thr Cys Ser Ala Ser
                      450                      455                      460

Pro Val Thr Gly Thr Tyr Lys Ala Ala Thr Ser Thr Phe Pro Thr Ser
465                      470                      475                      480

Thr Ala Gly Cys Val Pro Ala Thr Asp Val Val Pro Ile Thr Phe Tyr
                      485                      490                      495

Leu Ile Glu Asn Thr Tyr Tyr Gly Glu Asn Val Tyr Met Thr Gly Asn
                    500                      505                      510

Ile Ser Ala Leu Gly Asn Trp Asp Thr Ser Asp Gly Leu Ala Leu Asp
                    515                      520                      525

Ala Gly Leu Tyr Thr Glu Thr Asp Asn Leu Trp Phe Gly Thr Leu Glu
                    530                      535                      540

Leu Val Thr Ala Gly Thr Pro Phe Glu Tyr Lys Tyr Tyr Lys Ile Glu
545                      550                      555                      560

Pro Asn Gly Thr Val Thr Ile Trp Glu Ser Gly Asp Asn Arg Val Ser
                    565                      570                      575

Val Val Pro Thr Gly Cys Pro Ile Gln Pro Ser Leu His Asp Val Trp
                    580                      585                      590

Arg Ser

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<210> SEQ ID NO 10
<211> LENGTH: 591
<212> TYPE: PRT
<213> ORGANISM: Rasamsonia emersonii

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<400> SEQUENCE: 10

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Ala Thr Gly Ser Leu Asp Ser Phe Leu Ala Thr Glu Thr Pro Ile Ala
1                      5                      10                      15

Leu Gln Gly Val Leu Asn Asn Ile Gly Pro Asn Gly Ala Asp Val Ala
                20                      25                      30

Gly Ala Ser Ala Gly Ile Val Val Ala Ser Pro Ser Arg Ser Asp Pro
                35                      40                      45

Asn Tyr Phe Tyr Ser Trp Thr Arg Asp Ala Ala Leu Thr Ala Lys Tyr
                    50                      55                      60

Leu Val Asp Ala Phe Ile Ala Gly Asn Lys Asp Leu Glu Gln Thr Ile
65                      70                      75                      80

Gln Gln Tyr Ile Ser Ala Gln Ala Lys Val Gln Thr Ile Ser Asn Pro
                85                      90                      95

Ser Gly Asp Leu Ser Thr Gly Gly Leu Gly Glu Pro Lys Phe Asn Val
                100                      105                      110

Asn Glu Thr Ala Phe Thr Gly Pro Trp Gly Arg Pro Gln Arg Asp Gly
                115                      120                      125

Pro Ala Leu Arg Ala Thr Ala Leu Ile Ala Tyr Ala Asn Tyr Leu Ile
130                      135                      140

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Asp	Asn	Gly	Glu	Ala	Ser	Thr	Ala	Asp	Glu	Ile	Ile	Trp	Pro	Ile	Val
145					150					155					160
Gln	Asn	Asp	Leu	Ser	Tyr	Ile	Thr	Gln	Tyr	Trp	Asn	Ser	Ser	Thr	Phe
			165						170					175	
Asp	Leu	Trp	Glu	Glu	Val	Glu	Gly	Ser	Ser	Phe	Phe	Thr	Thr	Ala	Val
			180					185					190		
Gln	His	Arg	Ala	Leu	Val	Glu	Gly	Asn	Ala	Leu	Ala	Thr	Arg	Leu	Asn
		195					200					205			
His	Thr	Cys	Ser	Asn	Cys	Val	Ser	Gln	Ala	Pro	Gln	Val	Leu	Cys	Phe
	210					215					220				
Leu	Gln	Ser	Tyr	Trp	Thr	Gly	Ser	Tyr	Val	Leu	Ala	Asn	Phe	Gly	Gly
225					230					235					240
Ser	Gly	Arg	Ser	Gly	Lys	Asp	Val	Asn	Ser	Ile	Leu	Gly	Ser	Ile	His
				245					250					255	
Thr	Phe	Asp	Pro	Ala	Gly	Gly	Cys	Asp	Asp	Ser	Thr	Phe	Gln	Pro	Cys
			260					265					270		
Ser	Ala	Arg	Ala	Leu	Ala	Asn	His	Lys	Val	Val	Thr	Asp	Ser	Phe	Arg
		275					280					285			
Ser	Ile	Tyr	Ala	Ile	Asn	Ser	Gly	Ile	Ala	Glu	Gly	Ser	Ala	Val	Ala
	290					295					300				
Val	Gly	Arg	Tyr	Pro	Glu	Asp	Val	Tyr	Gln	Gly	Gly	Asn	Pro	Trp	Tyr
305					310					315					320
Leu	Ala	Thr	Ala	Ala	Ala	Glu	Gln	Leu	Tyr	Asp	Ala	Ile	Tyr	Gln	
				325				330					335		
Trp	Lys	Lys	Ile	Gly	Ser	Ile	Ser	Ile	Thr	Asp	Val	Ser	Leu	Pro	Phe
			340					345					350		
Phe	Gln	Asp	Ile	Tyr	Pro	Ser	Ala	Ala	Val	Gly	Thr	Tyr	Asn	Ser	Gly
		355					360					365			
Ser	Thr	Thr	Phe	Asn	Asp	Ile	Ile	Ser	Ala	Val	Gln	Thr	Tyr	Gly	Asp
	370					375					380				
Gly	Tyr	Leu	Ser	Ile	Val	Glu	Lys	Tyr	Thr	Pro	Ser	Asp	Gly	Ser	Leu
385					390					395					400
Thr	Glu	Gln	Phe	Ser	Arg	Thr	Asp	Gly	Thr	Pro	Leu	Ser	Ala	Ser	Ala
				405				410						415	
Leu	Thr	Trp	Ser	Tyr	Ala	Ser	Leu	Leu	Thr	Ala	Ser	Ala	Arg	Arg	Gln
			420					425					430		
Ser	Val	Val	Pro	Ala	Ser	Trp	Gly	Glu	Ser	Ser	Ala	Ser	Ser	Val	Pro
		435					440					445			
Ala	Val	Cys	Ser	Ala	Thr	Ser	Ala	Thr	Gly	Pro	Tyr	Ser	Thr	Ala	Thr
	450					455					460				
Asn	Thr	Val	Trp	Pro	Ser	Ser	Gly	Ser	Gly	Ser	Ser	Thr	Thr	Thr	Ser
465					470					475					480
Ser	Ala	Pro	Cys	Thr	Thr	Pro	Thr	Ser	Val	Ala	Val	Thr	Phe	Asp	Glu
			485					490						495	
Ile	Val	Ser	Thr	Ser	Tyr	Gly	Glu	Thr	Ile	Tyr	Leu	Ala	Gly	Ser	Ile
			500					505					510		
Pro	Glu	Leu	Gly	Asn	Trp	Ser	Thr	Ala	Ser	Ala	Ile	Pro	Leu	Arg	Ala
		515					520					525			
Asp	Ala	Tyr	Thr	Asn	Ser	Asn	Pro	Leu	Trp	Tyr	Val	Thr	Val	Asn	Leu
	530					535					540				
Pro	Pro	Gly	Thr	Ser	Phe	Glu	Tyr	Lys	Phe	Phe	Lys	Asn	Gln	Thr	Asp

-continued

545	550	555	560
Gly Thr Ile Val Trp Glu Asp Asp Pro Asn Arg Ser Tyr Thr Val Pro			
	565	570	575
Ala Tyr Cys Gly Gln Thr Thr Ala Ile Leu Asp Asp Ser Trp Gln			
	580	585	590

1. A method of producing a baked or par-baked product, said method comprising:

- a) providing a dough comprising a mature thermostable variant of a parent glucoamylase at least 70% identical to SEQ ID NO:1, SEQ ID NO:6, SEQ ID NO:7 or SEQ ID NO:8; and
- b) baking or par-baking the dough to produce a baked or par-baked product.

2. The method according to claim 1, wherein the baked or par-baked product is a type of bread, preferably a pan bread, toast bread, open bread, buns, Fino bread, Hammam bread, Samoli bread, baguettes, brioche, hamburger buns, rolls, brown bread, whole meal bread, rich bread, bran bread, flat bread, tortilla or biscuit, cake, or a patisserie.

3. The method according to claim 1, wherein the parent glucoamylase is from a species of *Penicillium*, preferably from *Penicillium oxycalum*, *Penicillium miczynskii*, *Penicillium russellii* or *Penicillium glabrum*.

4. The method according to claim 1, wherein the mature variant comprises at least one amino acid modification in one or more or all of the positions corresponding to positions 1, 2, 4, 6, 7, 11, 31, 34, 65, 79, 103, 132, 327, 445, 447, 481, 566, 568, 594 and 595 in SEQ ID NO:1.

5. The method according to claim 4, wherein the at least one amino acid modification comprises a substitution in one or more or all of the positions corresponding to positions 1, 2, 4, 11, 65, 79 and 327 in SEQ ID NO:1.

6. The method according to claim 4, wherein the at least one amino acid modification comprises a substitution in one or more or all of the positions corresponding to positions 1, 6, 7, 31, 34, 79, 103, 132, 445, 447, 481, 566, 568, 594 and 595 in SEQ ID NO:1.

7. The method according to claim 1, wherein the at least one amino acid modification comprises a substitution in one or more or all of the positions corresponding to positions 1, 6, 7, 31, 34, 50, 79, 103, 132, 445, 447, 481, 484, 501, 539, 566, 568, 594 and 595 in SEQ ID NO:1.

8. The method according to claim 1, wherein the mature thermostable variant has a thermostability improvement (Td) over its parent of at least 3° C.

9. The method according to claim 1, wherein the mature thermostable variant has a relative activity at 91° C. of at least 150 compared to its parent.

10. The method according to claim 1, wherein the baked or par-baked product after final bake-off has a reduced initial firmness and/or an increased initial elasticity, and/or a reduced increase in firmness and/or a higher elasticity after 1, 7 or 14 days, when cooled to room temperature, packed in a sealed container and stored at room temperature until analysis, compared to a control made without any added glucoamylase.

11. The method according to claim 1, wherein the baked or par-baked product after final bake-off has at least the same sweetness as a control product made with double the amount

of the mature glucoamylase the amino acid sequence of which is shown in SEQ ID NO:10.

12. The method according to claim 1, wherein the mature thermostable variant glucoamylase enzyme is comprised in the dough in an amount of 0.01-1,000 mg enzyme protein (mgEP) per kg flour.

13. The method according to claim 1, wherein the dough also comprises one or more additional enzyme selected from the group consisting of a alpha-amylase, maltogenic amylase, raw-starch degrading alpha-amylase, beta amylase, aminopeptidase, carboxypeptidase, catalase, chitinase, cellulolytic enzyme, cutinase, cyclodextrin glycosyltransferase, deoxyribonuclease, esterase, glucan 1,4-alpha-maltotetrahydrolase, glucanase, galactanase, alpha-galactosidase, beta-galactosidase, glucose oxidase, alpha-glucosidase, beta-glucosidase, haloperoxidase, hemicellulolytic enzyme, invertase, laccase, lipase, mannanase, mannosidase, oxidase, pectinolytic enzymes, peptidoglutaminase, peroxidase, phospholipase, phytase, polyphenoloxidase, proteolytic enzyme, ribonuclease, transglutaminase, and xylanase; wherein the one or more additional enzyme is comprised in an amount of 0.01-1,000 mg enzyme protein (mgEP) per kg flour.

14. A baking composition comprising a mature thermostable variant of a parent glucoamylase as defined in claim 1.

15. The baking composition of claim 14, which also comprises one or more additional enzyme selected from the group consisting of a alpha-amylase, maltogenic amylase, beta amylase, aminopeptidase, carboxypeptidase, catalase, cellulolytic enzyme, chitinase, cutinase, cyclodextrin glycosyltransferase, deoxyribonuclease, esterase, glucan 1,4-alpha-maltotetrahydrolase, glucanase, galactanase, alpha-galactosidase, beta-galactosidase, glucose oxidase, alpha-glucosidase, beta-glucosidase, haloperoxidase, hemicellulolytic enzyme, invertase, laccase, lipase, mannanase, mannosidase, oxidase, pectinolytic enzymes, peptidoglutaminase, peroxidase, phospholipase, phytase, polyphenoloxidase, proteolytic enzyme, ribonuclease, transglutaminase, and xylanase.

16. The baking composition of claim 14, which also comprises flour, sugar, yeast, salt and/or fat.

17. (canceled)

18. (canceled)

19. The method according to claim 4, wherein the at least one amino acid modification comprises a substitution in one or more or all of the positions corresponding to R1A, P2N, P4S, P11F, T65A, K79V and Q327F in SEQ ID NO:1.

20. The method according to claim 4, wherein the at least one amino acid modification comprises a substitution in one or more or all of the positions corresponding to R1A, G6S, G7T, R31F, K34Y, K79V, S103N, A132P, D445N, V447S, S481P, D566T, T568V, Q594R and F595S in SEQ ID NO:1.

21. The method according to claim 4, wherein the at least one amino acid modification comprises a substitution in one

or more or all of the positions corresponding to R1A, G6S, G7T, R31F, K34Y, E50R, K79V, S103N, A132P, D445N, V447S, S481P, T484P, E501A, N539P, D566T, T568V, Q594R and F595S in SEQ ID NO:1.

* * * * *